Yellow Taxi Pickup Prediction - New York

July 1, 2018

1 Taxi demand prediction in New York City

In [1]:
```
#Importing Libraries
# pip3 install graphviz
#pip3 install dask
#pip3 install toolz
#pip3 install cloudpickle
# https://www.youtube.com/watch?v=ieW3G7ZzRZ0
# https://github.com/dask/dask-tutorial
# please do go through this python notebook: https://github.com/dask/dask-tutorial/blo
#import dask.dataframe as dd#similar to pandas

import pandas as pd#pandas to create small dataframes

# pip3 install foliun
# if this doesnt work refere install_folium.JPG in drive
import folium #open street map

# unix time: https://www.unixtimestamp.com/
import datetime #Convert to unix time

import time #Convert to unix time

# if numpy is not installed already : pip3 install numpy
import numpy as np#Do aritmetic operations on arrays

# matplotlib: used to plot graphs
import matplotlib
# matplotlib.use('nbagg') : matplotlib uses this protocall which makes plots more user
matplotlib.use('nbagg')
import matplotlib.pylab as plt
import seaborn as sns#Plots
from matplotlib import rcParams#Size of plots

# this lib is used while we calculate the stight line distance between two (lat,lon) p
import gpxpy.geo #Get the haversine distance
```

```
from sklearn.cluster import MiniBatchKMeans, KMeans#Clustering
import math
import pickle
import os

# download migwin: https://mingw-w64.org/doku.php/download/mingw-builds
# install it in your system and keep the path, migw_path ='installed path'
#mingw_path = 'C:\\Program Files\\mingw-w64\\x86_64-5.3.0-posix-seh-rt_v4-rev0\\mingw6
#os.environ['PATH'] = mingw_path + ';' + os.environ['PATH']

# to install xgboost: pip3 install xgboost
# if it didnt happen check install_xgboost.JPG
import xgboost as xgb

# to install sklearn: pip install -U scikit-learn
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
from sklearn.metrics import mean_absolute_error
import warnings
warnings.filterwarnings("ignore")
```

2 Data Information

Ge the data from : http://www.nyc.gov/html/tlc/html/about/trip_record_data.shtml (2016 data) The data used in the attached datasets were collected and provided to the NYC Taxi and Limousine Commission (TLC)

2.1 Information on taxis:

Yellow Taxi: Yellow Medallion Taxicabs

These are the famous NYC yellow taxis that provide transportation exclusively through street-hails. The number of taxicabs is limited by a finite number of medallions issued by the TLC. You access this mode of transportation by standing in the street and hailing an available taxi with your hand. The pickups are not pre-arranged.

For Hire Vehicles (FHVs)

FHV transportation is accessed by a pre-arrangement with a dispatcher or limo company. These FHVs are not permitted to pick up passengers via street hails, as those rides are not considered pre-arranged.

Green Taxi: Street Hail Livery (SHL)

The SHL program will allow livery vehicle owners to license and outfit their vehicles with green borough taxi branding, meters, credit card machines, and ultimately the right to accept street hails in addition to pre-arranged rides.

Credits: Quora

Footnote:

In this notebook we are considering only the yellow taxis for the time period between Jan - Mar 2015 & Jan - Mar 2016

3 Data Collection

We Have collected all yellow taxi trips data from jan-2015 to dec-2016(Will be using only 2015 data)

file name
file name size
number of records
number of features
yellow_tripdata_2016-01

1. 59G
   10906858
   19

yellow_tripdata_2016-02

1. 66G
   11382049
   19
   yellow_tripdata_2016-03

   1. 78G
      12210952
      19
      yellow_tripdata_2016-04

      1. 74G
         11934338
         19

yellow_tripdata_2016-05

1. 73G
   11836853
   19

yellow_tripdata_2016-06

1. 62G
   11135470
   19

yellow_tripdata_2016-07
884Mb
10294080
17
yellow_tripdata_2016-08
854Mb
9942263
17

yellow_tripdata_2016-09
870Mb
10116018
17
yellow_tripdata_2016-10
933Mb
10854626
17
yellow_tripdata_2016-11
868Mb
10102128
17
yellow_tripdata_2016-12
897Mb
10449408
17
yellow_tripdata_2015-01
1.84Gb
12748986
19
yellow_tripdata_2015-02
1.81Gb
12450521
19
yellow_tripdata_2015-03
1.94Gb
13351609
19
yellow_tripdata_2015-04
1.90Gb
13071789
19
yellow_tripdata_2015-05
1.91Gb
13158262
19
yellow_tripdata_2015-06
1.79Gb
12324935
19
yellow_tripdata_2015-07
1.68Gb
11562783
19
yellow_tripdata_2015-08
1.62Gb
11130304
19

yellow_tripdata_2015-09
1.63Gb
11225063
19
yellow_tripdata_2015-10
1.79Gb
12315488
19
yellow_tripdata_2015-11
1.65Gb
11312676
19
yellow_tripdata_2015-12
1.67Gb
11460573
19

```
In [2]: #Looking at the features
        month = pd.read_csv('yellow_tripdata_2015-01.csv')
        print(month.columns)
```

```
Index(['VendorID', 'tpep_pickup_datetime', 'tpep_dropoff_datetime',
       'passenger_count', 'trip_distance', 'pickup_longitude',
       'pickup_latitude', 'RateCodeID', 'store_and_fwd_flag',
       'dropoff_longitude', 'dropoff_latitude', 'payment_type', 'fare_amount',
       'extra', 'mta_tax', 'tip_amount', 'tolls_amount',
       'improvement_surcharge', 'total_amount'],
      dtype='object')
```

3.1 Features in the dataset:

```
<tr>
    <th>Field Name</th>
    <th>Description</th>
</tr>
<tr>
    <td>VendorID</td>
    <td>
    A code indicating the TPEP provider that provided the record.
    <ol>
        <li>Creative Mobile Technologies</li>
        <li>VeriFone Inc.</li>
    </ol>
    </td>
</tr>
<tr>
    <td>tpep_pickup_datetime</td>
```

```
    <td>The date and time when the meter was engaged.</td>
</tr>
<tr>
    <td>tpep_dropoff_datetime</td>
    <td>The date and time when the meter was disengaged.</td>
</tr>
<tr>
    <td>Passenger_count</td>
    <td>The number of passengers in the vehicle. This is a driver-entered value.</td>
</tr>
<tr>
    <td>Trip_distance</td>
    <td>The elapsed trip distance in miles reported by the taximeter.</td>
</tr>
<tr>
    <td>Pickup_longitude</td>
    <td>Longitude where the meter was engaged.</td>
</tr>
<tr>
    <td>Pickup_latitude</td>
    <td>Latitude where the meter was engaged.</td>
</tr>
<tr>
    <td>RateCodeID</td>
    <td>The final rate code in effect at the end of the trip.
    <ol>
        <li> Standard rate </li>
        <li> JFK </li>
        <li> Newark </li>
        <li> Nassau or Westchester</li>
        <li> Negotiated fare </li>
        <li> Group ride</li>
    </ol>
    </td>
</tr>
<tr>
    <td>Store_and_fwd_flag</td>
    <td>This flag indicates whether the trip record was held in vehicle memory before sending t
    <br\>Y= store and forward trip
    <br\>N= not a store and forward trip
    </td>
</tr>

<tr>
    <td>Dropoff_longitude</td>
    <td>Longitude where the meter was disengaged.</td>
</tr>
<tr>
```

```
    <td>Dropoff_ latitude</td>
    <td>Latitude where the meter was disengaged.</td>
</tr>
<tr>
    <td>Payment_type</td>
    <td>A numeric code signifying how the passenger paid for the trip.
    <ol>
        <li> Credit card </li>
        <li> Cash </li>
        <li> No charge </li>
        <li> Dispute</li>
        <li> Unknown </li>
        <li> Voided trip</li>
    </ol>
    </td>
</tr>
<tr>
    <td>Fare_amount</td>
    <td>The time-and-distance fare calculated by the meter.</td>
</tr>
<tr>
    <td>Extra</td>
    <td>Miscellaneous extras and surcharges. Currently, this only includes. the $0.50 and $1 ru
</tr>
<tr>
    <td>MTA_tax</td>
    <td>0.50 MTA tax that is automatically triggered based on the metered rate in use.</td>
</tr>
<tr>
    <td>Improvement_surcharge</td>
    <td>0.30 improvement surcharge assessed trips at the flag drop. the improvement surcharge l
</tr>
<tr>
    <td>Tip_amount</td>
    <td>Tip amount  This field is automatically populated for credit card tips.Cash tips are no
</tr>
<tr>
    <td>Tolls_amount</td>
    <td>Total amount of all tolls paid in trip.</td>
</tr>
<tr>
    <td>Total_amount</td>
    <td>The total amount charged to passengers. Does not include cash tips.</td>
</tr>
```

4 ML Problem Formulation

Time-series forecasting and Regression

- To find number of pickups, given location cordinates(latitude and longitude) and time, in the query reigion and surrounding regions.

To solve the above we would be using data collected in Jan - Mar 2015 to predict the pickups in Jan - Mar 2016.

5 Performance metrics

1. Mean Absolute percentage error.
2. Mean Squared error.

5.1 Data Cleaning

In this section we will be doing univariate analysis and removing outlier/illegitimate values which may be caused due to some error

```
In [3]: #table below shows few datapoints along with all our features
        month.head(5)
```

```
Out[3]:    VendorID tpep_pickup_datetime tpep_dropoff_datetime  passenger_count  \
        0         2  2015-01-15 19:05:39   2015-01-15 19:23:42                1
        1         1  2015-01-10 20:33:38   2015-01-10 20:53:28                1
        2         1  2015-01-10 20:33:38   2015-01-10 20:43:41                1
        3         1  2015-01-10 20:33:39   2015-01-10 20:35:31                1
        4         1  2015-01-10 20:33:39   2015-01-10 20:52:58                1

           trip_distance  pickup_longitude  pickup_latitude  RateCodeID  \
        0           1.59        -73.993896        40.750111           1
        1           3.30        -74.001648        40.724243           1
        2           1.80        -73.963341        40.802788           1
        3           0.50        -74.009087        40.713818           1
        4           3.00        -73.971176        40.762428           1

          store_and_fwd_flag  dropoff_longitude  dropoff_latitude  payment_type  \
        0                  N         -73.974785         40.750618             1
        1                  N         -73.994415         40.759109             1
        2                  N         -73.951820         40.824413             2
        3                  N         -74.004326         40.719986             2
        4                  N         -74.004181         40.742653             2

           fare_amount  extra  mta_tax  tip_amount  tolls_amount  \
        0         12.0    1.0      0.5        3.25           0.0
        1         14.5    0.5      0.5        2.00           0.0
        2          9.5    0.5      0.5        0.00           0.0
        3          3.5    0.5      0.5        0.00           0.0
        4         15.0    0.5      0.5        0.00           0.0
```

```
   improvement_surcharge  total_amount
0                    0.3         17.05
1                    0.3         17.80
2                    0.3         10.80
3                    0.3          4.80
4                    0.3         16.30
```

5.1.1 1. Pickup Latitude and Pickup Longitude

It is inferred from the source https://www.flickr.com/places/info/2459115 that New York is bounded by the location cordinates(lat,long) - (40.5774, -74.15) & (40.9176,-73.7004) so hence any cordinates not within these cordinates are not considered by us as we are only concerned with pickups which originate within New York.

In [4]:
```
# Plotting pickup cordinates which are outside the bounding box of New-York
# we will collect all the points outside the bounding box of newyork city to outlier_l
outlier_locations = month[((month.pickup_longitude <= -74.15) | (month.pickup_latitude
                    (month.pickup_longitude >= -73.7004) | (month.pickup_latitude >= 40

# creating a map with the a base location
# read more about the folium here: http://folium.readthedocs.io/en/latest/quickstart.h

# note: you dont need to remember any of these, you dont need indeepth knowledge on th

map_osm = folium.Map(location=[40.734695, -73.990372], tiles='Stamen Toner')

# we will spot only first 100 outliers on the map, plotting all the outliers will take
sample_locations = outlier_locations.head(10000)
for i,j in sample_locations.iterrows():
    if int(j['pickup_latitude']) != 0:
        folium.Marker(list((j['pickup_latitude'],j['pickup_longitude']))).add_to(map_os
map_osm
```

Out[4]:
```
<folium.folium.Map at 0x154725e733c8>
```

5.1.2 2. Dropoff Latitude & Dropoff Longitude

It is inferred from the source https://www.flickr.com/places/info/2459115 that New York is bounded by the location cordinates(lat,long) - (40.5774, -74.15) & (40.9176,-73.7004) so hence any cordinates not within these cordinates are not considered by us as we are only concerned with dropoffs which are within New York.

In [5]:
```
# Plotting dropoff cordinates which are outside the bounding box of New-York
# we will collect all the points outside the bounding box of newyork city to outlier_l
outlier_locations = month[((month.dropoff_longitude <= -74.15) | (month.dropoff_latitu
                    (month.dropoff_longitude >= -73.7004) | (month.dropoff_latitude >= 4

# creating a map with the a base location
```

```
# read more about the folium here: http://folium.readthedocs.io/en/latest/quickstart.h

# note: you dont need to remember any of these, you dont need indeepth knowledge on th

map_osm = folium.Map(location=[40.734695, -73.990372], tiles='Stamen Toner')

# we will spot only first 100 outliers on the map, plotting all the outliers will take
sample_locations = outlier_locations.head(10000)
for i,j in sample_locations.iterrows():
    if int(j['pickup_latitude']) != 0:
        folium.Marker(list((j['dropoff_latitude'],j['dropoff_longitude']))).add_to(map_
map_osm
```

Out[5]: <folium.folium.Map at 0x15473eda58d0>

Observation:- The observations here are similar to those obtained while analysing pickup latitude and longitude

5.1.3 3. Trip Durations:

According to NYC Taxi & Limousine Commision Regulations the maximum allowed trip duration in a 24 hour interval is 12 hours.

In [6]:
```
os.environ['TZ'] = 'EST'
time.tzset()
time.tzname
```

Out[6]: ('EST', 'EST')

In [7]:
```
#The timestamps are converted to unix so as to get duration(trip-time) & speed also pi

# in out data we have time in the formate "YYYY-MM-DD HH:MM:SS" we convert thiss sting
# https://stackoverflow.com/a/27914405
def convert_to_unix(s):
    return time.mktime(datetime.datetime.strptime(s, "%Y-%m-%d %H:%M:%S").timetuple())



# we return a data frame which contains the columns
# 1.'passenger_count' : self explanatory
# 2.'trip_distance' : self explanatory
# 3.'pickup_longitude' : self explanatory
# 4.'pickup_latitude' : self explanatory
# 5.'dropoff_longitude' : self explanatory
# 6.'dropoff_latitude' : self explanatory
# 7.'total_amount' : total fair that was paid
# 8.'trip_times' : duration of each trip
# 9.'pickup_times : pickup time converted into unix time
# 10.'Speed' : velocity of each trip
```

```
def return_with_trip_times(month):
    duration = month[['tpep_pickup_datetime','tpep_dropoff_datetime']]#.compute()
    #pickups and dropoffs to unix time
    duration_pickup = [convert_to_unix(x) for x in duration['tpep_pickup_datetime'].val
    duration_drop = [convert_to_unix(x) for x in duration['tpep_dropoff_datetime'].valu
    #calculate duration of trips
    durations = (np.array(duration_drop) - np.array(duration_pickup))/float(60)

    #append durations of trips and speed in miles/hr to a new dataframe
    new_frame = month[['passenger_count','trip_distance','pickup_longitude','pickup_lat

    new_frame['trip_times'] = durations
    new_frame['pickup_times'] = duration_pickup
    new_frame['Speed'] = 60*(new_frame['trip_distance']/new_frame['trip_times'])

    return new_frame

# print(frame_with_durations.head())
#  passenger_count        trip_distance        pickup_longitude        pickup_latitude
#   1                  1.59              -73.993896                40.750111
#   1                       3.30            -74.001648              40.724243
#   1                       1.80             -73.963341              40.802788
#   1                       0.50            -74.009087              40.713818
#   1                       3.00            -73.971176              40.762428
frame_with_durations = return_with_trip_times(month)
```

In [8]:
```
frame_with_durations.head()
```

Out[8]:
```
   passenger_count  trip_distance  pickup_longitude  pickup_latitude  \
0                1           1.59        -73.993896        40.750111
1                1           3.30        -74.001648        40.724243
2                1           1.80        -73.963341        40.802788
3                1           0.50        -74.009087        40.713818
4                1           3.00        -73.971176        40.762428

   dropoff_longitude  dropoff_latitude  total_amount  trip_times  \
0         -73.974785         40.750618         17.05   18.050000
1         -73.994415         40.759109         17.80   19.833333
2         -73.951820         40.824413         10.80   10.050000
3         -74.004326         40.719986          4.80    1.866667
4         -74.004181         40.742653         16.30   19.316667

   pickup_times      Speed
0  1.421367e+09   5.285319
1  1.420940e+09   9.983193
2  1.420940e+09  10.746269
3  1.420940e+09  16.071429
4  1.420940e+09   9.318378
```

```
In [9]: # the skewed box plot shows us the presence of outliers
        sns.boxplot(y="trip_times", data =frame_with_durations)
        plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

```
In [10]: #calculating 0-100th percentile to find a the correct percentile value for removal of
         var =frame_with_durations["trip_times"].values
         var = np.sort(var,axis = None)
         for i in range(0,100,10):
             print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
         print ("100 percentile value is ",var[-1])
```

```
0 percentile value is -1211.0166666666667
10 percentile value is 3.8333333333333335
20 percentile value is 5.383333333333334
30 percentile value is 6.816666666666666
40 percentile value is 8.3
50 percentile value is 9.95
60 percentile value is 11.866666666666667
70 percentile value is 14.283333333333333
80 percentile value is 17.633333333333333
90 percentile value is 23.45
100 percentile value is  548555.6333333333
```

```
In [11]: #looking further from the 99th percecntile
         for i in range(90,100):
             print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
         print ("100 percentile value is ",var[-1])
```

```
90 percentile value is 23.45
91 percentile value is 24.35
92 percentile value is 25.383333333333333
93 percentile value is 26.55
94 percentile value is 27.933333333333334
95 percentile value is 29.583333333333332
96 percentile value is 31.683333333333334
97 percentile value is 34.46666666666667
98 percentile value is 38.71666666666667
99 percentile value is 46.75
100 percentile value is  548555.6333333333
```

```
In [12]: ### No of points that are trip time lessthan 1
         len(var[var<1])
```

Out[12]: 103426

In [9]:
```
#removing data based on our analysis and TLC regulations
frame_with_durations_modified=frame_with_durations[(frame_with_durations.trip_times>1)
```

In [14]:
```
#box-plot after removal of outliers
sns.boxplot(y="trip_times", data =frame_with_durations_modified)
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

In [15]:
```
#pdf of trip-times after removing the outliers
sns.FacetGrid(frame_with_durations_modified,size=6) \
      .map(sns.kdeplot,"trip_times") \
      .add_legend();
plt.show();
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

In [10]:
```
#converting the values to log-values to chec for log-normal
import math
frame_with_durations_modified['log_times']=[math.log(i) for i in frame_with_durations_
```

In [17]:
```
#pdf of log-values
sns.FacetGrid(frame_with_durations_modified,size=6) \
      .map(sns.kdeplot,"log_times") \
      .add_legend();
plt.show();
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

In [18]:
```
#Q-Q plot for checking if trip-times is log-normal
import scipy
scipy.stats.probplot(frame_with_durations_modified['log_times'].values, plot=plt)
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

5.1.4 4. Speed

In [20]:
```
# check for any outliers in the data after trip duration outliers removed
# box-plot for speeds with outliers
frame_with_durations_modified['Speed'] = 60*(frame_with_durations_modified['trip_dista
sns.boxplot(y="Speed", data =frame_with_durations_modified)
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

In [21]:
```
#calculating speed values at each percntile 0,10,20,30,40,50,60,70,80,90,100
var =frame_with_durations_modified["Speed"].values
var = np.sort(var,axis = None)
for i in range(0,100,10):
    print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
print("100 percentile value is ",var[-1])
```

```
0 percentile value is 0.0
10 percentile value is 6.409495548961425
20 percentile value is 7.80952380952381
30 percentile value is 8.929133858267717
40 percentile value is 9.98019801980198
50 percentile value is 11.06865671641791
60 percentile value is 12.286689419795222
70 percentile value is 13.796407185628745
80 percentile value is 15.963224893917962
90 percentile value is 20.186915887850468
100 percentile value is  192857142.85714284
```

In [22]:
```
#calculating speed values at each percntile 90,91,92,93,94,95,96,97,98,99,100
for i in range(90,100):
    print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
print("100 percentile value is ",var[-1])
```

```
90 percentile value is 20.186915887850468
91 percentile value is 20.91645569620253
92 percentile value is 21.752988047808763
93 percentile value is 22.721893491124263
94 percentile value is 23.844155844155843
95 percentile value is 25.182552504038775
96 percentile value is 26.80851063829787
97 percentile value is 28.84304932735426
98 percentile value is 31.591128254580514
99 percentile value is 35.7513566847558
100 percentile value is  192857142.85714284
```

```
In [23]: #calculating speed values at each percntile 99.0,99.1,99.2,99.3,99.4,99.5,99.6,99.7,9
         for i in np.arange(0.0, 1.0, 0.1):
             print("{} percentile value is {}".format(99+i,var[int(len(var)*(float(99+i)/100))]
         print("100 percentile value is ",var[-1])
```

```
99.0 percentile value is 35.7513566847558
99.1 percentile value is 36.31084727468969
99.2 percentile value is 36.91470054446461
99.3 percentile value is 37.588235294117645
99.4 percentile value is 38.33035714285714
99.5 percentile value is 39.17580340264651
99.6 percentile value is 40.15384615384615
99.7 percentile value is 41.338301043219076
99.8 percentile value is 42.86631016042781
99.9 percentile value is 45.3107822410148
100 percentile value is  192857142.85714284
```

```
In [12]: #removing further outliers based on the 99.9th percentile value
         frame_with_durations_modified=frame_with_durations[(frame_with_durations.Speed>0) & (
```

```
In [13]: #avg.speed of cabs in New-York
         frame_with_durations_modified.Speed.mean()
```

```
Out[13]: 12.45017399603333
```

The avg speed in Newyork speed is 12.45miles/hr, so a cab driver can travel 2 miles per 10min on avg.

5.1.5 4. Trip Distance

```
In [26]: # up to now we have removed the outliers based on trip durations and cab speeds
         # lets try if there are any outliers in trip distances
         # box-plot showing outliers in trip-distance values
         sns.boxplot(y="trip_distance", data =frame_with_durations_modified)
         plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

```
In [27]: #calculating trip distance values at each percntile 0,10,20,30,40,50,60,70,80,90,100
         var =frame_with_durations_modified["trip_distance"].values
         var = np.sort(var,axis = None)
         for i in range(0,100,10):
             print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
         print("100 percentile value is ",var[-1])
```

```
0 percentile value is 0.01
10 percentile value is 0.66
20 percentile value is 0.9
30 percentile value is 1.1
40 percentile value is 1.39
50 percentile value is 1.69
60 percentile value is 2.07
70 percentile value is 2.6
80 percentile value is 3.6
90 percentile value is 5.97
100 percentile value is  258.9
```

```
In [28]: #calculating trip distance values at each percntile 90,91,92,93,94,95,96,97,98,99,100
         for i in range(90,100):
             print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
         print("100 percentile value is ",var[-1])
```

```
90 percentile value is 5.97
91 percentile value is 6.45
92 percentile value is 7.07
93 percentile value is 7.85
94 percentile value is 8.72
95 percentile value is 9.6
96 percentile value is 10.6
97 percentile value is 12.1
98 percentile value is 16.03
99 percentile value is 18.17
100 percentile value is  258.9
```

```
In [29]: #calculating trip distance values at each percntile 99.0,99.1,99.2,99.3,99.4,99.5,99.
         for i in np.arange(0.0, 1.0, 0.1):
             print("{} percentile value is {}".format(99+i,var[int(len(var)*(float(99+i)/100))]
         print("100 percentile value is ",var[-1])
```

```
99.0 percentile value is 18.17
99.1 percentile value is 18.37
99.2 percentile value is 18.6
99.3 percentile value is 18.83
99.4 percentile value is 19.13
99.5 percentile value is 19.5
99.6 percentile value is 19.96
99.7 percentile value is 20.5
99.8 percentile value is 21.22
99.9 percentile value is 22.57
100 percentile value is  258.9
```

In [14]:
```
#removing further outliers based on the 99.9th percentile value
frame_with_durations_modified=frame_with_durations[(frame_with_durations.trip_distanc
```

In [31]:
```
#box-plot after removal of outliers
sns.boxplot(y="trip_distance", data = frame_with_durations_modified)
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

5.1.6 5. Total Fare

In [32]:
```
# up to now we have removed the outliers based on trip durations, cab speeds, and tri
# lets try if there are any outliers in based on the total_amount
# box-plot showing outliers in fare
sns.boxplot(y="total_amount", data =frame_with_durations_modified)
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

In [33]:
```
#calculating total fare amount values at each percntile 0,10,20,30,40,50,60,70,80,90,
var = frame_with_durations_modified["total_amount"].values
var = np.sort(var,axis = None)
for i in range(0,100,10):
    print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
print("100 percentile value is ",var[-1])
```

```
0 percentile value is -242.55
10 percentile value is 6.3
20 percentile value is 7.8
30 percentile value is 8.8
40 percentile value is 9.8
50 percentile value is 11.16
60 percentile value is 12.8
70 percentile value is 14.8
80 percentile value is 18.3
90 percentile value is 25.8
100 percentile value is  3950611.6
```

In [34]:
```
#calculating total fare amount values at each percntile 90,91,92,93,94,95,96,97,98,99
for i in range(90,100):
    print("{} percentile value is {}".format(i,var[int(len(var)*(float(i)/100))]))
print("100 percentile value is ",var[-1])
```

```
90 percentile value is 25.8
91 percentile value is 27.3
92 percentile value is 29.3
93 percentile value is 31.8
94 percentile value is 34.8
95 percentile value is 38.53
96 percentile value is 42.6
97 percentile value is 48.13
98 percentile value is 58.13
99 percentile value is 66.13
100 percentile value is  3950611.6
```

```
In [35]: #calculating total fare amount values at each percntile 99.0,99.1,99.2,99.3,99.4,99.5
         for i in np.arange(0.0, 1.0, 0.1):
             print("{} percentile value is {}".format(99+i,var[int(len(var)*(float(99+i)/100))]
         print("100 percentile value is ",var[-1])
```

```
99.0 percentile value is 66.13
99.1 percentile value is 68.13
99.2 percentile value is 69.6
99.3 percentile value is 69.6
99.4 percentile value is 69.73
99.5 percentile value is 69.75
99.6 percentile value is 69.76
99.7 percentile value is 72.58
99.8 percentile value is 75.35
99.9 percentile value is 88.28
100 percentile value is  3950611.6
```

Observation:- As even the 99.9th percentile value doesnt look like an outlier,as there is not much difference between the 99.8th percentile and 99.9th percentile, we move on to do graphical analyis

```
In [36]: #below plot shows us the fare values(sorted) to find a sharp increase to remove those
         # plot the fare amount excluding last two values in sorted data
         plt.plot(var[:-2])
         plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

```
In [37]: # a very sharp increase in fare values can be seen
         # plotting last three total fare values, and we can observe there is share increase i
         plt.plot(var[-3:])
         plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

```
In [38]: #now looking at values not including the last two points we again find a drastic incr
         # we plot last 50 values excluding last two values
         plt.plot(var[-50:-2])
         plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

5.2 Remove all outliers/erronous points.

```
In [15]: #removing all outliers based on our univariate analysis above
         def remove_outliers(new_frame):


             a = new_frame.shape[0]
             print ("Number of pickup records = ",a)
             temp_frame = new_frame[((new_frame.dropoff_longitude >= -74.15) & (new_frame.dropo
                                 (new_frame.dropoff_latitude >= 40.5774) & (new_frame.dropoff_la
                                 ((new_frame.pickup_longitude >= -74.15) & (new_frame.pickup_lat
                                 (new_frame.pickup_longitude <= -73.7004) & (new_frame.pickup_la
             b = temp_frame.shape[0]
             print ("Number of outlier coordinates lying outside NY boundaries:",(a-b))


             temp_frame = new_frame[(new_frame.trip_times > 0) & (new_frame.trip_times < 720)]
             c = temp_frame.shape[0]
             print ("Number of outliers from trip times analysis:",(a-c))


             temp_frame = new_frame[(new_frame.trip_distance > 0) & (new_frame.trip_distance <
             d = temp_frame.shape[0]
             print ("Number of outliers from trip distance analysis:",(a-d))

             temp_frame = new_frame[(new_frame.Speed <= 65) & (new_frame.Speed >= 0)]
             e = temp_frame.shape[0]
             print ("Number of outliers from speed analysis:",(a-e))

             temp_frame = new_frame[(new_frame.total_amount <1000) & (new_frame.total_amount >0
             f = temp_frame.shape[0]
             print ("Number of outliers from fare analysis:",(a-f))
```

```
    new_frame = new_frame[((new_frame.dropoff_longitude >= -74.15) & (new_frame.dropo
                           (new_frame.dropoff_latitude >= 40.5774) & (new_frame.dropoff_la
                           ((new_frame.pickup_longitude >= -74.15) & (new_frame.pickup_lat
                           (new_frame.pickup_longitude <= -73.7004) & (new_frame.pickup_la

    new_frame = new_frame[(new_frame.trip_times > 0) & (new_frame.trip_times < 720)]
    new_frame = new_frame[(new_frame.trip_distance > 0) & (new_frame.trip_distance < 2
    new_frame = new_frame[(new_frame.Speed < 45.31) & (new_frame.Speed > 0)]
    new_frame = new_frame[(new_frame.total_amount <1000) & (new_frame.total_amount >0)

    print ("Total outliers removed",a - new_frame.shape[0])
    print ("---")
    return new_frame
```

```
In [16]: print ("Removing outliers in the month of Jan-2015")
         print ("----")
         frame_with_durations_outliers_removed = remove_outliers(frame_with_durations)
         print("fraction of data points that remain after removing outliers", float(len(frame_w
```

```
Removing outliers in the month of Jan-2015
----
Number of pickup records =  12748986
Number of outlier coordinates lying outside NY boundaries: 293919
Number of outliers from trip times analysis: 23889
Number of outliers from trip distance analysis: 92597
Number of outliers from speed analysis: 24473
Number of outliers from fare analysis: 5275
Total outliers removed 377910
---
fraction of data points that remain after removing outliers 0.9703576425607495
```

6 Data-preperation

6.1 Clustering/Segmentation

```
In [43]: #trying different cluster sizes to choose the right K in K-means
         coords = frame_with_durations_outliers_removed[['pickup_latitude', 'pickup_longitude']
         neighbours=[]

         def find_min_distance(cluster_centers, cluster_len):
             nice_points = 0
             wrong_points = 0
             less2 = []
             more2 = []
             min_dist=1000
```

```
        for i in range(0, cluster_len):
            nice_points = 0
            wrong_points = 0
            for j in range(0, cluster_len):
                if j!=i:
                    distance = gpxpy.geo.haversine_distance(cluster_centers[i][0], cluste
                    min_dist = min(min_dist,distance/(1.60934*1000))
                    if (distance/(1.60934*1000)) <= 2:
                        nice_points +=1
                    else:
                        wrong_points += 1
            less2.append(nice_points)
            more2.append(wrong_points)
        neighbours.append(less2)
        print ("On choosing a cluster size of ",cluster_len,"\nAvg. Number of Clusters wi

    def find_clusters(increment):
        kmeans = MiniBatchKMeans(n_clusters=increment, batch_size=10000,random_state=42).
        frame_with_durations_outliers_removed['pickup_cluster'] = kmeans.predict(frame_wi
        cluster_centers = kmeans.cluster_centers_
        cluster_len = len(cluster_centers)
        return cluster_centers, cluster_len

    # we need to choose number of clusters so that, there are more number of cluster regi
    #that are close to any cluster center
    # and make sure that the minimum inter cluster should not be very less
    for increment in range(10, 100, 10):
        cluster_centers, cluster_len = find_clusters(increment)
        find_min_distance(cluster_centers, cluster_len)
```

```
On choosing a cluster size of  10
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 2.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 8.0
Min inter-cluster distance =  1.0933194607372518
---
On choosing a cluster size of  20
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 4.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 16.0
Min inter-cluster distance =  0.7123318236197774
---
On choosing a cluster size of  30
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 8.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 22.0
Min inter-cluster distance =  0.5179286172497254
---
On choosing a cluster size of  40
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 9.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 31.0
```

```
Min inter-cluster distance =  0.5064095487015859
---
On choosing a cluster size of  50
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 12.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 38.0
Min inter-cluster distance =  0.36495419250817024
---
On choosing a cluster size of  60
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 14.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 46.0
Min inter-cluster distance =  0.346654501371586
---
On choosing a cluster size of  70
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 16.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 54.0
Min inter-cluster distance =  0.30468071844965394
---
On choosing a cluster size of  80
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 18.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 62.0
Min inter-cluster distance =  0.29187627608454664
---
On choosing a cluster size of  90
Avg. Number of Clusters within the vicinity (i.e. intercluster-distance < 2): 21.0
Avg. Number of Clusters outside the vicinity (i.e. intercluster-distance > 2): 69.0
Min inter-cluster distance =  0.18237562550345013
---
```

6.1.1 Inference:

- The main objective was to find a optimal min. distance(Which roughly estimates to the radius of a cluster) between the clusters which we got was 40

```
In [18]: # if check for the 50 clusters you can observe that there are two clusters with only
         # so we choose 40 clusters for solve the further problem

         # Getting 40 clusters using the kmeans
         kmeans = MiniBatchKMeans(n_clusters=40, batch_size=10000,random_state=0).fit(coords)
         frame_with_durations_outliers_removed['pickup_cluster'] = kmeans.predict(frame_with_du
```

6.1.2 Plotting the cluster centers:

```
In [20]: # Plotting the cluster centers on OSM
         cluster_centers = kmeans.cluster_centers_
         cluster_len = len(cluster_centers)
         map_osm = folium.Map(location=[40.734695, -73.990372], tiles='Stamen Toner')
         for i in range(cluster_len):
```

```
        folium.Marker(list((cluster_centers[i][0],cluster_centers[i][1])), popup=(str(clus
    map_osm
```

Out[20]: <folium.folium.Map at 0x147d7a89d240>

6.1.3 Plotting the clusters:

In [20]:
```
#Visualising the clusters on a map
def plot_clusters(frame):
    city_long_border = (-74.03, -73.75)
    city_lat_border = (40.63, 40.85)
    fig, ax = plt.subplots(ncols=1, nrows=1)
    ax.scatter(frame.pickup_longitude.values[:100000], frame.pickup_latitude.values[:
               c=frame.pickup_cluster.values[:100000], cmap='tab20', alpha=0.2)
    ax.set_xlim(city_long_border)
    ax.set_ylim(city_lat_border)
    ax.set_xlabel('Longitude')
    ax.set_ylabel('Latitude')
    plt.show()

plot_clusters(frame_with_durations_outliers_removed)
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

6.2 Time-binning

In [19]:
```
#Refer:https://www.unixtimestamp.com/
# 1420070400 : 2015-01-01 00:00:00
# 1422748800 : 2015-02-01 00:00:00
# 1425168000 : 2015-03-01 00:00:00
# 1427846400 : 2015-04-01 00:00:00
# 1430438400 : 2015-05-01 00:00:00
# 1433116800 : 2015-06-01 00:00:00

# 1451606400 : 2016-01-01 00:00:00
# 1454284800 : 2016-02-01 00:00:00
# 1456790400 : 2016-03-01 00:00:00
# 1459468800 : 2016-04-01 00:00:00
# 1462060800 : 2016-05-01 00:00:00
# 1464739200 : 2016-06-01 00:00:00

def add_pickup_bins(frame,month,year):
    unix_pickup_times=[i for i in frame['pickup_times'].values]
    unix_times = np.array([[1420070400,1422748800,1425168000,1427846400,1430438400,143
                           [1451606400,1454284800,1456790400,1459468800,1462060800,1464739200
```

```
    unix_times = list(unix_times + 18000)

    start_pickup_unix=unix_times[year-2015][month-1]
    # https://www.timeanddate.com/time/zones/est
    # (int((i-start_pickup_unix)/600)+33) : our unix time is in gmt to we are convert
    tenminutewise_binned_unix_pickup_times=[(int((i-start_pickup_unix)/600)) for i in
    frame['pickup_bins'] = np.array(tenminutewise_binned_unix_pickup_times)
    return frame
```

In [20]:
```
# clustering, making pickup bins and grouping by pickup cluster and pickup bins
frame_with_durations_outliers_removed['pickup_cluster'] = kmeans.predict(frame_with_du
jan_2015_frame = add_pickup_bins(frame_with_durations_outliers_removed,1,2015)
jan_2015_groupby = jan_2015_frame[['pickup_cluster','pickup_bins','trip_distance']].gr
```

In [21]:
```
# we add two more columns 'pickup_cluster'(to which cluster it belogns to)
# and 'pickup_bins' (to which 10min intravel the trip belongs to)
jan_2015_frame.head()
```

Out[21]:
```
   passenger_count  trip_distance  pickup_longitude  pickup_latitude  \
0                1           1.59        -73.993896        40.750111
1                1           3.30        -74.001648        40.724243
2                1           1.80        -73.963341        40.802788
3                1           0.50        -74.009087        40.713818
4                1           3.00        -73.971176        40.762428

   dropoff_longitude  dropoff_latitude  total_amount  trip_times  \
0         -73.974785         40.750618         17.05   18.050000
1         -73.994415         40.759109         17.80   19.833333
2         -73.951820         40.824413         10.80   10.050000
3         -74.004326         40.719986          4.80    1.866667
4         -74.004181         40.742653         16.30   19.316667

   pickup_times      Speed  pickup_cluster  pickup_bins
0  1.421367e+09   5.285319              34         2130
1  1.420940e+09   9.983193               2         1419
2  1.420940e+09  10.746269              16         1419
3  1.420940e+09  16.071429              38         1419
4  1.420940e+09   9.318378              22         1419
```

In [22]:
```
# hear the trip_distance represents the number of pickups that are happend in that pa
# this data frame has two indices
# primary index: pickup_cluster (cluster number)
# secondary index : pickup_bins (we devid whole months time into 10min intravels 24*3
jan_2015_groupby.head()
```

Out[22]:
```
                            trip_distance
pickup_cluster pickup_bins
0              0                      104
               1                      200
```

```
2                    208
3                    141
4                    155
```

In [27]:
```
# upto now we cleaned data and prepared data for the month 2015,

# now do the same operations for months Jan, Feb, March of 2016
# 1. get the dataframe which inlcudes only required colums
# 2. adding trip times, speed, unix time stamp of pickup_time
# 4. remove the outliers based on trip_times, speed, trip_duration, total_amount
# 5. add pickup_cluster to each data point
# 6. add pickup_bin (index of 10min intravel to which that trip belongs to)
# 7. group by data, based on 'pickup_cluster' and 'pickuo_bin'

# Data Preparation for the months of Jan,Feb and March 2016
def datapreparation(month,kmeans,month_no,year_no):

    print ("Return with trip times..")

    frame_with_durations = return_with_trip_times(month)

    print ("Remove outliers..")
    frame_with_durations_outliers_removed = remove_outliers(frame_with_durations)

    print ("Estimating clusters..")
    frame_with_durations_outliers_removed['pickup_cluster'] = kmeans.predict(frame_wi
    #frame_with_durations_outliers_removed_2016['pickup_cluster'] = kmeans.predict(fr

    print ("Final groupbying..")
    final_updated_frame = add_pickup_bins(frame_with_durations_outliers_removed,month_
    final_groupby_frame = final_updated_frame[['pickup_cluster','pickup_bins','trip_di

    return final_updated_frame,final_groupby_frame

month_jan_2016 = pd.read_csv('yellow_tripdata_2016-01.csv')
month_feb_2016 = pd.read_csv('yellow_tripdata_2016-02.csv')
month_mar_2016 = pd.read_csv('yellow_tripdata_2016-03.csv')

jan_2016_frame,jan_2016_groupby = datapreparation(month_jan_2016,kmeans,1,2016)
feb_2016_frame,feb_2016_groupby = datapreparation(month_feb_2016,kmeans,2,2016)
mar_2016_frame,mar_2016_groupby = datapreparation(month_mar_2016,kmeans,3,2016)
```

```
Return with trip times..
Remove outliers..
Number of pickup records =  10906858
Number of outlier coordinates lying outside NY boundaries: 214677
Number of outliers from trip times analysis: 27190
Number of outliers from trip distance analysis: 79742
```

```
Number of outliers from speed analysis: 21047
Number of outliers from fare analysis: 4991
Total outliers removed 297784
---
Estimating clusters..
Final groupbying..
Return with trip times..
Remove outliers..
Number of pickup records =  11382049
Number of outlier coordinates lying outside NY boundaries: 223161
Number of outliers from trip times analysis: 27670
Number of outliers from trip distance analysis: 81902
Number of outliers from speed analysis: 22437
Number of outliers from fare analysis: 5476
Total outliers removed 308177
---
Estimating clusters..
Final groupbying..
Return with trip times..
Remove outliers..
Number of pickup records =  12210952
Number of outlier coordinates lying outside NY boundaries: 232444
Number of outliers from trip times analysis: 30868
Number of outliers from trip distance analysis: 87318
Number of outliers from speed analysis: 23889
Number of outliers from fare analysis: 5859
Total outliers removed 324635
---
Estimating clusters..
Final groupbying..
```

6.3 Smoothing

In [26]:
```
# Gets the unique bins where pickup values are present for each each reigion

# for each cluster region we will collect all the indices of 10min intravels in which
# we got an observation that there are some pickpbins that doesnt have any pickups
def return_unq_pickup_bins(frame):
    values = []
    for i in range(0,40):
        new = frame[frame['pickup_cluster'] == i]
        list_unq = list(set(new['pickup_bins']))
        list_unq.sort()
        values.append(list_unq)
    return values
```

In [27]:
```
# for every month we get all indices of 10min intravels in which atleast one pickup g
```

```
#jan
jan_2015_unique = return_unq_pickup_bins(jan_2015_frame)
jan_2016_unique = return_unq_pickup_bins(jan_2016_frame)

#feb
feb_2016_unique = return_unq_pickup_bins(feb_2016_frame)

#march
mar_2016_unique = return_unq_pickup_bins(mar_2016_frame)
```

In [28]:
```
# for each cluster number of 10min intravels with 0 pickups
for i in range(40):
    print("for the ",i,"th cluster number of 10min intavels with zero pickups: ",4464
    print('-'*60)
```

```
for the  0 th cluster number of 10min intavels with zero pickups:  40
------------------------------------------------------------
for the  1 th cluster number of 10min intavels with zero pickups:  1985
------------------------------------------------------------
for the  2 th cluster number of 10min intavels with zero pickups:  29
------------------------------------------------------------
for the  3 th cluster number of 10min intavels with zero pickups:  354
------------------------------------------------------------
for the  4 th cluster number of 10min intavels with zero pickups:  37
------------------------------------------------------------
for the  5 th cluster number of 10min intavels with zero pickups:  153
------------------------------------------------------------
for the  6 th cluster number of 10min intavels with zero pickups:  34
------------------------------------------------------------
for the  7 th cluster number of 10min intavels with zero pickups:  34
------------------------------------------------------------
for the  8 th cluster number of 10min intavels with zero pickups:  117
------------------------------------------------------------
for the  9 th cluster number of 10min intavels with zero pickups:  40
------------------------------------------------------------
for the  10 th cluster number of 10min intavels with zero pickups:  25
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for the  11 th cluster number of 10min intavels with zero pickups:  44
------------------------------------------------------------
for the  12 th cluster number of 10min intavels with zero pickups:  42
------------------------------------------------------------
for the  13 th cluster number of 10min intavels with zero pickups:  28
------------------------------------------------------------
for the  14 th cluster number of 10min intavels with zero pickups:  26
------------------------------------------------------------
for the  15 th cluster number of 10min intavels with zero pickups:  31
------------------------------------------------------------
```

```
for the  16 th cluster number of 10min intavels with zero pickups:  40
------------------------------------------------------------
for the  17 th cluster number of 10min intavels with zero pickups:  58
------------------------------------------------------------
for the  18 th cluster number of 10min intavels with zero pickups:  1190
------------------------------------------------------------
for the  19 th cluster number of 10min intavels with zero pickups:  1357
------------------------------------------------------------
for the  20 th cluster number of 10min intavels with zero pickups:  53
------------------------------------------------------------
for the  21 th cluster number of 10min intavels with zero pickups:  29
------------------------------------------------------------
for the  22 th cluster number of 10min intavels with zero pickups:  29
------------------------------------------------------------
for the  23 th cluster number of 10min intavels with zero pickups:  163
------------------------------------------------------------
for the  24 th cluster number of 10min intavels with zero pickups:  35
------------------------------------------------------------
for the  25 th cluster number of 10min intavels with zero pickups:  41
------------------------------------------------------------
for the  26 th cluster number of 10min intavels with zero pickups:  31
------------------------------------------------------------
for the  27 th cluster number of 10min intavels with zero pickups:  214
------------------------------------------------------------
for the  28 th cluster number of 10min intavels with zero pickups:  36
------------------------------------------------------------
for the  29 th cluster number of 10min intavels with zero pickups:  41
------------------------------------------------------------
for the  30 th cluster number of 10min intavels with zero pickups:  1180
------------------------------------------------------------
for the  31 th cluster number of 10min intavels with zero pickups:  42
------------------------------------------------------------
for the  32 th cluster number of 10min intavels with zero pickups:  44
------------------------------------------------------------
for the  33 th cluster number of 10min intavels with zero pickups:  43
------------------------------------------------------------
for the  34 th cluster number of 10min intavels with zero pickups:  39
------------------------------------------------------------
for the  35 th cluster number of 10min intavels with zero pickups:  42
------------------------------------------------------------
for the  36 th cluster number of 10min intavels with zero pickups:  36
------------------------------------------------------------
for the  37 th cluster number of 10min intavels with zero pickups:  321
------------------------------------------------------------
for the  38 th cluster number of 10min intavels with zero pickups:  36
------------------------------------------------------------
for the  39 th cluster number of 10min intavels with zero pickups:  43
------------------------------------------------------------
```

there are two ways to fill up these values
Fill the missing value with 0's
Fill the missing values with the avg values
Case 1:(values missing at the start) Ex1: _ _ _ x =>ceil(x/4), ceil(x/4), ceil(x/4), ceil(x/4) Ex2: _ _ x => ceil(x/3), ceil(x/3), ceil(x/3)
Case 2:(values missing in middle) Ex1: x _ _ y => ceil((x+y)/4), ceil((x+y)/4), ceil((x+y)/4), ceil((x+y)/4) Ex2: x _ _ _ y => ceil((x+y)/5), ceil((x+y)/5), ceil((x+y)/5), ceil((x+y)/5), ceil((x+y)/5)
Case 3:(values missing at the end) Ex1: x _ _ _ => ceil(x/4), ceil(x/4), ceil(x/4), ceil(x/4) Ex2: x _ => ceil(x/2), ceil(x/2)

In [29]:
```
# Fills a value of zero for every bin where no pickup data is present
# the count_values: number pickps that are happened in each region for each 10min int
# there wont be any value if there are no picksups.
# values: number of unique bins

# for every 10min intravel(pickup_bin) we will check it is there in our unique bin,
# if it is there we will add the count_values[index] to smoothed data
# if not we add 0 to the smoothed data
# we finally return smoothed data
def fill_missing(count_values,values):
    smoothed_regions=[]
    ind=0
    for r in range(0,40):
        smoothed_bins=[]
        for i in range(4464):
            if i in values[r]:
                smoothed_bins.append(count_values[ind])
                ind+=1
            else:
                smoothed_bins.append(0)
        smoothed_regions.extend(smoothed_bins)
    return smoothed_regions
```

In [30]:
```
# Fills a value of zero for every bin where no pickup data is present
# the count_values: number pickps that are happened in each region for each 10min int
# there wont be any value if there are no picksups.
# values: number of unique bins

# for every 10min intravel(pickup_bin) we will check it is there in our unique bin,
# if it is there we will add the count_values[index] to smoothed data
# if not we add smoothed data (which is calculated based on the methods that are disc
# we finally return smoothed data
def smoothing(count_values,values):
    smoothed_regions=[] # stores list of final smoothed values of each reigion
    ind=0
    repeat=0
```

```
smoothed_value=0
for r in range(0,40):
    smoothed_bins=[] #stores the final smoothed values
    repeat=0
    for i in range(4464):
        if repeat!=0: # prevents iteration for a value which is already visited/r
            repeat-=1
            continue
        if i in values[r]: #checks if the pickup-bin exists
            smoothed_bins.append(count_values[ind]) # appends the value of the pi
        else:
            if i!=0:
                right_hand_limit=0
                for j in range(i,4464):
                    if  j not in values[r]: #searches for the left-limit or the p
                        continue
                    else:
                        right_hand_limit=j
                        break
                if right_hand_limit==0:
                #Case 1: When we have the last/last few values are found to be mi
                    smoothed_value=count_values[ind-1]*1.0/((4463-i)+2)*1.0
                    for j in range(i,4464):
                        smoothed_bins.append(math.ceil(smoothed_value))
                    smoothed_bins[i-1] = math.ceil(smoothed_value)
                    repeat=(4463-i)
                    ind-=1
                else:
                #Case 2: When we have the missing values between two known values
                    smoothed_value=(count_values[ind-1]+count_values[ind])*1.0/((i
                    for j in range(i,right_hand_limit+1):
                        smoothed_bins.append(math.ceil(smoothed_value))
                    smoothed_bins[i-1] = math.ceil(smoothed_value)
                    repeat=(right_hand_limit-i)
            else:
                #Case 3: When we have the first/first few values are found to be
                right_hand_limit=0
                for j in range(i,4464):
                    if  j not in values[r]:
                        continue
                    else:
                        right_hand_limit=j
                        break
                smoothed_value=count_values[ind]*1.0/((right_hand_limit-i)+1)*1.0
                for j in range(i,right_hand_limit+1):
                        smoothed_bins.append(math.ceil(smoothed_value))
                repeat=(right_hand_limit-i)
        ind+=1
```

```
            smoothed_regions.extend(smoothed_bins)
        return smoothed_regions
```

In [31]:
```
#Filling Missing values of Jan-2015 with 0
# here in jan_2015_groupby dataframe the trip_distance represents the number of pickup
jan_2015_fill = fill_missing(jan_2015_groupby['trip_distance'].values,jan_2015_unique

#Smoothing Missing values of Jan-2015
jan_2015_smooth = smoothing(jan_2015_groupby['trip_distance'].values,jan_2015_unique)
```

In [32]:
```
# number of 10min indices for jan 2015= 24*31*60/10 = 4464
# number of 10min indices for jan 2016 = 24*31*60/10 = 4464
# number of 10min indices for feb 2016 = 24*29*60/10 = 4176
# number of 10min indices for march 2016 = 24*30*60/10 = 4320
# for each cluster we will have 4464 values, therefore 40*4464 = 178560 (length of th
print("number of 10min intravels among all the clusters ",len(jan_2015_fill))
```

```
number of 10min intravels among all the clusters  178560
```

In [33]:
```
# Smoothing vs Filling
# sample plot that shows two variations of filling missing values
# we have taken the number of pickups for cluster region 2
plt.figure(figsize=(10,5))
plt.plot(jan_2015_fill[4464:8920], label="zero filled values")
plt.plot(jan_2015_smooth[4464:8920], label="filled with avg values")
plt.legend()
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

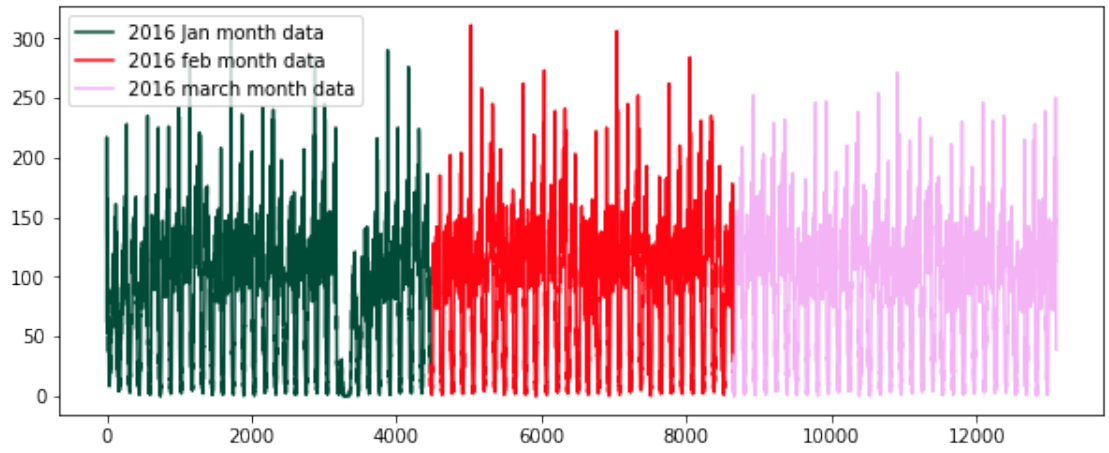
In [34]:
```
# why we choose, these methods and which method is used for which data?

# Ans: consider we have data of some month in 2015 jan 1st, 10 _ _ _ 20, i.e there ar
# 10st 10min intravel, 0 pickups happened in 2nd 10mins intravel, 0 pickups happened
# and 20 pickups happened in 4th 10min intravel.
# in fill_missing method we replace these values like 10, 0, 0, 20
# where as in smoothing method we replace these values as 6,6,6,6,6, if you can check
# that are happened in the first 40min are same in both cases, but if you can observe
# wheen you are using smoothing we are looking at the future number of pickups which

# so we use smoothing for jan 2015th data since it acts as our training data
# and we use simple fill_misssing method for 2016th data.
```

In [35]:
```
# Jan-2015 data is smoothed, Jan,Feb & March 2016 data missing values are filled with
jan_2015_smooth = smoothing(jan_2015_groupby['trip_distance'].values,jan_2015_unique)
```

```
jan_2016_smooth = fill_missing(jan_2016_groupby['trip_distance'].values,jan_2016_uniqu
feb_2016_smooth = fill_missing(feb_2016_groupby['trip_distance'].values,feb_2016_uniqu
mar_2016_smooth = fill_missing(mar_2016_groupby['trip_distance'].values,mar_2016_uniqu

# Making list of all the values of pickup data in every bin for a period of 3 months
regions_cum = []

# a =[1,2,3]
# b = [2,3,4]
# a+b = [1, 2, 3, 2, 3, 4]

# number of 10min indices for jan 2015= 24*31*60/10 = 4464
# number of 10min indices for jan 2016 = 24*31*60/10 = 4464
# number of 10min indices for feb 2016 = 24*29*60/10 = 4176
# number of 10min indices for march 2016 = 24*31*60/10 = 4464
# regions_cum: it will contain 40 lists, each list will contain 4464+4176+4464 values
# that are happened for three months in 2016 data

for i in range(0,40):
    regions_cum.append(jan_2016_smooth[4464*i:4464*(i+1)]+feb_2016_smooth[4176*i:4176

# print(len(regions_cum))
# 40
# print(len(regions_cum[0]))
# 13104
```

6.4 Time series and Fourier Transforms

```
In [37]: %matplotlib inline
         def uniqueish_color():
             """There're better ways to generate unique colors, but this isn't awful."""
             return plt.cm.gist_ncar(np.random.random())
         first_x = list(range(0,4464))
         second_x = list(range(4464,8640))
         third_x = list(range(8640,13104))
         for i in range(40):
             plt.figure(figsize=(10,4))
             plt.plot(first_x,regions_cum[i][:4464], color=uniqueish_color(), label='2016 Jan r
             plt.plot(second_x,regions_cum[i][4464:8640], color=uniqueish_color(), label='2016
             plt.plot(third_x,regions_cum[i][8640:], color=uniqueish_color(), label='2016 marc
             plt.legend()
             plt.show()
```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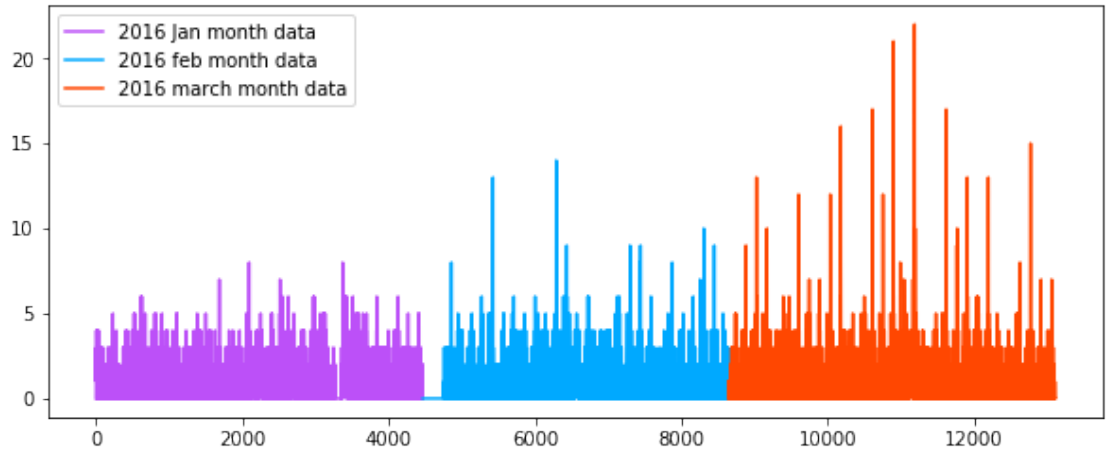
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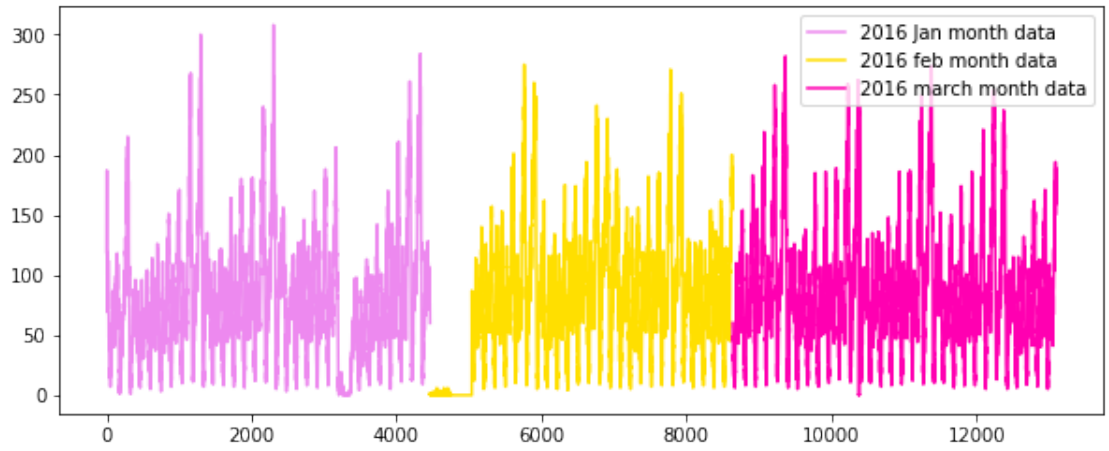
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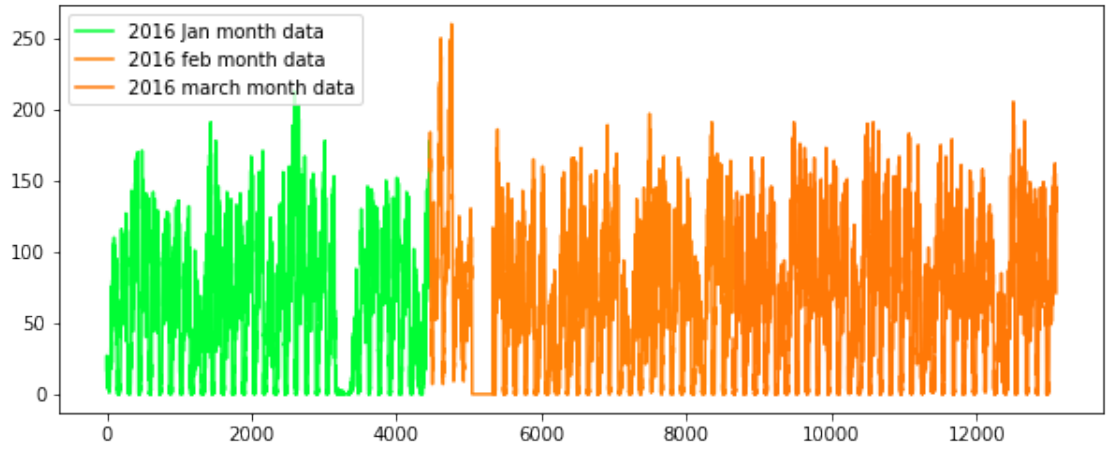
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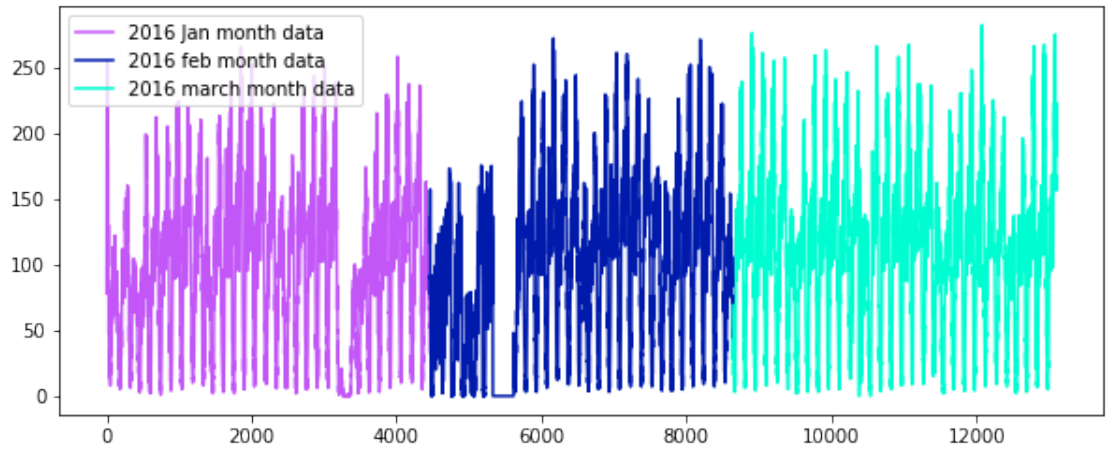
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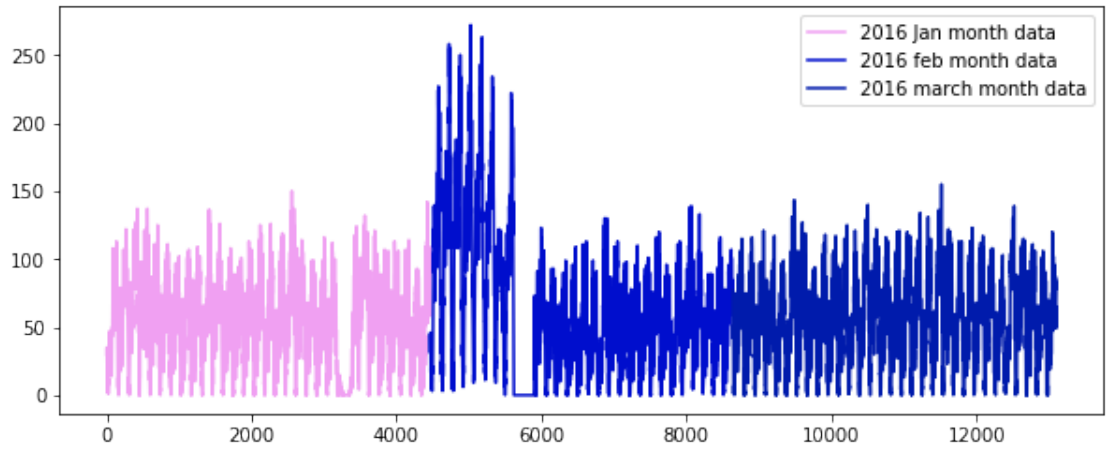
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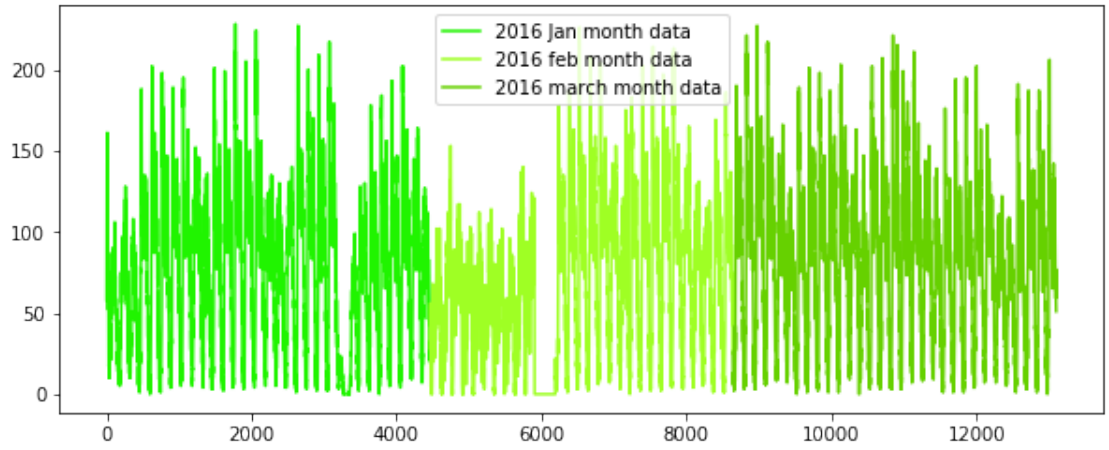
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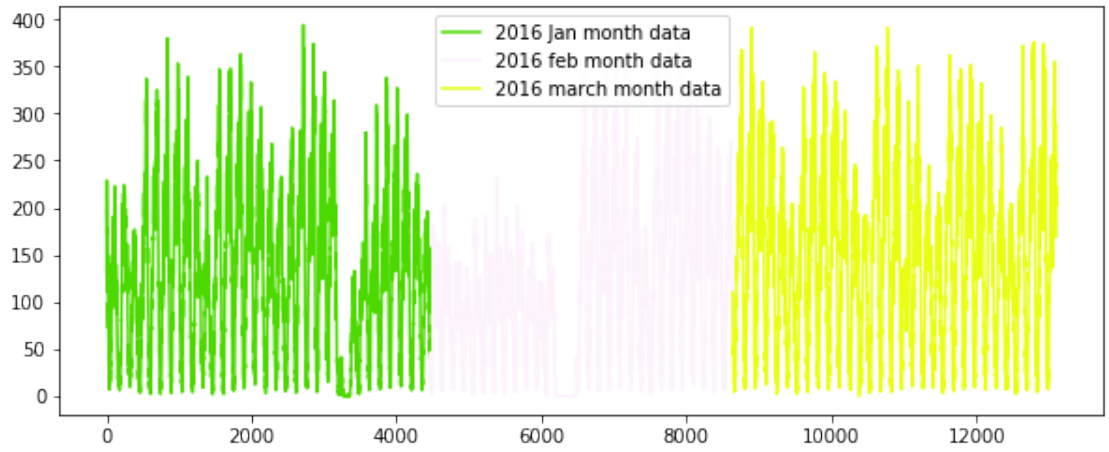
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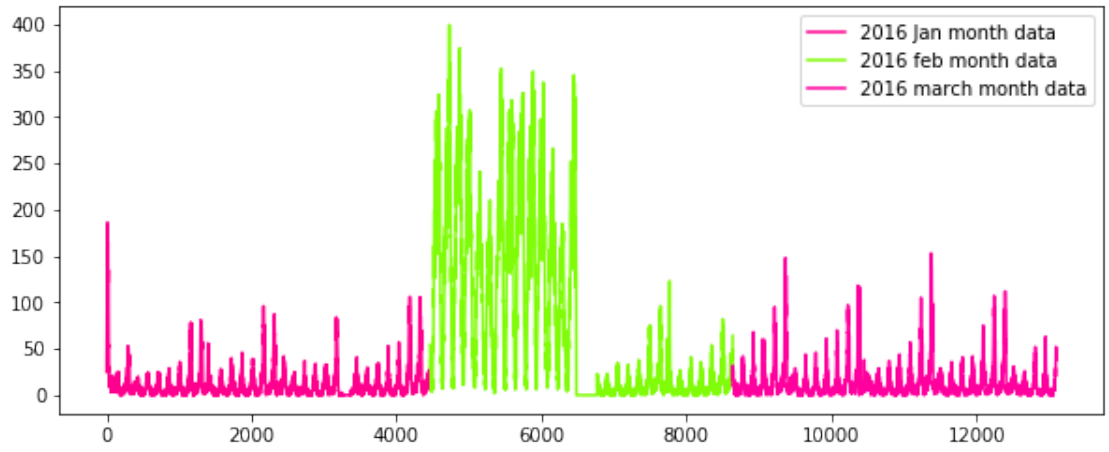
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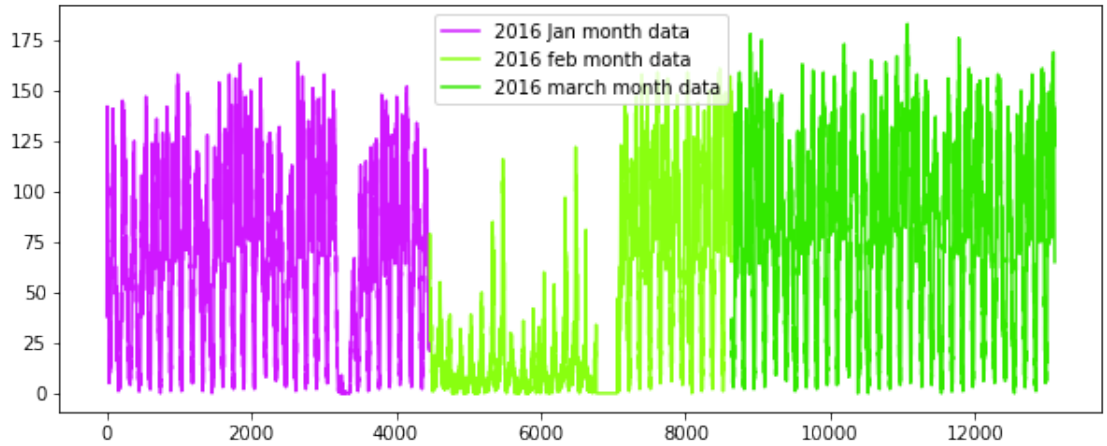
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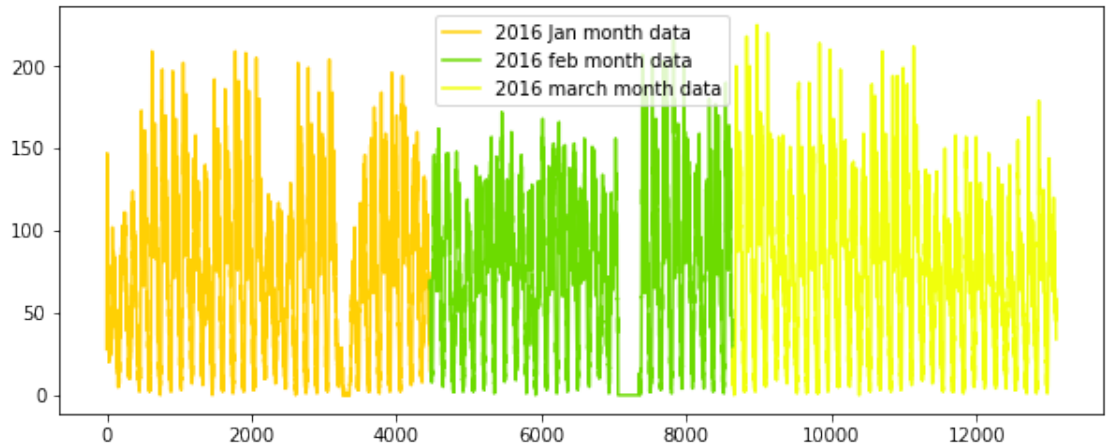
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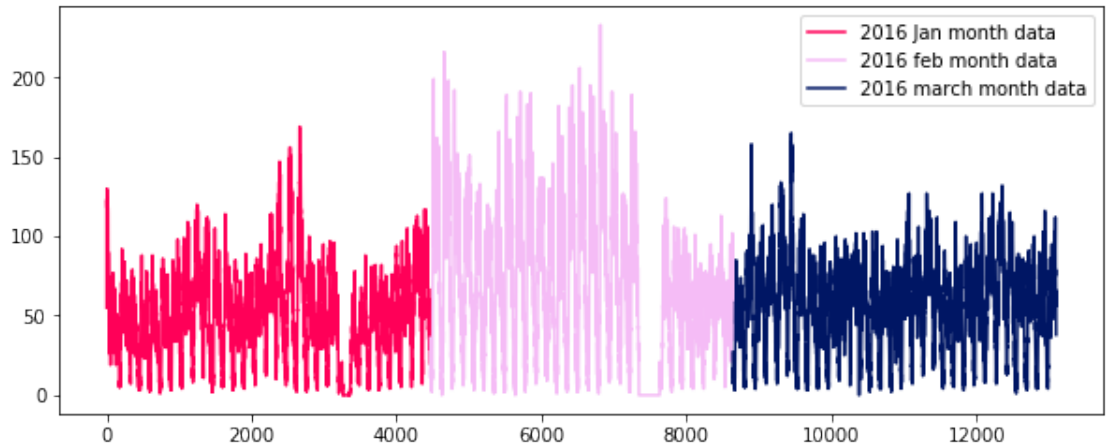
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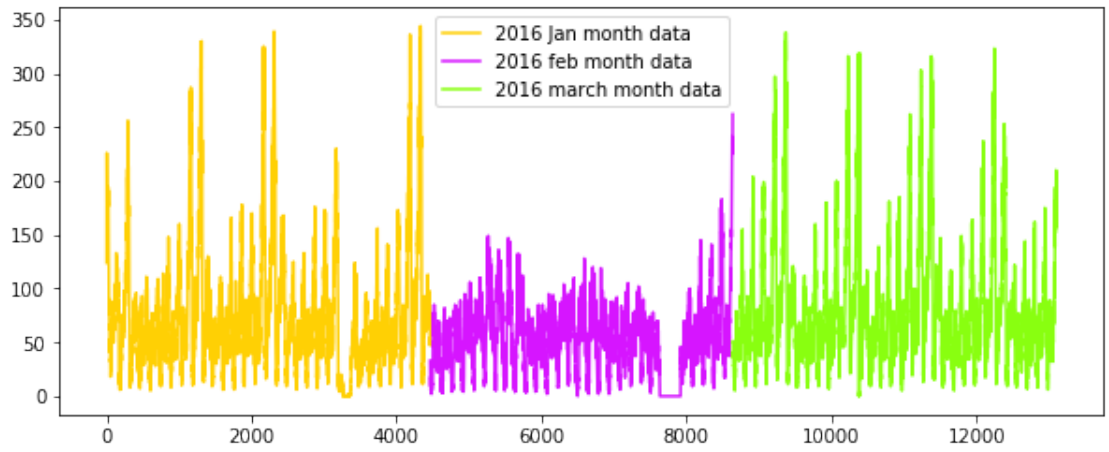
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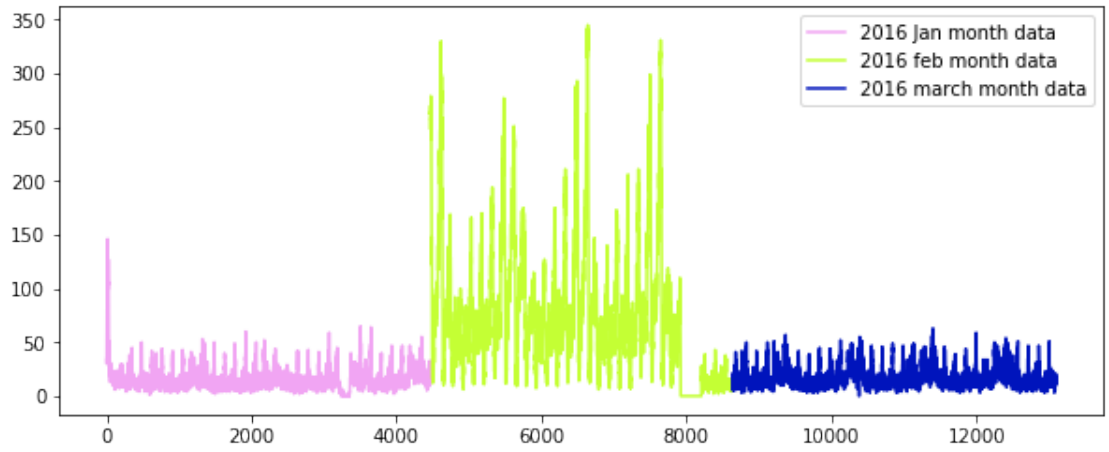
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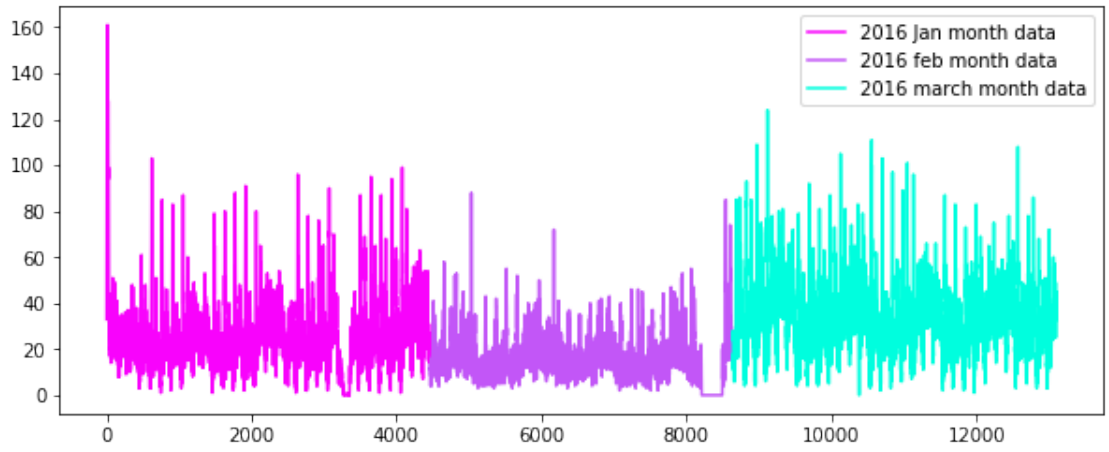
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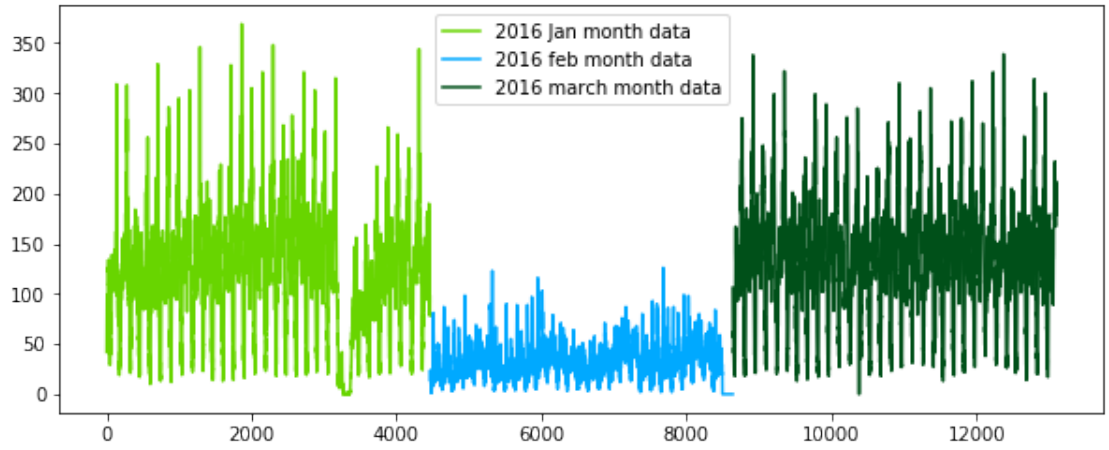
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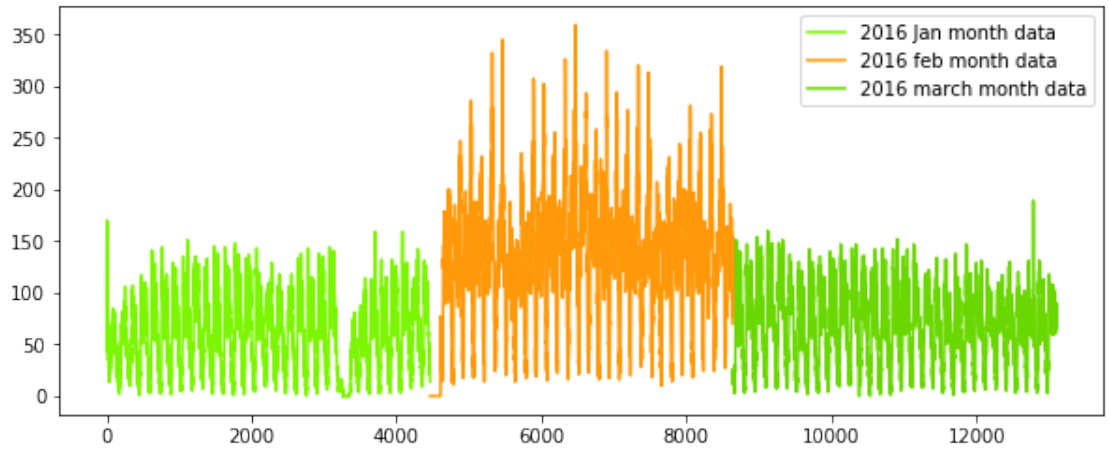
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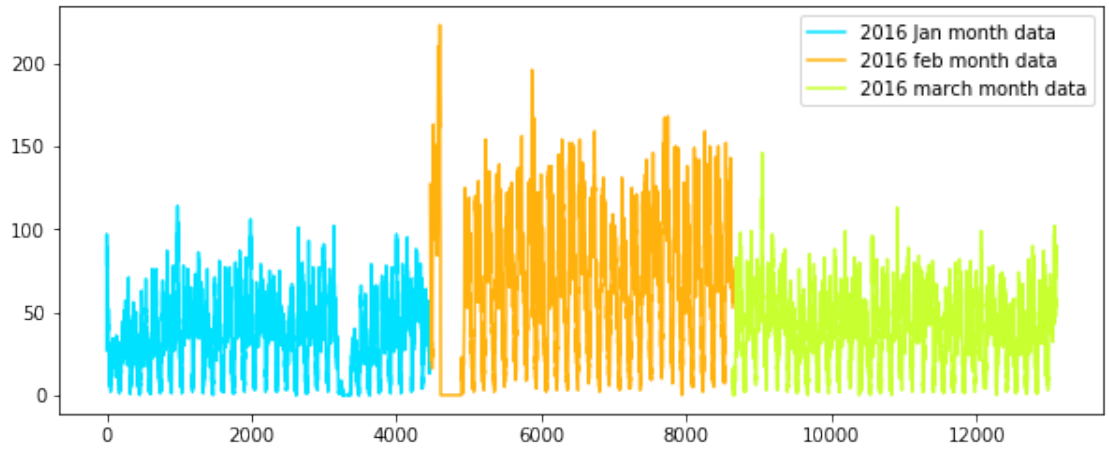
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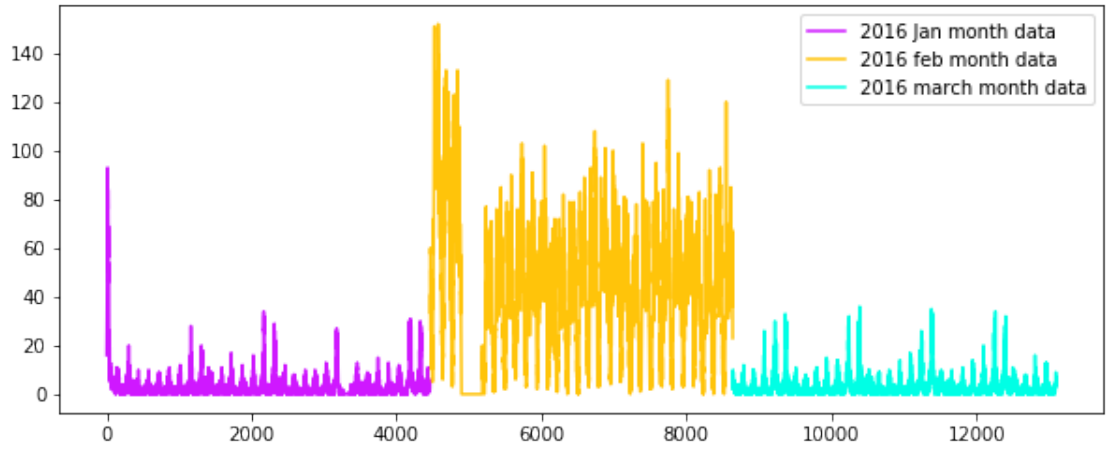
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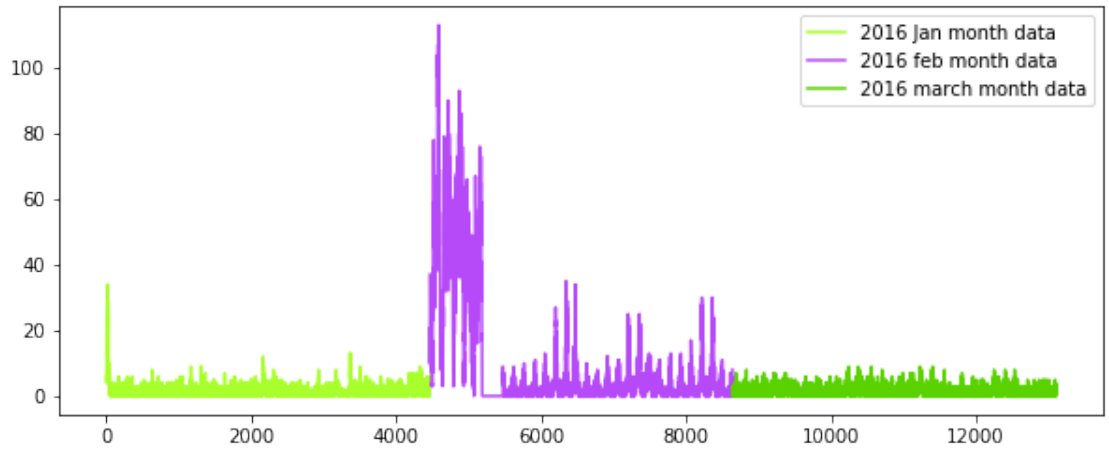
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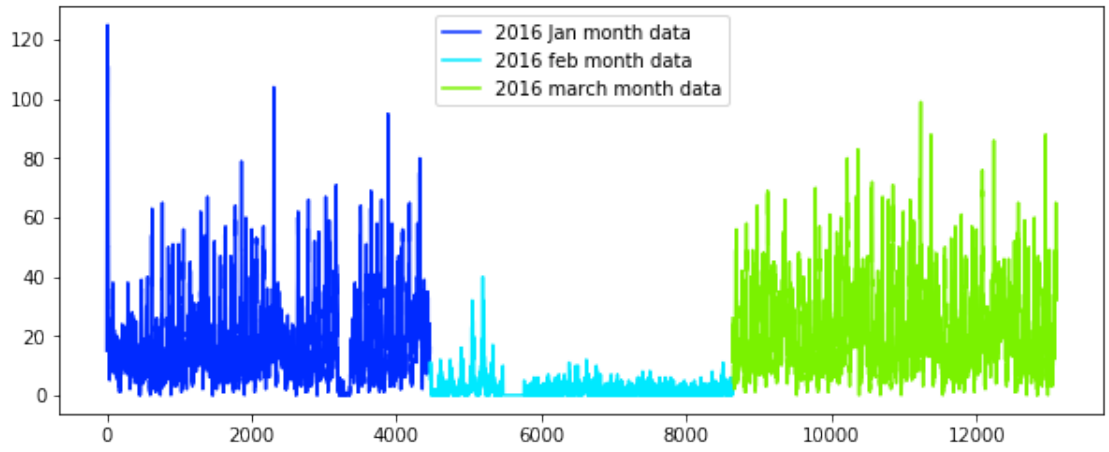
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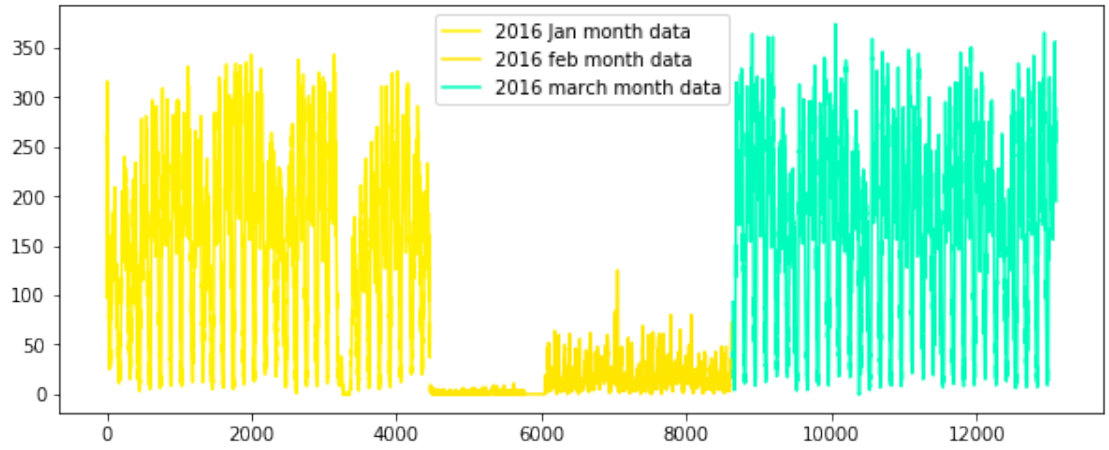
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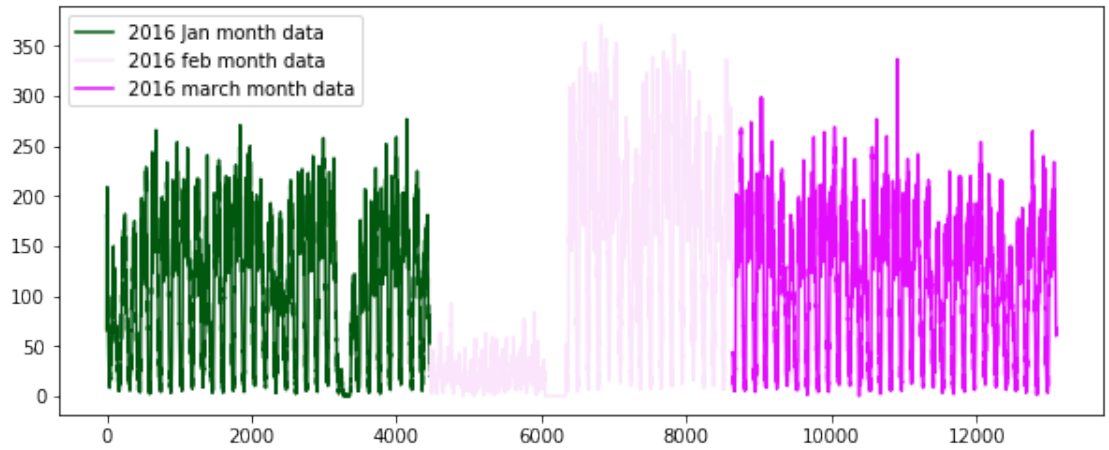
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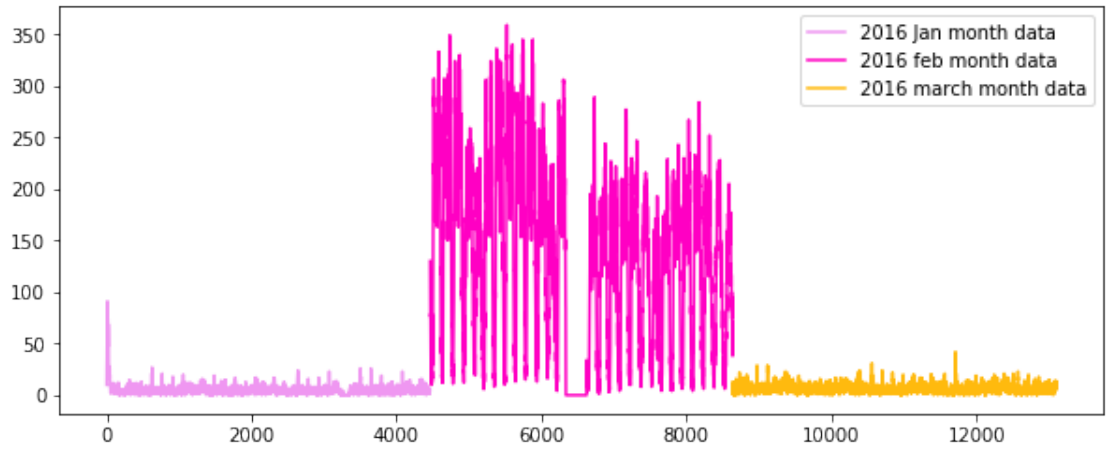
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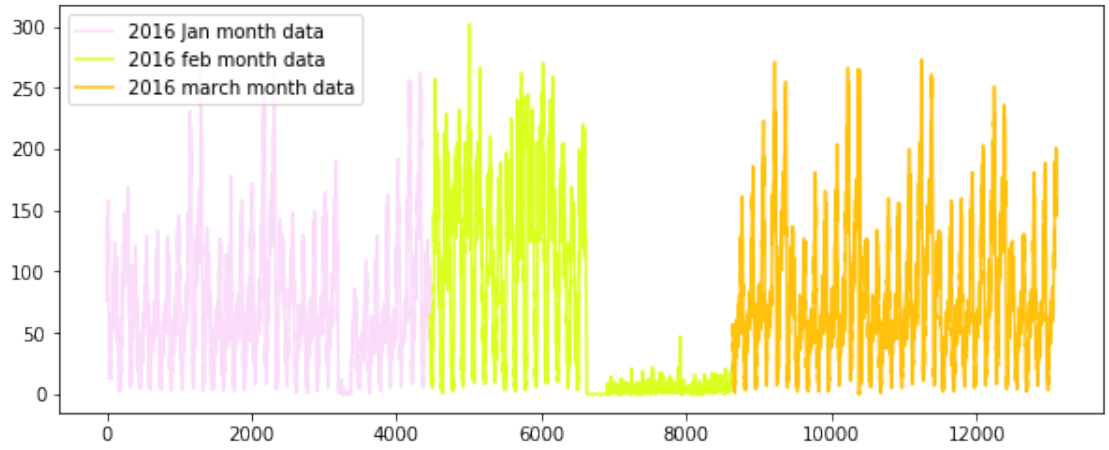
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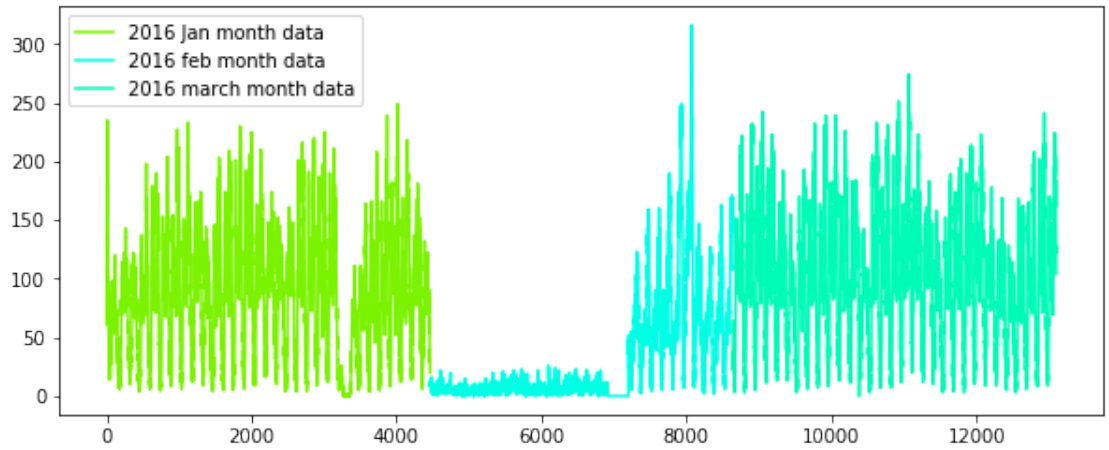
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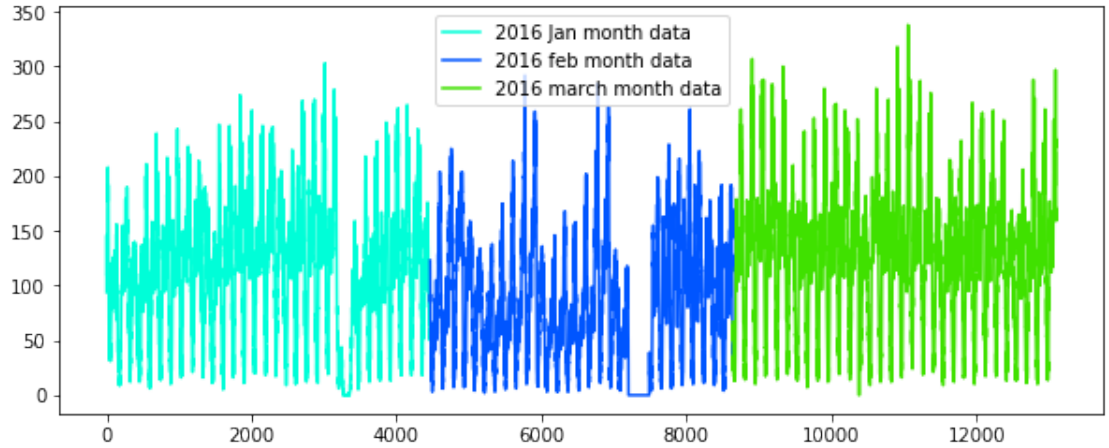
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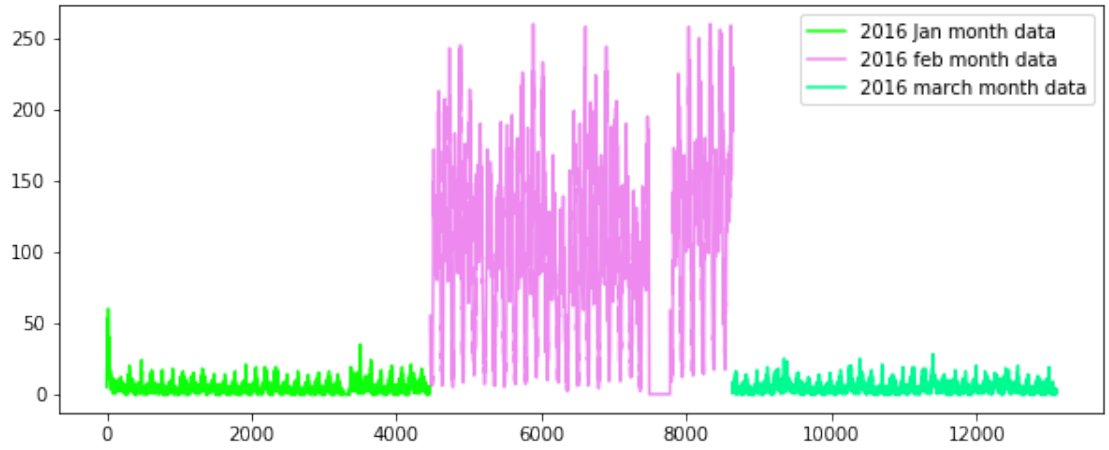
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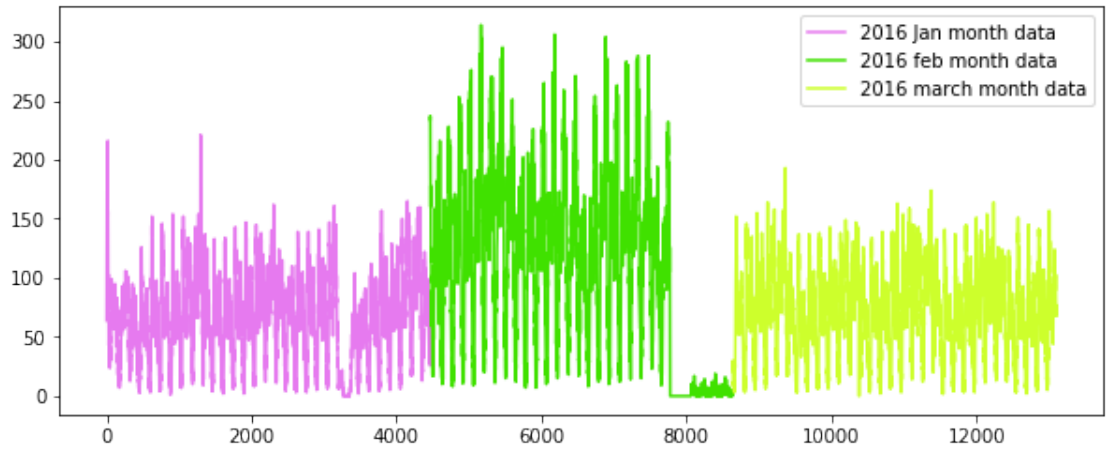
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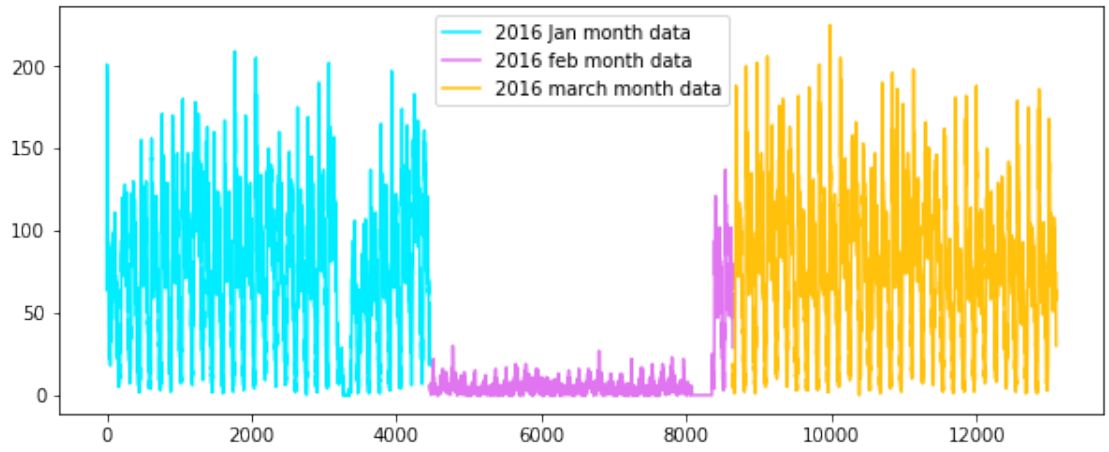
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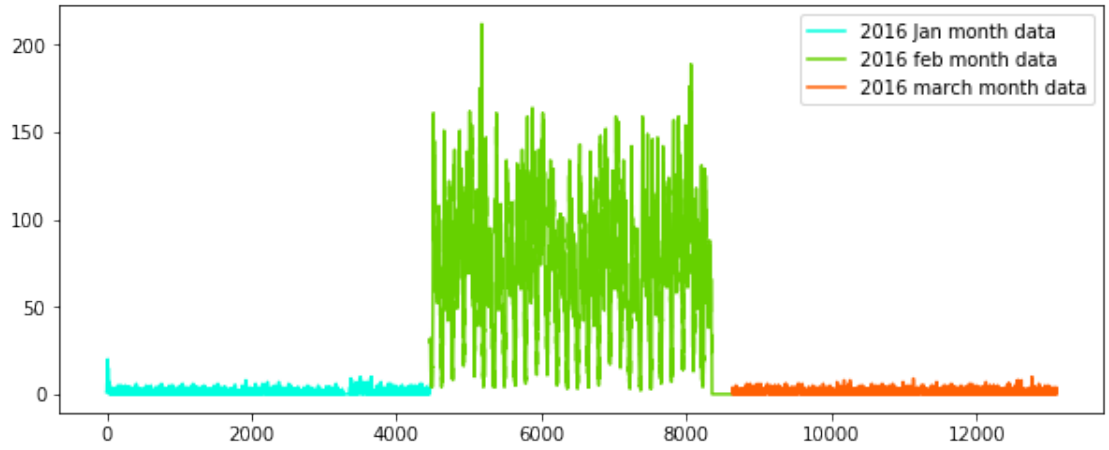
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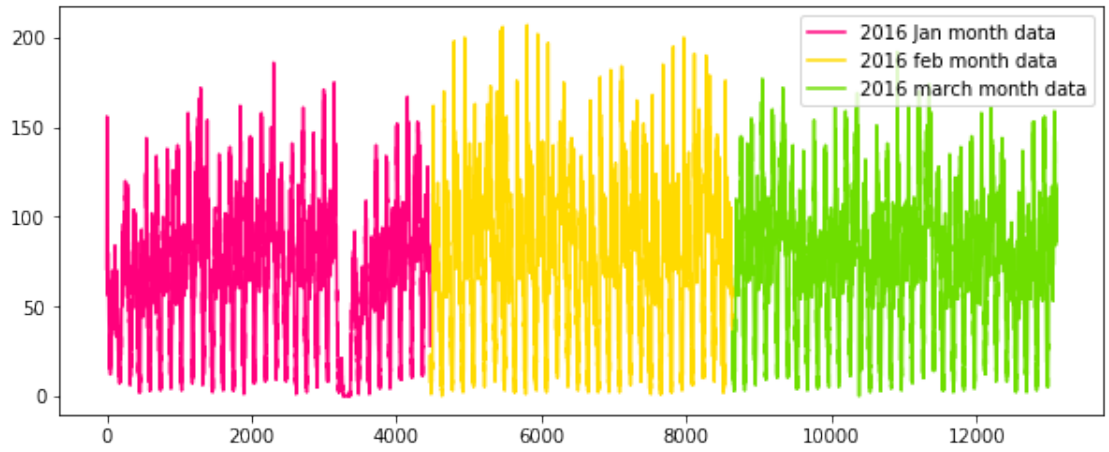
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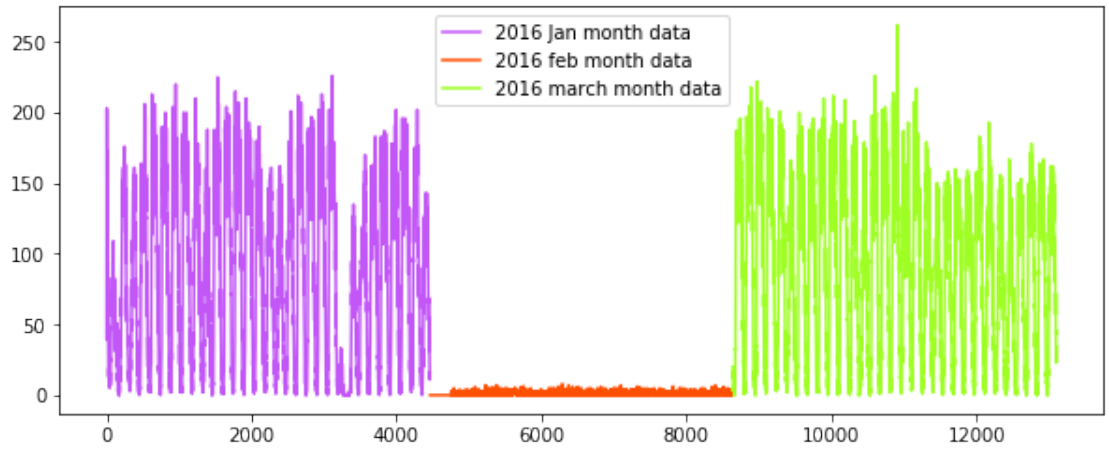
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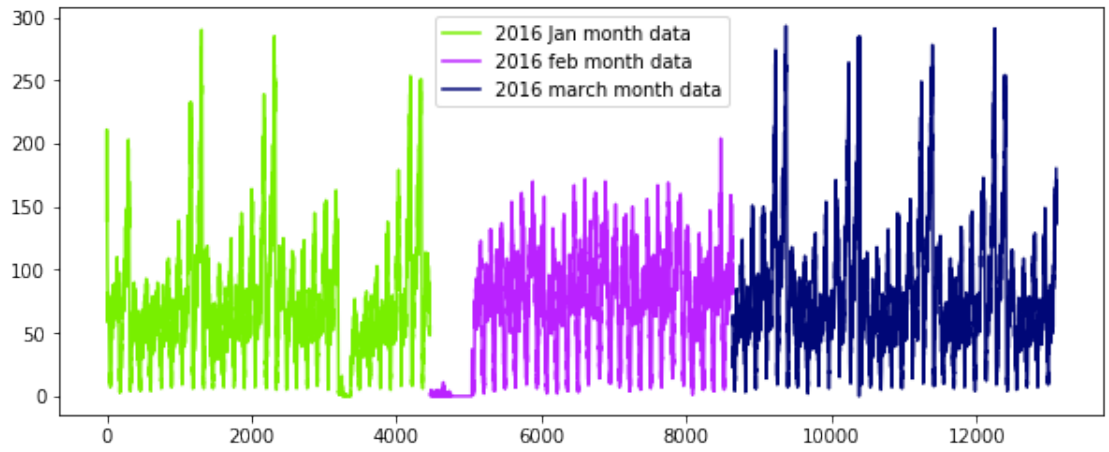
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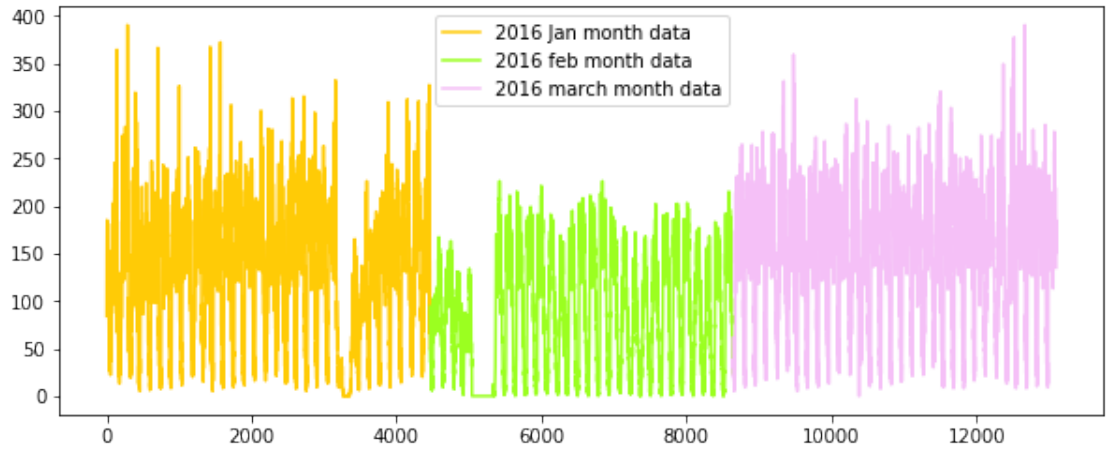
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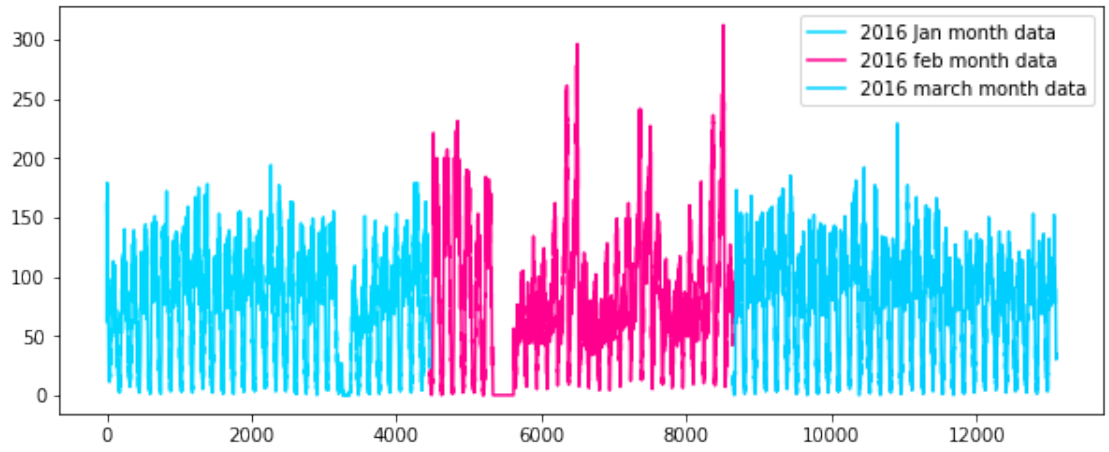
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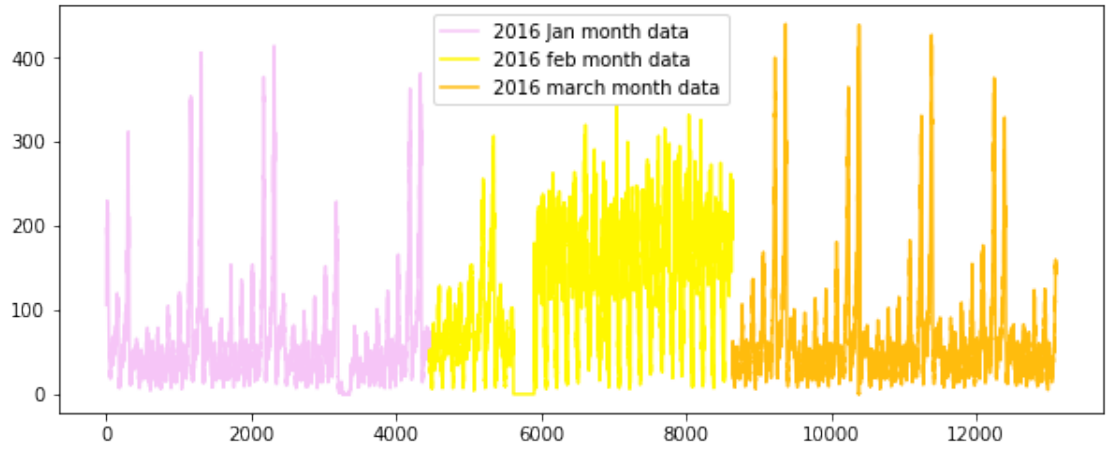
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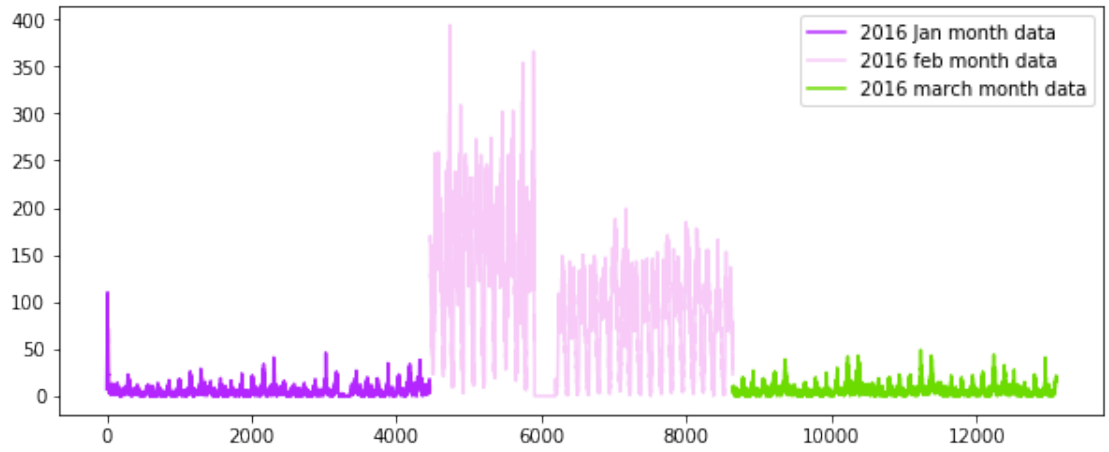
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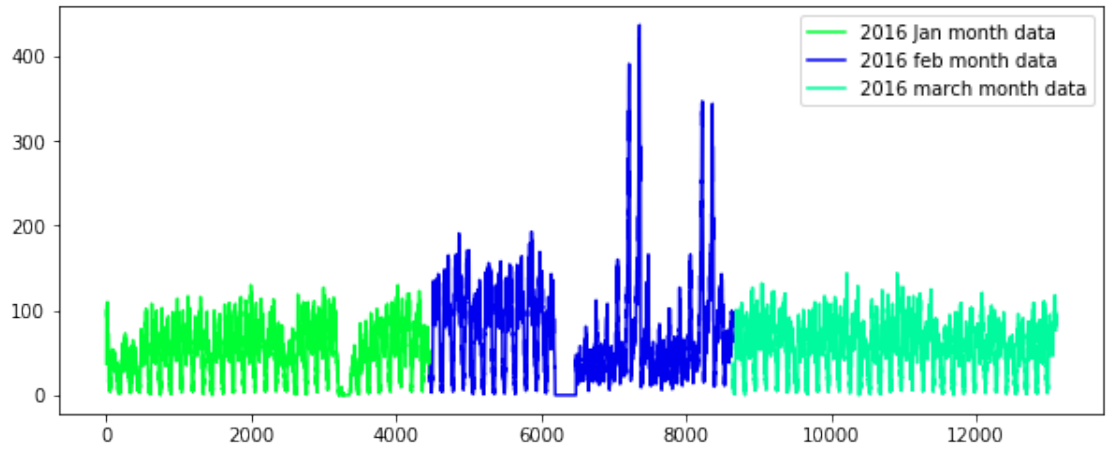
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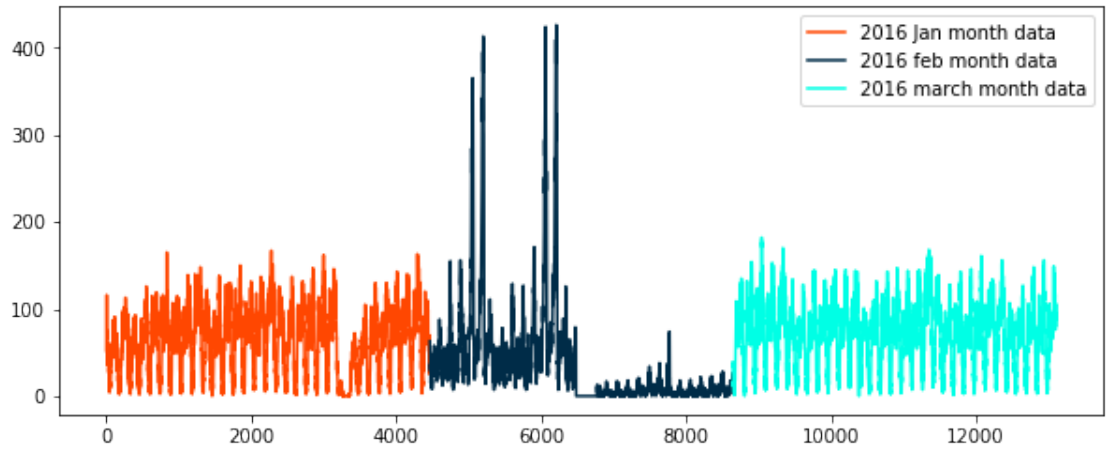


```
In [89]: # getting peaks: https://blog.ytotech.com/2015/11/01/findpeaks-in-python/
         # read more about fft function : https://docs.scipy.org/doc/numpy/reference/generated/
         Y    = np.fft.fft(np.array(jan_2016_smooth)[0:4460])
         # read more about the fftfreq: https://docs.scipy.org/doc/numpy/reference/generated/n
         freq = np.fft.fftfreq(4460, 1)
         n = len(freq)
         plt.figure()
         plt.plot( freq[:int(n/2)], np.abs(Y)[:int(n/2)] )
         plt.xlabel("Frequency")
         plt.ylabel("Amplitude")
         plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
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```

```
In [36]: #Preparing the Dataframe only with x(i) values as jan-2015 data and y(i) values as ja
         ratios_jan = pd.DataFrame()
         ratios_jan['Given']=jan_2015_smooth
         ratios_jan['Prediction']=jan_2016_smooth
         ratios_jan['Ratios']=ratios_jan['Prediction']*1.0/ratios_jan['Given']*1.0
```

6.5 Modelling: Baseline Models

Now we get into modelling in order to forecast the pickup densities for the months of Jan, Feb and March of 2016 for which we are using multiple models with two variations 1. Using Ratios of the 2016 data to the 2015 data i.e $R_t = P_t^{2016}/P_t^{2015}$ 2. Using Previous known values of the 2016 data itself to predict the future values

6.5.1 Simple Moving Averages

The First Model used is the Moving Averages Model which uses the previous n values in order to predict the next value

Using Ratio Values - $R_t = (R_{t-1} + R_{t-2} + R_{t-3}....R_{t-n})/n$

```
In [37]: def MA_R_Predictions(ratios,month):
             predicted_ratio=(ratios['Ratios'].values)[0]
             error=[]
             predicted_values=[]
             window_size=3
             predicted_ratio_values=[]
             for i in range(0,4464*40):
                 if i%4464==0:
                     predicted_ratio_values.append(0)
                     predicted_values.append(0)
                     error.append(0)
                     continue
                 predicted_ratio_values.append(predicted_ratio)
                 predicted_values.append(int(((ratios['Given'].values)[i])*predicted_ratio))
                 error.append(abs((math.pow(int(((ratios['Given'].values)[i])*predicted_ratio)-
                 if i+1>=window_size:
                     predicted_ratio=sum((ratios['Ratios'].values)[(i+1)-window_size:(i+1)])/wi
                 else:
                     predicted_ratio=sum((ratios['Ratios'].values)[0:(i+1)])/(i+1)


             ratios['MA_R_Predicted'] = predicted_values
             ratios['MA_R_Error'] = error
             mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['P
             mse_err = sum([e**2 for e in error])/len(error)
             return ratios,mape_err,mse_err
```

For the above the Hyperparameter is the window-size (n) which is tuned manually and it is found that the window-size of 3 is optimal for getting the best results using Moving Averages using previous Ratio values therefore we get $R_t = (R_{t-1} + R_{t-2} + R_{t-3})/3$

Next we use the Moving averages of the 2016 values itself to predict the future value using $P_t = (P_{t-1} + P_{t-2} + P_{t-3}....P_{t-n})/n$

```
In [38]: def MA_P_Predictions(ratios,month):
             predicted_value=(ratios['Prediction'].values)[0]
             error=[]
             predicted_values=[]
             window_size=1
             predicted_ratio_values=[]
             for i in range(0,4464*40):
                 predicted_values.append(predicted_value)
                 error.append(abs((math.pow(predicted_value-(ratios['Prediction'].values)[i],1)
                 if i+1>=window_size:
```

```
            predicted_value=int(sum((ratios['Prediction'].values)[(i+1)-window_size:(i
        else:
            predicted_value=int(sum((ratios['Prediction'].values)[0:(i+1)])/(i+1))

    ratios['MA_P_Predicted'] = predicted_values
    ratios['MA_P_Error'] = error
    mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['
    mse_err = sum([e**2 for e in error])/len(error)
    return ratios,mape_err,mse_err
```

For the above the Hyperparameter is the window-size (n) which is tuned manually and it is found that the window-size of 1 is optimal for getting the best results using Moving Averages using previous 2016 values therefore we get $P_t = P_{t-1}$

6.5.2 Weighted Moving Averages

The Moving Avergaes Model used gave equal importance to all the values in the window used, but we know intuitively that the future is more likely to be similar to the latest values and less similar to the older values. Weighted Averages converts this analogy into a mathematical relationship giving the highest weight while computing the averages to the latest previous value and decreasing weights to the subsequent older ones

Weighted Moving Averages using Ratio Values - $R_t = (N * R_{t-1} + (N-1) * R_{t-2} + (N-2) * R_{t-3} .... 1 * R_{t-n})$

```
In [39]: def WA_R_Predictions(ratios,month):
             predicted_ratio=(ratios['Ratios'].values)[0]
             alpha=0.5
             error=[]
             predicted_values=[]
             window_size=5
             predicted_ratio_values=[]
             for i in range(0,4464*40):
                 if i%4464==0:
                     predicted_ratio_values.append(0)
                     predicted_values.append(0)
                     error.append(0)
                     continue
                 predicted_ratio_values.append(predicted_ratio)
                 predicted_values.append(int(((ratios['Given'].values)[i])*predicted_ratio))
                 error.append(abs((math.pow(int(((ratios['Given'].values)[i])*predicted_ratio)-
                 if i+1>=window_size:
                     sum_values=0
                     sum_of_coeff=0
                     for j in range(window_size,0,-1):
                         sum_values += j*(ratios['Ratios'].values)[i-window_size+j]
                         sum_of_coeff+=j
                     predicted_ratio=sum_values/sum_of_coeff
                 else:
                     sum_values=0
```

```
            sum_of_coeff=0
            for j in range(i+1,0,-1):
                sum_values += j*(ratios['Ratios'].values)[j-1]
                sum_of_coeff+=j
            predicted_ratio=sum_values/sum_of_coeff

        ratios['WA_R_Predicted'] = predicted_values
        ratios['WA_R_Error'] = error
        mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['
        mse_err = sum([e**2 for e in error])/len(error)
        return ratios,mape_err,mse_err
```

For the above the Hyperparameter is the window-size (n) which is tuned manually and it is found that the window-size of 5 is optimal for getting the best results using Weighted Moving Averages using previous Ratio values therefore we get $R_t = (5 * R_{t-1} + 4 * R_{t-2} + 3 * R_{t-3} + 2 * R_{t-4} + R_{t-5})/15$

Weighted Moving Averages using Previous 2016 Values - $P_t = (N * P_{t-1} + (N-1) * P_{t-2} + (N-2) * P_{t-3} .... 1 * P_{t-n})/(N * (N+1)/2)$

```
In [40]: def WA_P_Predictions(ratios,month):
             predicted_value=(ratios['Prediction'].values)[0]
             error=[]
             predicted_values=[]
             window_size=2
             for i in range(0,4464*40):
                 predicted_values.append(predicted_value)
                 error.append(abs((math.pow(predicted_value-(ratios['Prediction'].values)[i],1
                 if i+1>=window_size:
                     sum_values=0
                     sum_of_coeff=0
                     for j in range(window_size,0,-1):
                         sum_values += j*(ratios['Prediction'].values)[i-window_size+j]
                         sum_of_coeff+=j
                     predicted_value=int(sum_values/sum_of_coeff)

                 else:
                     sum_values=0
                     sum_of_coeff=0
                     for j in range(i+1,0,-1):
                         sum_values += j*(ratios['Prediction'].values)[j-1]
                         sum_of_coeff+=j
                     predicted_value=int(sum_values/sum_of_coeff)

             ratios['WA_P_Predicted'] = predicted_values
             ratios['WA_P_Error'] = error
             mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['
             mse_err = sum([e**2 for e in error])/len(error)
             return ratios,mape_err,mse_err
```

For the above the Hyperparameter is the window-size (n) which is tuned manually and it is found that the window-size of 2 is optimal for getting the best results using Weighted Moving Averages using previous 2016 values therefore we get $P_t = (2 * P_{t-1} + P_{t-2})/3$

6.5.3 Exponential Weighted Moving Averages

https://en.wikipedia.org/wiki/Moving_average#Exponential_moving_average Through weighted averaged we have satisfied the analogy of giving higher weights to the latest value and decreasing weights to the subsequent ones but we still do not know which is the correct weighting scheme as there are infinetly many possibilities in which we can assign weights in a non-increasing order and tune the the hyperparameter window-size. To simplify this process we use Exponential Moving Averages which is a more logical way towards assigning weights and at the same time also using an optimal window-size.

In exponential moving averages we use a single hyperparameter alpha (α) which is a value between 0 & 1 and based on the value of the hyperparameter alpha the weights and the window sizes are configured. For eg. If $\alpha = 0.9$ then the number of days on which the value of the current iteration is based is~$1/(1-\alpha) = 10$ i.e. we consider values 10 days prior before we predict the value for the current iteration. Also the weights are assigned using $2/(N+1) = 0.18$,where N = number of prior values being considered, hence from this it is implied that the first or latest value is assigned a weight of 0.18 which keeps exponentially decreasing for the subsequent values.

$R_t^{'} = \alpha * R_{t-1} + (1-\alpha) * R_{t-1}^{'}$

```
In [41]: def EA_R1_Predictions(ratios,month):
             predicted_ratio=(ratios['Ratios'].values)[0]
             alpha=0.6
             error=[]
             predicted_values=[]
             predicted_ratio_values=[]
             for i in range(0,4464*40):
                 if i%4464==0:
                     predicted_ratio_values.append(0)
                     predicted_values.append(0)
                     error.append(0)
                     continue
                 predicted_ratio_values.append(predicted_ratio)
                 predicted_values.append(int(((ratios['Given'].values)[i])*predicted_ratio))
                 error.append(abs((math.pow(int(((ratios['Given'].values)[i])*predicted_ratio)-
                 predicted_ratio = (alpha*predicted_ratio) + (1-alpha)*((ratios['Ratios'].valu

             ratios['EA_R1_Predicted'] = predicted_values
             ratios['EA_R1_Error'] = error
             mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['
             mse_err = sum([e**2 for e in error])/len(error)
             return ratios,mape_err,mse_err
```

$P_t^{'} = \alpha * P_{t-1} + (1-\alpha) * P_{t-1}^{'}$

```
In [42]: def EA_P1_Predictions(ratios,month):
             predicted_value= (ratios['Prediction'].values)[0]
```

```
    alpha=0.3
    error=[]
    predicted_values=[]
    for i in range(0,4464*40):
        if i%4464==0:
            predicted_values.append(0)
            error.append(0)
            continue
        predicted_values.append(predicted_value)
        error.append(abs((math.pow(predicted_value-(ratios['Prediction'].values)[i],1)
        predicted_value =int((alpha*predicted_value) + (1-alpha)*((ratios['Prediction

    ratios['EA_P1_Predicted'] = predicted_values
    ratios['EA_P1_Error'] = error
    mape_err = (sum(error)/len(error))/(sum(ratios['Prediction'].values)/len(ratios['
    mse_err = sum([e**2 for e in error])/len(error)
    return ratios,mape_err,mse_err
```

```
In [43]: mean_err=[0]*10
         median_err=[0]*10
         ratios_jan,mean_err[0],median_err[0]=MA_R_Predictions(ratios_jan,'jan')
         ratios_jan,mean_err[1],median_err[1]=MA_P_Predictions(ratios_jan,'jan')
         ratios_jan,mean_err[2],median_err[2]=WA_R_Predictions(ratios_jan,'jan')
         ratios_jan,mean_err[3],median_err[3]=WA_P_Predictions(ratios_jan,'jan')
         ratios_jan,mean_err[4],median_err[4]=EA_R1_Predictions(ratios_jan,'jan')
         ratios_jan,mean_err[5],median_err[5]=EA_P1_Predictions(ratios_jan,'jan')
```

6.6 Comparison between baseline models

We have chosen our error metric for comparison between models as MAPE (Mean Absolute Percentage Error) so that we can know that on an average how good is our model with predictions and MSE (Mean Squared Error) is also used so that we have a clearer understanding as to how well our forecasting model performs with outliers so that we make sure that there is not much of a error margin between our prediction and the actual value

```
In [45]: print ("Error Metric Matrix (Forecasting Methods) - MAPE & MSE")
         print ("-------------------------------------------------------------------------------
         print ("Moving Averages (Ratios) -                          MAPE: ",mean_err[0],"
         print ("Moving Averages (2016 Values) -                     MAPE: ",mean_err[1],"
         print ("-------------------------------------------------------------------------------
         print ("Weighted Moving Averages (Ratios) -                 MAPE: ",mean_err[2],"
         print ("Weighted Moving Averages (2016 Values) -            MAPE: ",mean_err[3],"
         print ("-------------------------------------------------------------------------------
         print ("Exponential Moving Averages (Ratios) -           MAPE: ",mean_err[4],"
         print ("Exponential Moving Averages (2016 Values) -      MAPE: ",mean_err[5],"
```

```
Error Metric Matrix (Forecasting Methods) - MAPE & MSE
-------------------------------------------------------------------------------------------
Moving Averages (Ratios) -                          MAPE:  0.18210684551733733       MSE:  4
```

```
Moving Averages (2016 Values) -                        MAPE:  0.14259943893312463        MSE:
-----------------------------------------------------------------------------------------------
Weighted Moving Averages (Ratios) -                    MAPE:  0.17856610294168937       MSE:  3
Weighted Moving Averages (2016 Values) -               MAPE:  0.13518503122892725       MSE:  1
-----------------------------------------------------------------------------------------------
Exponential Moving Averages (Ratios) -              MAPE:  0.17801742169014942       MSE:  383
Exponential Moving Averages (2016 Values) -         MAPE:  0.13489103761553553       MSE:  159
```

Plese Note:- The above comparisons are made using Jan 2015 and Jan 2016 only

From the above matrix it is inferred that the best forecasting model for our prediction would be:- $P_t^{'} = \alpha * P_{t-1} + (1-\alpha) * P_{t-1}^{'}$ i.e Exponential Moving Averages using 2016 Values

6.7 Regression Models

6.7.1 Train-Test Split

Before we start predictions using the tree based regression models we take 3 months of 2016 pickup data and split it such that for every region we have 70% data in train and 30% in test, ordered date-wise for every region

In [46]:
```
# Preparing data to be split into train and test, The below prepares data in cumulati
# number of 10min indices for jan 2015= 24*31*60/10 = 4464
# number of 10min indices for jan 2016 = 24*31*60/10 = 4464
# number of 10min indices for feb 2016 = 24*29*60/10 = 4176
# number of 10min indices for march 2016 = 24*31*60/10 = 4464
# regions_cum: it will contain 40 lists, each list will contain 4464+4176+4464 values
# that are happened for three months in 2016 data

# print(len(regions_cum))
# 40
# print(len(regions_cum[0]))
# 12960

# we take number of pickups that are happened in last 5 10min intravels
number_of_time_stamps = 5

# output varaible
# it is list of lists
# it will contain number of pickups 13099 for each cluster
output = []


# tsne_lat will contain 13104-5=13099 times lattitude of cluster center for every clu
# Ex: [[cent_lat 13099times],[cent_lat 13099times], [cent_lat 13099times].... 40 list
# it is list of lists
tsne_lat = []
```

```
# tsne_lon will contain 13104-5=13099 times logitude of cluster center for every clus
# Ex: [[cent_long 13099times],[cent_long 13099times], [cent_long 13099times].... 40 l
# it is list of lists
tsne_lon = []

# we will code each day
# sunday = 0, monday=1, tue = 2, wed=3, thur=4, fri=5,sat=6
# for every cluster we will be adding 13099 values, each value represent to which day
# it is list of lists
tsne_weekday = []

# its an numbpy array, of shape (523960, 5)
# each row corresponds to an entry in out data
# for the first row we will have [f0,f1,f2,f3,f4] fi=number of pickups happened in i+
# the second row will have [f1,f2,f3,f4,f5]
# the third row will have [f2,f3,f4,f5,f6]
# and so on...
tsne_feature = []


tsne_feature = [0]*number_of_time_stamps
for i in range(0,40):
    tsne_lat.append([kmeans.cluster_centers_[i][0]]*13099)
    tsne_lon.append([kmeans.cluster_centers_[i][1]]*13099)
    # jan 1st 2016 is thursday, so we start our day from 4: "(int(k/144))%7+4"
    # our prediction start from 5th 10min intravel since we need to have number of pi
    tsne_weekday.append([int(((int(k/144))%7+4)%7) for k in range(5,4464+4176+4464)])
    # regions_cum is a list of lists [[x1,x2,x3..x13104], [x1,x2,x3..x13104], [x1,x2,
    tsne_feature = np.vstack((tsne_feature, [regions_cum[i][r:r+number_of_time_stamps]
    output.append(regions_cum[i][5:])
tsne_feature = tsne_feature[1:]
```

In [48]:
```
len(tsne_lat[0])*len(tsne_lat) == tsne_feature.shape[0] == len(tsne_weekday)*len(tsne_
```

Out[48]: True

In [49]:
```
# Getting the predictions of exponential moving averages to be used as a feature in c

# upto now we computed 8 features for every data point that starts from 50th min of t
# 1. cluster center lattitude
# 2. cluster center longitude
# 3. day of the week
# 4. f_t_1: number of pickups that are happened previous t-1th 10min intravel
# 5. f_t_2: number of pickups that are happened previous t-2th 10min intravel
# 6. f_t_3: number of pickups that are happened previous t-3th 10min intravel
# 7. f_t_4: number of pickups that are happened previous t-4th 10min intravel
# 8. f_t_5: number of pickups that are happened previous t-5th 10min intravel
```

```
# from the baseline models we said the exponential weighted moving avarage gives us t
# we will try to add the same exponential weighted moving avarage at t as a feature t
# exponential weighted moving avarage => p'(t) = alpha*p'(t-1) + (1-alpha)*P(t-1)
alpha=0.3

# it is a temporary array that store exponential weighted moving avarage for each 10m
# for each cluster it will get reset
# for every cluster it contains 13104 values
predicted_values=[]

# it is similar like tsne_lat
# it is list of lists
# predict_list is a list of lists [[x5,x6,x7..x13104], [x5,x6,x7..x13104], [x5,x6,x7.
predict_list = []
tsne_flat_exp_avg = []
for r in range(0,40):
    for i in range(0,13104):
        if i==0:
            predicted_value= regions_cum[r][0]
            predicted_values.append(0)
            continue
        predicted_values.append(predicted_value)
        predicted_value =int((alpha*predicted_value) + (1-alpha)*(regions_cum[r][i]))
    predict_list.append(predicted_values[5:])
    predicted_values=[]
```

In [50]:
```
# train, test split : 70% 30% split
# Before we start predictions using the tree based regression models we take 3 months
# and split it such that for every region we have 70% data in train and 30% in test,
# ordered date-wise for every region
print("size of train data :", int(13099*0.7))
print("size of test data :", int(13099*0.3))
```

```
size of train data : 9169
size of test data : 3929
```

In [51]:
```
# extracting first 9169 timestamp values i.e 70% of 13099 (total timestamps) for our
train_features =  [tsne_feature[i*13099:(13099*i+9169)] for i in range(0,40)]
# temp = [0]*(12955 - 9068)
test_features = [tsne_feature[(13099*(i))+9169:13099*(i+1)] for i in range(0,40)]
```

In [52]:
```
print("Number of data clusters",len(train_features), "Number of data points in trian d
print("Number of data clusters",len(train_features), "Number of data points in test da
```

```
Number of data clusters 40 Number of data points in trian data 9169 Each data point contains 5
Number of data clusters 40 Number of data points in test data 3930 Each data point contains 5 
```

In [53]:
```
# extracting first 9169 timestamp values i.e 70% of 13099 (total timestamps) for our
tsne_train_flat_lat = [i[:9169] for i in tsne_lat]
tsne_train_flat_lon = [i[:9169] for i in tsne_lon]
tsne_train_flat_weekday = [i[:9169] for i in tsne_weekday]
tsne_train_flat_output = [i[:9169] for i in output]
tsne_train_flat_exp_avg = [i[:9169] for i in predict_list]
```

In [54]:
```
# extracting the rest of the timestamp values i.e 30% of 12956 (total timestamps) for
tsne_test_flat_lat = [i[9169:] for i in tsne_lat]
tsne_test_flat_lon = [i[9169:] for i in tsne_lon]
tsne_test_flat_weekday = [i[9169:] for i in tsne_weekday]
tsne_test_flat_output = [i[9169:] for i in output]
tsne_test_flat_exp_avg = [i[9169:] for i in predict_list]
```

In [55]:
```
# the above contains values in the form of list of lists (i.e. list of values of each
train_new_features = []
for i in range(0,40):
    train_new_features.extend(train_features[i])
test_new_features = []
for i in range(0,40):
    test_new_features.extend(test_features[i])
```

In [56]:
```
# converting lists of lists into sinle list i.e flatten
# a  = [[1,2,3,4],[4,6,7,8]]
# print(sum(a,[]))
# [1, 2, 3, 4, 4, 6, 7, 8]

tsne_train_lat = sum(tsne_train_flat_lat, [])
tsne_train_lon = sum(tsne_train_flat_lon, [])
tsne_train_weekday = sum(tsne_train_flat_weekday, [])
tsne_train_output = sum(tsne_train_flat_output, [])
tsne_train_exp_avg = sum(tsne_train_flat_exp_avg,[])
```

In [57]:
```
# converting lists of lists into sinle list i.e flatten
# a  = [[1,2,3,4],[4,6,7,8]]
# print(sum(a,[]))
# [1, 2, 3, 4, 4, 6, 7, 8]

tsne_test_lat = sum(tsne_test_flat_lat, [])
tsne_test_lon = sum(tsne_test_flat_lon, [])
tsne_test_weekday = sum(tsne_test_flat_weekday, [])
tsne_test_output = sum(tsne_test_flat_output, [])
tsne_test_exp_avg = sum(tsne_test_flat_exp_avg,[])
```

In [58]:
```
# Preparing the data frame for our train data
columns = ['ft_5','ft_4','ft_3','ft_2','ft_1']
df_train = pd.DataFrame(data=train_new_features, columns=columns)
df_train['lat'] = tsne_train_lat
df_train['lon'] = tsne_train_lon
```

```
df_train['weekday'] = tsne_train_weekday
df_train['exp_avg'] = tsne_train_exp_avg

print(df_train.shape)
```

```
(366760, 9)
```

```
In [59]: # Preparing the data frame for our train data
         df_test = pd.DataFrame(data=test_new_features, columns=columns)
         df_test['lat'] = tsne_test_lat
         df_test['lon'] = tsne_test_lon
         df_test['weekday'] = tsne_test_weekday
         df_test['exp_avg'] = tsne_test_exp_avg
         print(df_test.shape)
```

```
(157200, 9)
```

```
In [60]: df_test.head()
```

Out[60]:

	ft_5	ft_4	ft_3	ft_2	ft_1	lat	lon	weekday	exp_avg
0	118	106	104	93	102	40.776228	-73.982119	4	100
1	106	104	93	102	101	40.776228	-73.982119	4	100
2	104	93	102	101	120	40.776228	-73.982119	4	114
3	93	102	101	120	131	40.776228	-73.982119	4	125
4	102	101	120	131	164	40.776228	-73.982119	4	152

```
In [86]: df_train = pd.read_csv('df_train.csv')
```

```
In [88]: df_test = pd.read_csv('df_test.csv')
```

6.7.2 Using Linear Regression

```
In [63]: # find more about LinearRegression function here http://scikit-learn.org/stable/modul
         # -------------------------
         # default paramters
         # sklearn.linear_model.LinearRegression(fit_intercept=True, normalize=False, copy_X=T

         # some of methods of LinearRegression()
         # fit(X, y[, sample_weight])        Fit linear model.
         # get_params([deep])        Get parameters for this estimator.
         # predict(X)        Predict using the linear model
         # score(X, y[, sample_weight])        Returns the coefficient of determination R^2 of
         # set_params(**params)        Set the parameters of this estimator.
         # -----------------------
         # video link: https://www.appliedaicourse.com/course/applied-ai-course-online/lessons
         # -----------------------
```

```
from sklearn.linear_model import LinearRegression
lr_reg=LinearRegression().fit(df_train, tsne_train_output)

y_pred = lr_reg.predict(df_test)
lr_test_predictions = [round(value) for value in y_pred]
y_pred = lr_reg.predict(df_train)
lr_train_predictions = [round(value) for value in y_pred]
print('Train MAPE = ',(mean_absolute_error(tsne_train_output, lr_train_predictions))/
print('Test MAPE = ',(mean_absolute_error(tsne_test_output, lr_test_predictions))/(su
```

```
Train MAPE =  0.1331672957224943
Test MAPE =  0.12912870954053876
```

```
In [111]: from sklearn.linear_model import Ridge
          for alpha in [10**i for i in range(-4,4)]:
              lr_reg = Ridge(alpha=alpha).fit(df_train, tsne_train_output)

              y_pred = lr_reg.predict(df_test)
              lr_ridge_test_predictions = [round(value) for value in y_pred]
              y_pred = lr_reg.predict(df_train)
              lr_ridge_train_predictions = [round(value) for value in y_pred]
              print('alpha = ',alpha)
              print('Train MAPE = ',(mean_absolute_error(tsne_train_output, lr_ridge_train_pre
              print('Test MAPE = ',(mean_absolute_error(tsne_test_output, lr_ridge_test_predic
              print('------')
```

```
alpha =  0.0001
Train MAPE =  0.1331672957224943
Test MAPE =  0.12912870954053876
------
alpha =  0.001
Train MAPE =  0.1331672957224943
Test MAPE =  0.12912870954053876
------
alpha =  0.01
Train MAPE =  0.1331672957224943
Test MAPE =  0.12912870954053876
------
alpha =  0.1
Train MAPE =  0.13316738445730014
Test MAPE =  0.1291285184010031
------
alpha =  1
Train MAPE =  0.13316738445730014
Test MAPE =  0.12912842283123527
------
alpha =  10
```

```
Train MAPE =  0.13316742882470306
Test MAPE =  0.12913052536612776
------
alpha =  100
Train MAPE =  0.13317341842409647
Test MAPE =  0.1291321500521811
------
alpha =  1000
Train MAPE =  0.13324649153669607
Test MAPE =  0.12920182041293785
------
```

6.7.3 Using Random Forest Regressor

```
In [61]: def MAPE(y,y_pred):
             y = np.array(y)
             y_pred = np.array(y_pred)
             y_pred = np.round(y_pred)
             return (mean_absolute_error(y,y_pred))/(sum(y.ravel())/len(y.ravel()))
         from sklearn.metrics import make_scorer
         mape_score = make_scorer(MAPE,greater_is_better=False)
```

```
In [135]: rf_reg = RandomForestRegressor(random_state=25,n_jobs=-1)
          rf_reg.fit(df_train, tsne_train_output)
          y_pred = rf_reg.predict(df_test)
          rndf_test_predictions = [round(value) for value in y_pred]
          y_pred = rf_reg.predict(df_train)
          rndf_train_predictions = [round(value) for value in y_pred]
          print('Train MAPE = ',MAPE(tsne_train_output,rndf_train_predictions))
          print('Test MAPE = ',MAPE(tsne_test_output, rndf_test_predictions))
```

```
Train MAPE =  0.05515742220064981
Test MAPE =  0.1352156436241585
```

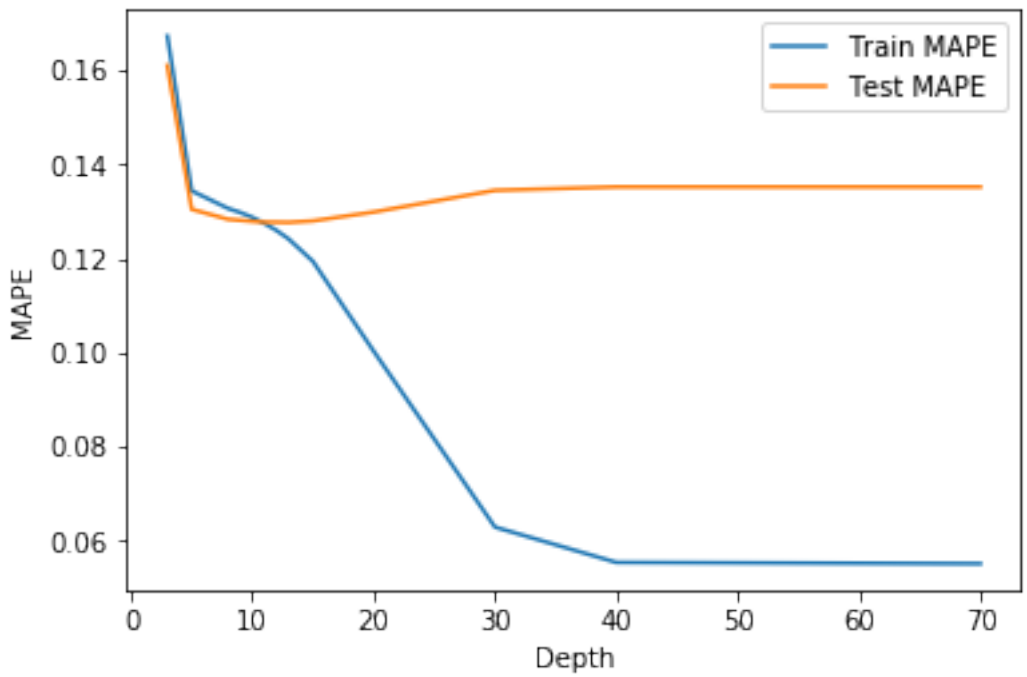
```
In [94]: %matplotlib inline
         train_errors = []
         test_errors = []
         depths = [3,5,8,9,10,11,12,13,15,20,30,40,70]
         for depth in depths:
             rf_reg = RandomForestRegressor(max_depth=depth,random_state=25,n_jobs=-1)
             rf_reg.fit(df_train, tsne_train_output)
             y_pred = rf_reg.predict(df_test)
             rndf_test_predictions = [round(value) for value in y_pred]
             y_pred = rf_reg.predict(df_train)
             rndf_train_predictions = [round(value) for value in y_pred]
             train_err = MAPE(tsne_train_output,rndf_train_predictions)
             test_err = MAPE(tsne_test_output, rndf_test_predictions)
```

```
            train_errors.append(train_err)
            test_errors.append(test_err)
            print('Depth = ',depth)
            print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
            print('---')
        plt.figure()
        plt.plot(depths,train_errors,label='Train MAPE')
        plt.plot(depths,test_errors,label='Test MAPE')
        plt.xlabel('Depth')
        plt.ylabel('MAPE')
        plt.legend()
```

```
Depth =  3
Train MAPE =  0.16728436443917163 Test MAPE =  0.16101068852283543
---
Depth =  5
Train MAPE =  0.13439156983850709 Test MAPE =  0.13046582616241506
---
Depth =  8
Train MAPE =  0.1306510428336218 Test MAPE =  0.1283376785721112
---
Depth =  9
Train MAPE =  0.12979111383033995 Test MAPE =  0.1281213086177171
---
Depth =  10
Train MAPE =  0.1287573977098434 Test MAPE =  0.12790589436100142
---
Depth =  11
Train MAPE =  0.12747287266067323 Test MAPE =  0.12773520675563574
---
Depth =  12
Train MAPE =  0.12595617299205336 Test MAPE =  0.12772478965094097
---
Depth =  13
Train MAPE =  0.12414789075148176 Test MAPE =  0.1276850326275187
---
Depth =  15
Train MAPE =  0.11938917621711989 Test MAPE =  0.12799458310555872
---
Depth =  20
Train MAPE =  0.10032530179816648 Test MAPE =  0.12987711639250885
---
Depth =  30
Train MAPE =  0.06292469032661951 Test MAPE =  0.1344909380746132
---
Depth =  40
Train MAPE =  0.05542109767616854 Test MAPE =  0.13521717274044398
---
```

```
Depth =  70
Train MAPE =  0.05515742220064981 Test MAPE =  0.13521554805439065
---
```

```
Out[94]: <matplotlib.legend.Legend at 0x1457da9c5668>
```



Depth of 10-13 may be best depth without overfitting to the data.

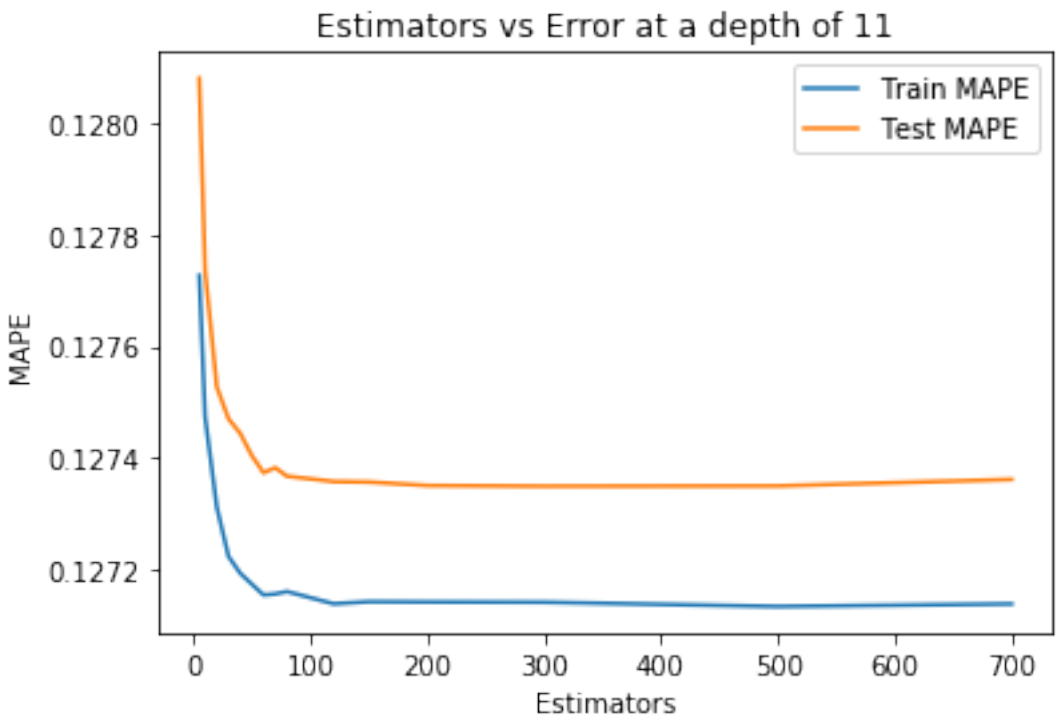
```
In [95]: train_errors = []
         test_errors = []
         estimators = [5,10,20,30,40,50,60,70,80,120,150,200,300,500,700]
         for i in estimators:
             rf_reg = RandomForestRegressor(n_estimators=i,max_depth=11,random_state=25,n_jobs=
             rf_reg.fit(df_train, tsne_train_output)
             y_pred = rf_reg.predict(df_test)
             rndf_test_predictions = [round(value) for value in y_pred]
             y_pred = rf_reg.predict(df_train)
             rndf_train_predictions = [round(value) for value in y_pred]
             train_err = MAPE(tsne_train_output,rndf_train_predictions)
             test_err = MAPE(tsne_test_output, rndf_test_predictions)
             train_errors.append(train_err)
             test_errors.append(test_err)
             print('Estimators = ',i)
             print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
             print('---')
```

```
plt.figure()
plt.plot(estimators,train_errors,label='Train MAPE')
plt.plot(estimators,test_errors,label='Test MAPE')
plt.xlabel('Estimators')
plt.ylabel('MAPE')
plt.title('Estimators vs Error at a depth of 11')
plt.legend()
```

```
Estimators =  5
Train MAPE =  0.12772723098157998 Test MAPE =  0.12808126488499133
---
Estimators =  10
Train MAPE =  0.12747287266067323 Test MAPE =  0.12773520675563574
---
Estimators =  20
Train MAPE =  0.12731110910964827 Test MAPE =  0.12752686466174035
---
Estimators =  30
Train MAPE =  0.12722277361044623 Test MAPE =  0.12746990508010658
---
Estimators =  40
Train MAPE =  0.12719260377646463 Test MAPE =  0.12744381453348572
---
Estimators =  50
Train MAPE =  0.12717286028216782 Test MAPE =  0.12740348409145644
---
Estimators =  60
Train MAPE =  0.12715356046190016 Test MAPE =  0.12737299733551485
---
Estimators =  70
Train MAPE =  0.12715542389282256 Test MAPE =  0.12738207646345984
---
Estimators =  80
Train MAPE =  0.12715977189830815 Test MAPE =  0.12736678530060513
---
Estimators =  120
Train MAPE =  0.1271377212990598 Test MAPE =  0.12735722832382093
---
Estimators =  150
Train MAPE =  0.12714153689571045 Test MAPE =  0.12735627262614252
---
Estimators =  200
Train MAPE =  0.12714104885427838 Test MAPE =  0.1273500605912328
---
Estimators =  300
Train MAPE =  0.12714056081284633 Test MAPE =  0.12734881818425084
---
Estimators =  500
```

```
Train MAPE =  0.1271329296195451 Test MAPE =  0.12734929603309006
---
Estimators =  700
Train MAPE =  0.12713754382944817 Test MAPE =  0.12736095554476679
---
```

```
Out[95]: <matplotlib.legend.Legend at 0x1457dad93470>
```



Even incresing no of Estimators also there is no such inmprovement in the train or test scores so may be 100-150 be better choice. we will decide this with randomsearch .

```
In [62]: ## for cross validation we need a custom cv folds because
         ##in every fold we have to include data from each cluster (Region)
         ## and we need to consider it as time series data so written a custom cv iterable
         ## for 10 fold cv
         from sklearn.model_selection import TimeSeriesSplit
         def custom_cv_10fold():
             '''
             Gives a iterable with train and test indeces for 10 fold cv
             '''
             tscv = TimeSeriesSplit(n_splits=10)
             train_cv= []
             test_cv = []
             for train, test in tscv.split(df_train[0:9169]):
```

```
            train_temp = []
            test_temp = []
            for i in range(0,40):
                train_temp.extend(list(np.array(train)+i*9169))
            for i in range(0,40):
                test_temp.extend(list(np.array(test)+i*9169))
            train_cv.append(np.array(train_temp))
            test_cv.append(np.array(test_temp))
        return zip(np.array(train_cv),np.array(test_cv))
cv_custom = custom_cv_10fold()
```

In [63]:
```
from sklearn.model_selection import RandomizedSearchCV
from scipy.stats import randint as sp_randint
from scipy.stats import uniform
param_dist = {"max_depth": sp_randint(8,14),
              "n_estimators":sp_randint(90,160),
              "max_features": sp_randint(3,9),
              "min_samples_split": sp_randint(2, 8),
              "min_samples_leaf": sp_randint(1, 12),
              "bootstrap": [True, False]}
model_rs_rf = RandomizedSearchCV(RandomForestRegressor(n_jobs=-1),
                                 param_distributions=param_dist,
                                 n_iter=50,scoring=mape_score,
                                 cv=cv_custom,n_jobs=-1)
model_rs_rf.fit(df_train,tsne_train_output)
model_rs_rf = pickle.dump(model_rs_rf,open('model_rs_rf.p','wb'))
```

In [119]:
```
dict_scores = []
idx = 0
for i in model_rs_rf.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['max_features'])
    dict_score.append(i[0]['min_samples_leaf'])
    dict_score.append(i[0]['min_samples_split'])
    dict_score.append(i[0]['bootstrap'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_rf.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df = pd.DataFrame(dict_scores,columns=['n_estimators','depth','max_features',
                                              'min_samples_split','bootstrap','Test_
                                              'Test_std','Train_score'])
```

In [127]:
```
scores_df.sort_values('Test_score',ascending=True).head(15)
```

Out[127]:
```
    n_estimators  depth  max_features  min_samples_leaf  min_samples_split  \
21           154     13             3                 5                  6
```

```
20          112     14            3                 8                  5
41           91     12            3                 6                  5
45           92     12            3                 2                  5
2           145     14            4                 8                  6
18          106     12            4                 9                  3
25          115     11            3                 3                  4
14          126     12            3                 6                  3
54           94     12            4                 5                  5
16          108     14            3                 8                  6
50          144     13            5                 6                  6
9           107     12            5                11                  7
47          104     14            5                10                  7
33          107     12            5                10                  7
36          112     14            5                10                  3

    bootstrap  Test_score  Test_std  Train_score
21       True    0.130042  0.002303     0.120083
20       True    0.130079  0.002313     0.119649
41       True    0.130096  0.002214     0.123596
45       True    0.130153  0.002315     0.121832
2        True    0.130199  0.002348     0.119241
18       True    0.130207  0.002281     0.124457
25       True    0.130213  0.002203     0.125369
14      False    0.130218  0.002405     0.121311
54       True    0.130238  0.002276     0.122998
16      False    0.130243  0.002465     0.115099
50       True    0.130265  0.002369     0.120540
9        True    0.130286  0.002270     0.124939
47       True    0.130297  0.002361     0.120400
33       True    0.130298  0.002280     0.124669
36       True    0.130306  0.002330     0.120406
```

In [122]:
```
#with max features = auto and bootstrap = True
model_rs_rf1 = pickle.load(open('/home/u15918/model_rs_rf1.p','rb'))
param_dist = {"max_depth": sp_randint(11,16),
              "n_estimators":sp_randint(100,220),
              "min_samples_split": sp_randint(3, 8),
              "min_samples_leaf": sp_randint(2, 12)}
model_rs_rf1 = RandomizedSearchCV(RandomForestRegressor(bootstrap=True,max_features=
                                                        n_jobs=-1), param_distributio
                                  n_iter=80,scoring=mape_score,cv=cv_custom,random_s
model_rs_rf1.fit(df_train, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_rf1,open('model_rs_rf1.p','wb'))
```

In [125]:
```
dict_scores = []
idx = 0
for i in model_rs_rf1.grid_scores_:
```

```
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['min_samples_leaf'])
    dict_score.append(i[0]['min_samples_split'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_rf1.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df1 = pd.DataFrame(dict_scores,columns=['n_estimators','depth','min_samples_le
                                               'min_samples_split','Test_score',
                                               'Test_std','Train_score'])
```

In [128]:
```
scores_df1.sort_values('Test_score',ascending=True).head(15)
```

Out[128]:

	n_estimators	depth	min_samples_leaf	min_samples_split	Test_score	\
37	198	12	10	5	0.130591	
19	186	13	9	4	0.130608	
68	174	12	11	5	0.130631	
55	144	12	11	3	0.130631	
38	184	13	7	4	0.130633	
32	163	13	10	6	0.130635	
49	193	13	11	3	0.130639	
78	157	13	8	7	0.130640	
50	130	11	10	3	0.130644	
61	140	12	8	6	0.130646	
67	154	13	7	6	0.130648	
41	168	11	8	3	0.130653	
52	162	13	9	4	0.130655	
70	174	13	6	6	0.130657	
72	190	14	11	7	0.130659	

	Test_std	Train_score
37	0.002364	0.124920
19	0.002398	0.122397
68	0.002323	0.125141
55	0.002358	0.125141
38	0.002435	0.121649
32	0.002398	0.122750
49	0.002377	0.123061
78	0.002420	0.122058
50	0.002271	0.127027
61	0.002359	0.124483
67	0.002433	0.121672
41	0.002307	0.126714
52	0.002418	0.122407
70	0.002417	0.121272
72	0.002443	0.121033

Got good cv scores at n_estimators = 154, depth = 13, max_features = 3, min_samples_leaf = 5, min_samples_split = 6, bootstrap = True and the Test Score = 0.130042, test std = 0.002303, Train score = 0.120083.

```
In [64]: regr1 = RandomForestRegressor(max_features=3,min_samples_leaf=5,
                                       min_samples_split=6,n_estimators=154,
                                       bootstrap = True,max_depth = 13,n_jobs=-1)
         regr1.fit(df_train, tsne_train_output)
```

```
Out[64]: RandomForestRegressor(bootstrap=True, criterion='mse', max_depth=13,
                    max_features=3, max_leaf_nodes=None, min_impurity_decrease=0.0,
                    min_impurity_split=None, min_samples_leaf=5,
                    min_samples_split=6, min_weight_fraction_leaf=0.0,
                    n_estimators=154, n_jobs=-1, oob_score=False, random_state=None,
                    verbose=0, warm_start=False)
```

```
In [65]: rndf_test_predictions = regr1.predict(df_test)
         rndf_train_predictions = regr1.predict(df_train)
         print('Train MAPE for Random Forest',MAPE(tsne_train_output,rndf_train_predictions))
         print('Test MAPE for Random Forest',MAPE(tsne_test_output,rndf_test_predictions))
```

```
Train MAPE for Random Forest 0.12262595572931803
Test MAPE for Random Forest 0.12646250415728488
```

```
In [66]: #feature importances based on analysis using random forest
         print (df_train.columns)
         print (regr1.feature_importances_)
```

```
Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
       'exp_avg'],
      dtype='object')
[0.02036286 0.08988435 0.10135364 0.17920857 0.26774517 0.00067306
 0.00104094 0.00034333 0.33938808]
```

6.7.4 Using XgBoost Regressor

```
In [64]: import xgboost as xgb
```

```
In [25]: #basic xgboost without any change
         xgb_reg = xgb.XGBRegressor(random_state=25,n_jobs=-1)
         xgb_reg.fit(df_train, tsne_train_output)
         xgb_test_predictions = xgb_reg.predict(df_test)
         xgb_train_predictions = xgb_reg.predict(df_train)
         print('Train MAPE = ',MAPE(tsne_train_output,xgb_train_predictions))
         print('Test MAPE = ',MAPE(tsne_test_output, xgb_test_predictions))
```

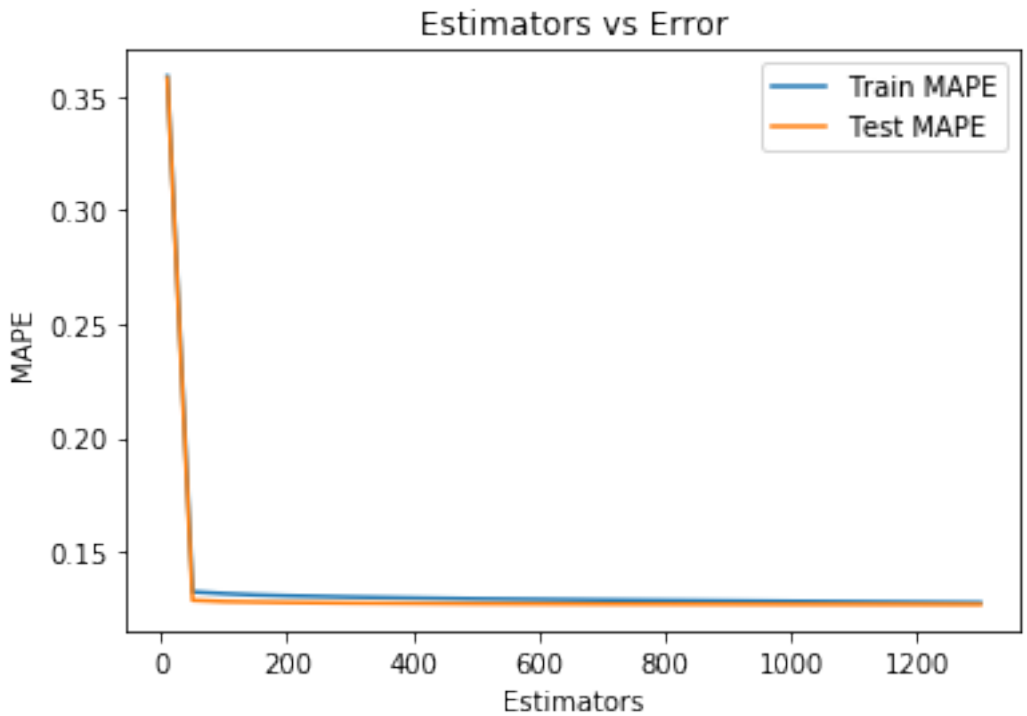
```
Train MAPE =  0.1314521406606395
Test MAPE =  0.12796734572172377
```

```
In [28]: %matplotlib inline
         train_errors = []
         test_errors = []
         estimators = [10,50,100,150,200,300,500,700,1000,1300]
         for i in estimators:
             xgb_reg = xgb.XGBRegressor(n_estimators=i,random_state=25,n_jobs=-1)
             xgb_reg.fit(df_train, tsne_train_output)
             xgb_test_predictions = xgb_reg.predict(df_test)
             xgb_train_predictions = xgb_reg.predict(df_train)
             train_err = MAPE(tsne_train_output,xgb_train_predictions)
             test_err = MAPE(tsne_test_output, xgb_test_predictions)
             train_errors.append(train_err)
             test_errors.append(test_err)
             print('Estimators = ',i)
             print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
             print('---')
         plt.figure()
         plt.plot(estimators,train_errors,label='Train MAPE')
         plt.plot(estimators,test_errors,label='Test MAPE')
         plt.xlabel('Estimators')
         plt.ylabel('MAPE')
         plt.title('Estimators vs Error')
         plt.legend()
```

```
Estimators =  10
Train MAPE =  0.3589861959699313 Test MAPE =  0.35803225670804195
---
Estimators =  50
Train MAPE =  0.13224330018940447 Test MAPE =  0.12853550799154398
---
Estimators =  100
Train MAPE =  0.1314521406606395 Test MAPE =  0.12796734572172377
---
Estimators =  150
Train MAPE =  0.13088268504423653 Test MAPE =  0.12775623210456097
---
Estimators =  200
Train MAPE =  0.13049003352844638 Test MAPE =  0.12756203433630617
---
Estimators =  300
Train MAPE =  0.12994653284274818 Test MAPE =  0.12732091181204103
---
Estimators =  500
Train MAPE =  0.12915905580842513 Test MAPE =  0.1270299974387302
---
Estimators =  700
Train MAPE =  0.12866648890127233 Test MAPE =  0.1268873117753422
---
```

```
Estimators =  1000
Train MAPE =  0.1280621161387759 Test MAPE =  0.1267545653678098
---
Estimators =  1300
Train MAPE =  0.1275066806216938 Test MAPE =  0.12667830069307193
---
```

Out[28]: <matplotlib.legend.Legend at 0x147b6bd448d0>

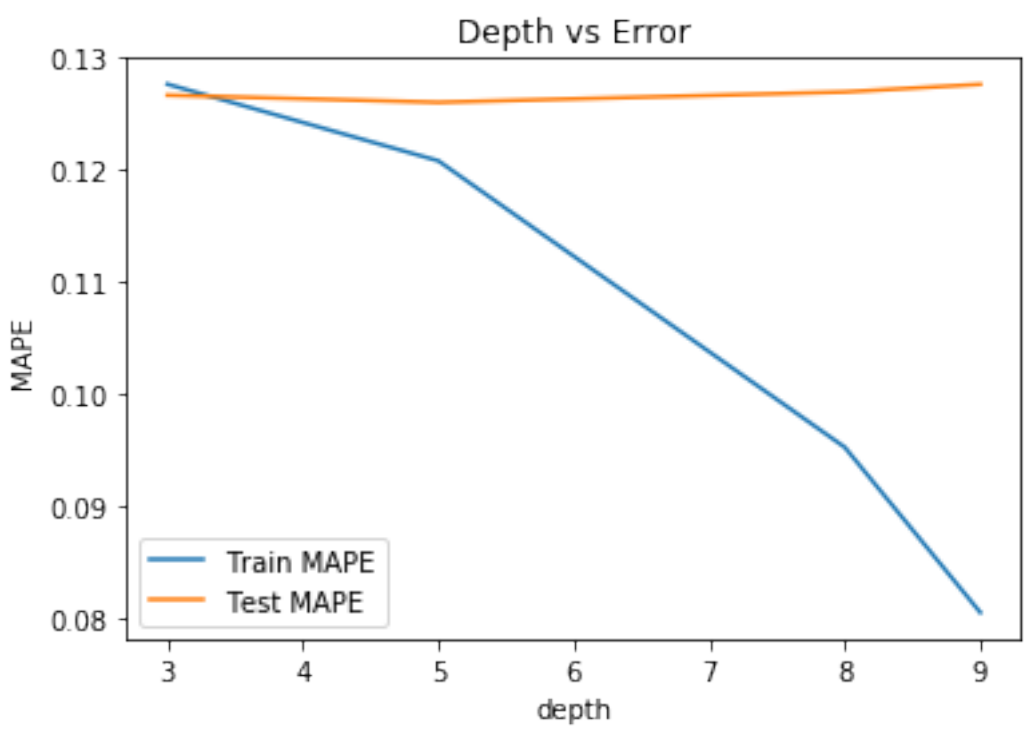


In [31]:
```
%matplotlib inline
train_errors = []
test_errors = []
depths = [3,5,8,9]
for i in depths:
    xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=i,random_state=25,n_jobs=-
    xgb_reg.fit(df_train, tsne_train_output)
    xgb_test_predictions = xgb_reg.predict(df_test)
    xgb_train_predictions = xgb_reg.predict(df_train)
    train_err = MAPE(tsne_train_output,xgb_train_predictions)
    test_err = MAPE(tsne_test_output, xgb_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('depth = ',i)
```

```
        print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
        print('---')
    plt.figure()
    plt.plot(depths,train_errors,label='Train MAPE')
    plt.plot(depths,test_errors,label='Test MAPE')
    plt.xlabel('depth')
    plt.ylabel('MAPE')
    plt.title('Depth vs Error')
    plt.legend()
```

```
depth =  3
Train MAPE =  0.12767709581628703 Test MAPE =  0.12669311400708744
---
depth =  5
Train MAPE =  0.120841676253723 Test MAPE =  0.126057383911403
---
depth =  8
Train MAPE =  0.09528787123869796 Test MAPE =  0.12698259483388064
---
depth =  9
Train MAPE =  0.08055212570882472 Test MAPE =  0.12766333829021861
---
```

```
Out[31]: <matplotlib.legend.Legend at 0x147b70dd35f8>
```

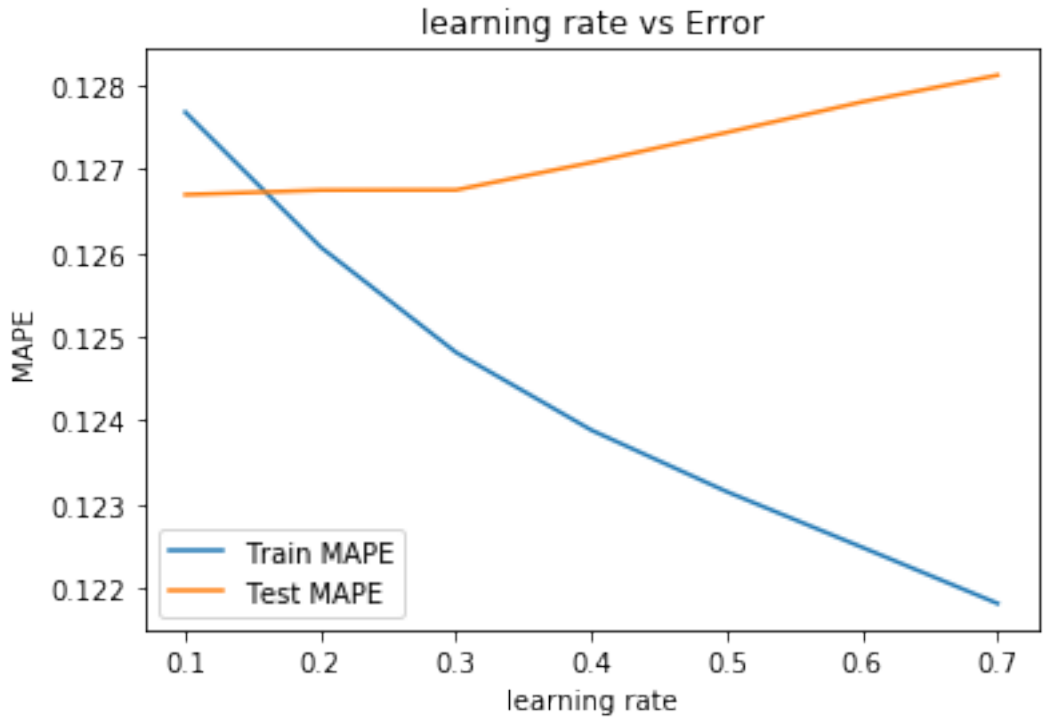


We can observe that, if depth is incresing then model is overfitting to the data.

```
In [33]: %matplotlib inline
         train_errors = []
         test_errors = []
         learning_rates = [0.1,0.2,0.3,0.4,0.5,0.6,0.7]
         for i in learning_rates:
             xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=3,learning_rate=i,random_st
             xgb_reg.fit(df_train, tsne_train_output)
             xgb_test_predictions = xgb_reg.predict(df_test)
             xgb_train_predictions = xgb_reg.predict(df_train)
             train_err = MAPE(tsne_train_output,xgb_train_predictions)
             test_err = MAPE(tsne_test_output, xgb_test_predictions)
             train_errors.append(train_err)
             test_errors.append(test_err)
             print('learning rate = ',i)
             print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
             print('---')
         plt.figure()
         plt.plot(learning_rates,train_errors,label='Train MAPE')
         plt.plot(learning_rates,test_errors,label='Test MAPE')
         plt.xlabel('learning rate')
         plt.ylabel('MAPE')
         plt.title('learning rate vs Error')
         plt.legend()
```

```
learning rate =  0.1
Train MAPE =  0.12767709581628703 Test MAPE =  0.12669311400708744
---
learning rate =  0.2
Train MAPE =  0.12606789011259117 Test MAPE =  0.12674691978638244
---
learning rate =  0.3
Train MAPE =  0.12481282501895599 Test MAPE =  0.1267493090305785
---
learning rate =  0.4
Train MAPE =  0.1238859456046767 Test MAPE =  0.12707711333427626
---
learning rate =  0.5
Train MAPE =  0.12315215312787972 Test MAPE =  0.12743530882414777
---
learning rate =  0.6
Train MAPE =  0.12248899355652208 Test MAPE =  0.12779952520939336
---
learning rate =  0.7
Train MAPE =  0.12181988875317394 Test MAPE =  0.1281174858270034
---
```

Out[33]: <matplotlib.legend.Legend at 0x147b70a36f60>

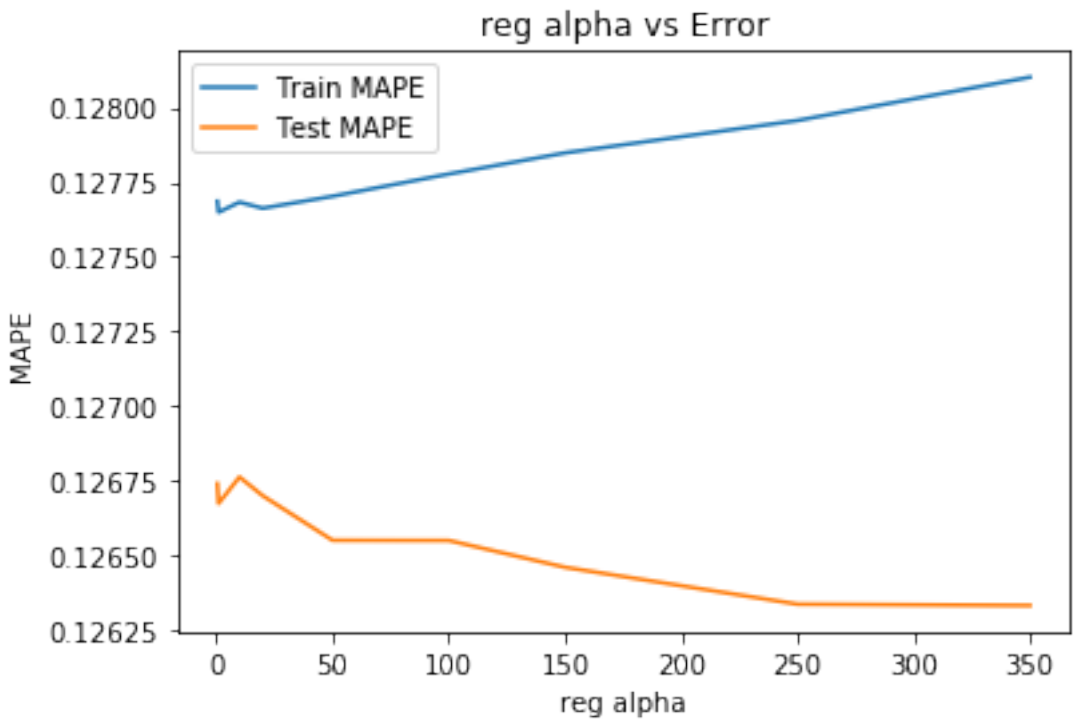


In [35]:
```
%matplotlib inline
train_errors = []
test_errors = []
alphas = [0.5,1,10,20,50,100,150,250,350]
for i in alphas:
    xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=3,learning_rate=0.1,reg_al
    xgb_reg.fit(df_train, tsne_train_output)
    xgb_test_predictions = xgb_reg.predict(df_test)
    xgb_train_predictions = xgb_reg.predict(df_train)
    train_err = MAPE(tsne_train_output,xgb_train_predictions)
    test_err = MAPE(tsne_test_output, xgb_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('alpha = ',i)
    print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
    print('---')
plt.figure()
plt.plot(alphas,train_errors,label='Train MAPE')
plt.plot(alphas,test_errors,label='Test MAPE')
plt.xlabel('reg alpha')
plt.ylabel('MAPE')
```

```
        plt.title('reg alpha vs Error')
        plt.legend()
```

```
alpha =  0.5
Train MAPE =  0.12768610239907857 Test MAPE =  0.12673879635611587
---
alpha =  1
Train MAPE =  0.1276490999850482 Test MAPE =  0.1266720886581622
---
alpha =  10
Train MAPE =  0.12768290794606876 Test MAPE =  0.12676096854225521
---
alpha =  20
Train MAPE =  0.1276625876755341 Test MAPE =  0.12669817920478307
---
alpha =  50
Train MAPE =  0.12770256270555974 Test MAPE =  0.12654880365764615
---
alpha =  100
Train MAPE =  0.12777732177947007 Test MAPE =  0.1265480390995034
---
alpha =  150
Train MAPE =  0.12784817652192393 Test MAPE =  0.12645810794796417
---
alpha =  250
Train MAPE =  0.12795687665906358 Test MAPE =  0.12633434509860886
---
alpha =  350
Train MAPE =  0.12810222427101028 Test MAPE =  0.12632994888928817
---
```

```
Out[35]: <matplotlib.legend.Legend at 0x147b7062cfd0>
```




```
In [91]: param_dist = {"max_depth": sp_randint(2,5),
                       "learning_rate":uniform(0,0.4),
                       "n_estimators":sp_randint(500,1500),
                       "min_child_weight": sp_randint(2, 8),
                       "gamma": uniform(0,4),
                       "subsample":uniform(0.7,0.3),
                       "colsample_bytree": uniform(0.7,0.3),
                       "reg_alpha":uniform(50,300),
                       "reg_lambda":uniform(50,300)}
         model_rs_xgb = RandomizedSearchCV(xgb.XGBRegressor(n_jobs=-1,random_state=25), param_d
                                           n_iter=60,scoring=mape_score,cv=cv_random,n_jobs=-1
         model_rs_xgb.fit(df_train, np.array(tsne_train_output).reshape(-1,1))

         pickle.dump(model_rs_xgb,open('model_rs_xgb.p','wb'))

In [94]: dict_scores = []
         idx = 0
         for i in model_rs_xgb.grid_scores_:
             dict_score = []
             dict_score.append(i[0]['n_estimators'])
             dict_score.append(i[0]['max_depth'])
             dict_score.append(i[0]['subsample'])
             dict_score.append(i[0]['min_child_weight'])
             dict_score.append(i[0]['learning_rate'])
```

```
        dict_score.append(i[0]['reg_alpha'])
        dict_score.append(i[0]['reg_lambda'])
        dict_score.append(i[0]['gamma'])
        dict_score.append(i[0]['colsample_bytree'])
        dict_score.append(-i[1])
        dict_score.append((np.abs(i[2]).std()))
        dict_score.append(-model_rs_xgb.cv_results_['mean_train_score'][idx])
        dict_scores.append(dict_score)
        idx = idx + 1
    scores_df = pd.DataFrame(dict_scores,columns=['n_estimators','depth','subsample','min_
                                                  'learning_rate','reg_alpha','reg_lambda
                                                  'gamma','Test_score',
                                                  'Test_std','Train_score'])
```

In [95]:
```
scores_df.sort_values('Test_score')
```

Out[95]:

	n_estimators	depth	subsample	min_child_weight	learning_rate	\
45	796	4	0.933216	2	0.081514	
39	1269	4	0.750671	5	0.019686	
47	613	4	0.935360	7	0.088536	
0	701	4	0.882286	7	0.114138	
53	1032	4	0.992537	4	0.159035	
49	1104	4	0.778829	6	0.052591	
2	608	3	0.944728	5	0.046327	
59	503	3	0.977901	7	0.081593	
25	1347	3	0.951024	5	0.023088	
33	1119	3	0.806651	3	0.124731	
41	568	3	0.962890	3	0.233031	
4	808	3	0.701262	5	0.110865	
6	513	4	0.866341	5	0.364493	
21	532	3	0.990889	4	0.324569	
28	1390	4	0.702220	7	0.158471	
48	1058	3	0.958791	3	0.134699	
36	1429	3	0.701496	6	0.176591	
54	1096	2	0.992240	4	0.350167	
24	891	4	0.922409	2	0.180075	
15	510	4	0.813808	3	0.213109	
40	598	3	0.837293	6	0.311250	
58	610	3	0.933162	2	0.393003	
52	1147	3	0.729421	2	0.173638	
35	1411	2	0.933978	5	0.059402	
37	1255	2	0.926283	7	0.043733	
51	868	3	0.870342	7	0.370501	
14	1398	3	0.853813	7	0.290587	
3	878	3	0.705963	5	0.261680	
17	735	3	0.798319	7	0.269010	
12	1087	4	0.729657	6	0.164418	
8	844	2	0.951920	4	0.054009	

38	1227	3	0.869399	7	0.276098
27	1462	4	0.722860	6	0.214687
10	1198	2	0.911253	2	0.059024
34	689	2	0.973945	6	0.322509
16	1076	3	0.998999	2	0.370001
31	1196	3	0.862370	4	0.291155
46	1026	3	0.883843	6	0.323351
43	1123	2	0.785158	4	0.112809
1	981	2	0.902372	3	0.234643
11	1212	2	0.866583	7	0.389662
57	874	2	0.792305	5	0.354907
30	612	2	0.886479	2	0.382674
44	750	2	0.941443	4	0.097955
5	1004	4	0.884257	7	0.372866
55	1369	2	0.767052	4	0.297252
23	990	3	0.751789	4	0.368227
13	1218	2	0.864430	6	0.240162
7	1451	3	0.717069	7	0.262637
26	937	2	0.800293	4	0.266372
18	1161	3	0.948821	6	0.307687
56	850	4	0.820970	2	0.365787
42	523	2	0.827874	2	0.095348
50	872	2	0.731666	3	0.162513
22	1314	4	0.707236	6	0.221981
32	666	4	0.776555	7	0.330375
9	937	4	0.708827	2	0.309224
20	1336	4	0.713797	2	0.363135
29	880	4	0.712936	7	0.004337
19	870	2	0.975263	6	0.003173

	reg_alpha	reg_lambda	colsample_bytree	gamma	Test_score	Test_std	\
45	175.862283	102.592314	2.507191	0.733809	0.130877	0.003307	
39	272.480590	79.577790	2.885866	0.864165	0.130880	0.002639	
47	305.132601	265.236226	1.858015	0.788401	0.130917	0.003128	
0	210.487735	282.589172	0.087005	0.762129	0.130968	0.003114	
53	142.162497	89.387013	1.397480	0.790416	0.131140	0.003586	
49	74.477874	273.228559	0.754762	0.799141	0.131147	0.003306	
2	310.231456	102.713242	1.297850	0.903477	0.131196	0.002373	
59	188.950882	88.681491	0.164319	0.731025	0.131200	0.002610	
25	266.332782	182.648103	3.300705	0.737611	0.131221	0.002353	
33	208.415269	215.577340	1.972931	0.960867	0.131263	0.003088	
41	98.789716	276.208857	3.952521	0.964486	0.131271	0.003041	
4	315.415834	65.565579	1.722700	0.994411	0.131293	0.002932	
6	296.703638	323.765237	0.211042	0.759489	0.131357	0.003231	
21	96.167065	123.438260	2.183110	0.787656	0.131400	0.003422	
28	278.889961	153.270885	1.456388	0.822320	0.131512	0.003656	
48	53.166925	326.987975	0.021381	0.797509	0.131516	0.003657	
36	234.321445	159.663834	2.235439	0.739189	0.131569	0.003322	

54	230.309207	118.760654	0.724383	0.990805	0.131577	0.003377
24	189.526679	147.724660	0.180776	0.813147	0.131578	0.004403
15	64.010087	127.502759	2.166771	0.784944	0.131582	0.003666
40	254.737916	84.032833	2.574325	0.822053	0.131613	0.003362
58	83.921797	273.309007	0.888909	0.742125	0.131637	0.003429
52	121.881177	99.876886	1.154507	0.994512	0.131651	0.003369
35	298.439421	228.997739	0.910128	0.840166	0.131667	0.002561
37	63.287400	155.008173	2.436820	0.893348	0.131692	0.002414
51	88.422607	286.298573	1.351296	0.861264	0.131698	0.003419
14	142.010235	269.289559	2.525042	0.804809	0.131708	0.003499
3	115.447056	298.356078	3.220455	0.825588	0.131714	0.003278
17	184.409888	82.863459	1.499316	0.880622	0.131730	0.003345
12	109.138793	64.299133	1.539584	0.810742	0.131746	0.003616
8	124.046970	128.233128	0.766270	0.858800	0.131762	0.002355
38	319.533975	319.834727	2.394713	0.900972	0.131767	0.004091
27	295.371924	315.962029	1.185236	0.829028	0.131781	0.003827
10	339.419926	147.651111	3.109954	0.739128	0.131785	0.002623
34	219.593468	116.068725	0.767892	0.791378	0.131800	0.003275
16	62.222314	201.541411	2.746905	0.825311	0.131807	0.004060
31	97.773235	202.353786	2.232289	0.893887	0.131820	0.003539
46	85.752295	202.597227	1.160845	0.825990	0.131854	0.003712
43	326.294355	69.002119	0.949731	0.885539	0.131859	0.002766
1	255.413100	97.025832	3.617841	0.808218	0.131882	0.003219
11	276.732276	109.296109	0.766848	0.888819	0.131902	0.003419
57	288.499735	78.032448	0.285719	0.796390	0.131938	0.003270
30	302.006350	109.083935	3.112972	0.935295	0.131955	0.003234
44	64.247456	172.161223	3.168630	0.798148	0.131956	0.002565
5	218.458572	259.591195	3.293809	0.833160	0.131978	0.003665
55	198.832263	188.554611	2.725353	0.997072	0.131984	0.003170
23	246.666985	142.157483	2.685759	0.732726	0.131985	0.003613
13	64.487489	180.799056	1.770170	0.761370	0.132012	0.003236
7	69.959603	294.758367	1.837242	0.790629	0.132043	0.003692
26	98.236564	97.815010	3.900030	0.886599	0.132058	0.003152
18	51.387334	80.136033	0.914039	0.913119	0.132111	0.003909
56	107.586382	281.257550	0.188861	0.716873	0.132128	0.003757
42	221.638377	319.776635	2.767584	0.822788	0.132142	0.002331
50	247.411404	186.234870	0.155888	0.853917	0.132147	0.002757
22	208.656755	227.168271	2.294937	0.919786	0.132174	0.004265
32	118.407703	338.847848	1.028724	0.909037	0.132291	0.004250
9	140.180808	282.113480	0.992376	0.783144	0.132308	0.004086
20	218.584237	201.705689	3.790497	0.742906	0.132849	0.003974
29	344.497576	323.547846	3.793414	0.922281	0.135900	0.004930
19	247.111483	279.612811	2.069298	0.779414	0.157314	0.005812

	Train_score
45	0.129074
39	0.131395
47	0.130402

0	0.129620
53	0.125089
49	0.130267
2	0.132628
59	0.132024
25	0.132770
33	0.130113
41	0.130316
4	0.130754
6	0.127733
21	0.128921
28	0.126230
48	0.130201
36	0.129068
54	0.130889
24	0.126307
15	0.127749
40	0.129290
58	0.128965
52	0.128836
35	0.133063
37	0.133260
51	0.128133
14	0.127434
3	0.129574
17	0.128961
12	0.125042
8	0.133500
38	0.128505
27	0.125803
10	0.133192
34	0.131867
16	0.126331
31	0.127345
46	0.127555
43	0.132663
1	0.131882
11	0.130780
57	0.131573
30	0.131950
44	0.133340
5	0.123860
55	0.131227
23	0.128076
13	0.131612
7	0.127919
26	0.131793
18	0.125595

```
56     0.124750
42     0.134076
50     0.132982
22     0.125077
32     0.126499
9      0.125405
20     0.123119
29     0.136746
19     0.156447
```

In [80]:
```
param_dist = {"max_depth": sp_randint(3,5),
              "learning_rate":uniform(0,0.25),
              "n_estimators":sp_randint(600,1500),
              "min_child_weight": sp_randint(2, 6),
              "gamma": uniform(1,5),
              "subsample":uniform(0.7,0.3),
              "colsample_bytree": uniform(0.7,0.3),
              "reg_alpha":uniform(50,300),
              "reg_lambda":uniform(50,300)}
model_rs_xgb1 = RandomizedSearchCV(xgb.XGBRegressor(n_jobs=-1,random_state=25), param_
                                   n_iter=60,scoring=mape_score,cv=cv_random,n_jobs=-1
model_rs_xgb1.fit(df_train, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_xgb1,open('model_rs_xgb1.p','wb'))
```

In [82]:
```
dict_scores = []
idx = 0
for i in model_rs_xgb1.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['subsample'])
    dict_score.append(i[0]['min_child_weight'])
    dict_score.append(i[0]['learning_rate'])
    dict_score.append(i[0]['reg_alpha'])
    dict_score.append(i[0]['reg_lambda'])
    dict_score.append(i[0]['gamma'])
    dict_score.append(i[0]['colsample_bytree'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_xgb1.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df1 = pd.DataFrame(dict_scores,columns=['n_estimators','depth','subsample','min
                                               'learning_rate','reg_alpha','reg_lambda
                                              'colsample_bytree','Test_score',
                                               'Test_std','Train_score'])
```

In [84]:
```
scores_df1.sort_values('Test_score').head(15)
```

Out[84]:

	n_estimators	depth	subsample	min_child_weight	learning_rate	\
54	1300	4	0.970102	4	0.046343	
53	840	4	0.772684	4	0.079339	
35	806	4	0.920082	5	0.094608	
51	1005	4	0.955422	4	0.087905	
36	703	4	0.762869	5	0.077343	
25	1082	4	0.899544	4	0.031255	
33	1260	4	0.897162	4	0.082450	
17	896	4	0.898376	3	0.075462	
10	1191	3	0.857143	4	0.034896	
4	1335	3	0.809783	2	0.084558	
45	1099	4	0.763994	5	0.107428	
46	1499	3	0.849022	5	0.124632	
59	1305	3	0.929542	3	0.050319	
48	1153	4	0.873212	5	0.126770	
34	774	3	0.747812	3	0.101997	

	reg_alpha	reg_lambda	gamma	colsample_bytree	Test_score	Test_std	\
54	180.790625	58.127168	2.626861	0.961831	0.130605	0.003077	
53	221.182717	112.644784	1.249390	0.800053	0.130846	0.002975	
35	235.892488	160.175510	2.773804	0.994072	0.130857	0.003270	
51	142.492381	141.716180	3.276192	0.999961	0.130918	0.003446	
36	132.485974	129.748329	4.773886	0.978133	0.130930	0.002931	
25	198.113250	185.534899	1.105037	0.996339	0.130931	0.002785	
33	65.511045	87.027911	5.718612	0.757984	0.130963	0.003282	
17	318.566059	299.690894	4.569821	0.885391	0.131026	0.003378	
10	271.281741	114.521138	1.050294	0.887457	0.131027	0.002500	
4	344.877534	143.344217	1.446662	0.934201	0.131052	0.002942	
45	237.694511	195.608075	3.833065	0.933837	0.131108	0.003277	
46	324.032336	134.870329	2.437720	0.780461	0.131129	0.003035	
59	266.114150	285.350913	5.157400	0.832989	0.131138	0.002839	
48	180.673541	188.886060	2.751861	0.794426	0.131139	0.003346	
34	288.012746	197.818088	4.001075	0.816006	0.131186	0.002750	

	Train_score
54	0.128536
53	0.129305
35	0.128856
51	0.128096
36	0.129785
25	0.131097
33	0.127267
17	0.129948
10	0.131993
4	0.130351
45	0.128067
46	0.129365
59	0.131592

```
48     0.127081
34     0.131372
```

got good cv scores at n_estimators = 1300, depth = 4, subsample =0.970102, min_child_weight = 4, learning_rate = 0.046343, reg_alpha = 180.790625, reg_lambda =58.127168, gamma =2.626861, colsample_bytree = 0.961831, Test_score = 0.130605, Test_std = 0.003077,Train_score = 0.128536

```
In [68]: regr_xgb = xgb.XGBRegressor(
          learning_rate =  0.046343,
          n_estimators=1300,
          max_depth=4,
          min_child_weight=4,
          gamma=2.626861,
          subsample=0.970102,
          reg_alpha=180.790625, reg_lambda=58.127168,
          colsample_bytree=0.961831,njobs=-1,random_state=25)
         regr_xgb.fit(df_train, tsne_train_output)
```

```
Out[68]: XGBRegressor(base_score=0.5, booster='gbtree', colsample_bylevel=1,
                colsample_bytree=0.961831, gamma=2.626861, learning_rate=0.046343,
                max_delta_step=0, max_depth=4, min_child_weight=4, missing=None,
                n_estimators=1300, n_jobs=1, njobs=-1, nthread=None,
                objective='reg:linear', random_state=25, reg_alpha=180.790625,
                reg_lambda=58.127168, scale_pos_weight=1, seed=None, silent=True,
                subsample=0.970102)
```

```
In [69]: xgb_test_predictions = regr_xgb.predict(df_test)
         xgb_train_predictions = regr_xgb.predict(df_train)
         print('Train MAPE for Random Forest',MAPE(tsne_train_output,xgb_train_predictions))
         print('Test MAPE for Random Forest',MAPE(tsne_test_output,xgb_test_predictions))
```

```
Train MAPE for Random Forest 0.12787426455483747
Test MAPE for Random Forest 0.12624422280753395
```

6.7.5 Calculating the error metric values for various models

```
In [70]: train_mape=[]
         test_mape=[]

         train_mape.append(MAPE(tsne_train_output,df_train['ft_1'].values))
         train_mape.append(MAPE(tsne_train_output,df_train['exp_avg'].values))
         train_mape.append(MAPE(tsne_train_output,rndf_train_predictions))
         train_mape.append(MAPE(tsne_train_output, xgb_train_predictions))
         train_mape.append(MAPE(tsne_train_output, lr_train_predictions))

         test_mape.append(MAPE(tsne_test_output, df_test['ft_1'].values))
         test_mape.append(MAPE(tsne_test_output, df_test['exp_avg'].values))
         test_mape.append(MAPE(tsne_test_output, rndf_test_predictions))
```

```
        test_mape.append(MAPE(tsne_test_output, xgb_test_predictions))
        test_mape.append(MAPE(tsne_test_output, lr_test_predictions))
```

```
In [71]: print ("Error Metric Matrix (Tree Based Regression Methods) -  MAPE")
         print ("-------------------------------------------------------------------------------
         print ("Baseline Model -                              Train: ",train_mape[0],"      Tes
         print ("Exponential Averages Forecasting -            Train: ",train_mape[1],"      Tes
         print ("Linear Regression -                          Train: ",train_mape[3],"      Test
         print ("Random Forest Regression -                    Train: ",train_mape[2],"     Test
```

```
Error Metric Matrix (Tree Based Regression Methods) -  MAPE
-----------------------------------------------------------------------------------------------
Baseline Model -                              Train:  0.13985452815932514       Test:  0.136531
Exponential Averages Forecasting -            Train:  0.13275578806046567       Test:  0.129361
Linear Regression -                          Train:  0.12787426455483747       Test:  0.1262442
Random Forest Regression -                    Train:  0.12262595572931803      Test:  0.1264625
```

6.7.6 Error Metric Matrix

```
In [72]: print ("Error Metric Matrix (Tree Based Regression Methods) -  MAPE")
         print ("-------------------------------------------------------------------------------
         print ("Baseline Model -                              Train: ",train_mape[0],"      Tes
         print ("Exponential Averages Forecasting -            Train: ",train_mape[1],"      Tes
         print ("Linear Regression -                          Train: ",train_mape[4],"      Test
         print ("Random Forest Regression -                    Train: ",train_mape[2],"     Test
         print ("XgBoost Regression -                          Train: ",train_mape[3],"      Tes
         print ("-------------------------------------------------------------------------------
```

```
Error Metric Matrix (Tree Based Regression Methods) -  MAPE
-----------------------------------------------------------------------------------------------
Baseline Model -                              Train:  0.13985452815932514       Test:  0.136531
Exponential Averages Forecasting -            Train:  0.13275578806046567       Test:  0.129361
Linear Regression -                          Train:  0.1331672957224943       Test:  0.12912870
Random Forest Regression -                    Train:  0.12262595572931803      Test:  0.1264625
XgBoost Regression -                          Train:  0.12787426455483747       Test:  0.126244
-----------------------------------------------------------------------------------------------
```

Fourier Analysis:

```
In [111]: ### Cluster 0 data
          plt.plot(regions_cum[0][5:])
          plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.HTML object>
```

We can observe that there is so much seasonal component i.e periodic in nature so for the test samples got only seasonal components and restored that seasonal somponent with four transforms and inverse of that so that i can predict seasonl term in the series.
First tried total values to decompose ito sum ofwaves using fourier for training samples but it is highly overfitting. so got seasonal components and decomposed and that series used as feature to predict.

```
In [93]: def fourier_transform(X,n_predict=0,n_wave=144):
             ##fft
             fft_x = np.fft.fft(X)
             ##fft freq
             freq = np.fft.fftfreq(len(X),1)
             idx = list(range(len(X)))
             ##sorting
             idx.sort(key=lambda i:np.absolute(freq[i]))
             t = np.arange(0,len(X)+n_predict)
             ##restoring series
             restored_series = np.zeros(t.size)
             for j in idx[:1+n_wave*2]:
                 #amplitude
                 ampl = np.absolute(fft_x[j])/len(X)
                 #phase
                 phase = np.angle(fft_x[j])
                 restored_series += ampl * np.cos(2 * np.pi * freq[j] * t + phase)
             #return np.absolute(np.fft.ifft(np.array(xf)[idx[:1+n_harm*2]]))
             return restored_series
```

```
In [94]: from statsmodels.tsa.seasonal import seasonal_decompose
         final_fourier_train = []
         final_fourier_test = []
         for i in range(40):
             #for each cluster
             temp_data = tsne_train_output[9169*i:9169*(i+1)]
             #sesonal decompose to get seasonal
             result = seasonal_decompose(x=temp_data, model='additive',freq=144)
             #getting top harmonics and predicting test samples aslo
             fourier_decom = fourier_transform(result.seasonal,n_predict=3930,n_wave=440)
             #training data
             train_fourier = fourier_decom[0:9169]
             #testing data
             test_fourier = fourier_decom[9169:]
             final_fourier_train.extend(train_fourier)
             final_fourier_test.extend(test_fourier)
```

```
In [95]: print('train',len(final_fourier_train))
         print('Test',len(final_fourier_test))
```

```
train 366760
Test 157200
```

```
In [96]: df_train['fourier_decom'] = final_fourier_train
         df_test['fourier_decom'] = final_fourier_test
```

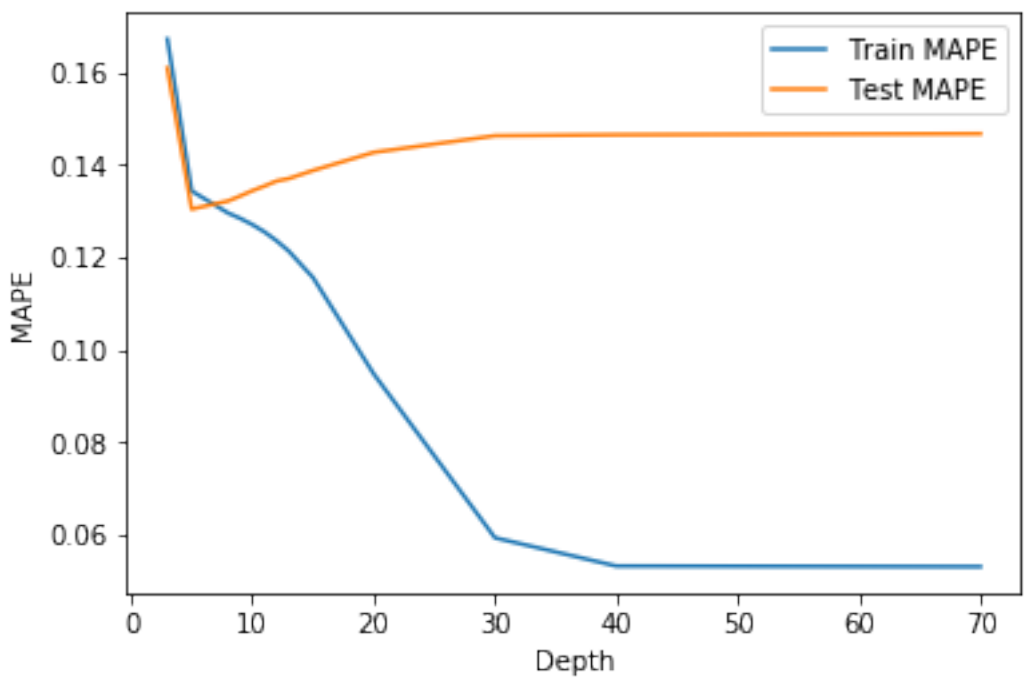
Random Forest:

```
In [114]: %matplotlib inline
          train_errors = []
          test_errors = []
          depths = [3,5,8,9,10,11,12,13,15,20,30,40,70]
          for depth in depths:
              rf_reg = RandomForestRegressor(max_depth=depth,random_state=25,n_jobs=-1)
              rf_reg.fit(df_train, tsne_train_output)
              rndf_test_predictions = rf_reg.predict(df_test)
              rndf_train_predictions = rf_reg.predict(df_train)
              train_err = MAPE(tsne_train_output,rndf_train_predictions)
              test_err = MAPE(tsne_test_output, rndf_test_predictions)
              train_errors.append(train_err)
              test_errors.append(test_err)
              print('Depth = ',depth)
              print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
              print('---')
          plt.figure()
          plt.plot(depths,train_errors,label='Train MAPE')
          plt.plot(depths,test_errors,label='Test MAPE')
          plt.xlabel('Depth')
          plt.ylabel('MAPE')
          plt.legend()
```

```
Depth =  3
Train MAPE =  0.16728436443917163 Test MAPE =  0.16101068852283543
---
Depth =  5
Train MAPE =  0.13439156983850709 Test MAPE =  0.13046582616241506
---
Depth =  8
Train MAPE =  0.12961825842858646 Test MAPE =  0.13221093012320853
---
Depth =  9
Train MAPE =  0.1284598699059012 Test MAPE =  0.13329144191842926
---
Depth =  10
Train MAPE =  0.12711815527437467 Test MAPE =  0.13445452599306545
---
Depth =  11
Train MAPE =  0.12553184315058252 Test MAPE =  0.13545007626467473
---
Depth =  12
Train MAPE =  0.12361623616236164 Test MAPE =  0.13654540137391097
```

```
---
Depth =  13
Train MAPE =  0.1213754161107801 Test MAPE =  0.13706816800400626
---
Depth =  15
Train MAPE =  0.11565534869007461 Test MAPE =  0.13882149096483415
---
Depth =  20
Train MAPE =  0.0948047545883659 Test MAPE =  0.142745203353352
---
Depth =  30
Train MAPE =  0.059301071428412976 Test MAPE =  0.14634197156608267
---
Depth =  40
Train MAPE =  0.05319305543662627 Test MAPE =  0.14655881936931597
---
Depth =  70
Train MAPE =  0.05305587142681575 Test MAPE =  0.14679449441681414
---
```

Out[114]: <matplotlib.legend.Legend at 0x1510e2fce588>



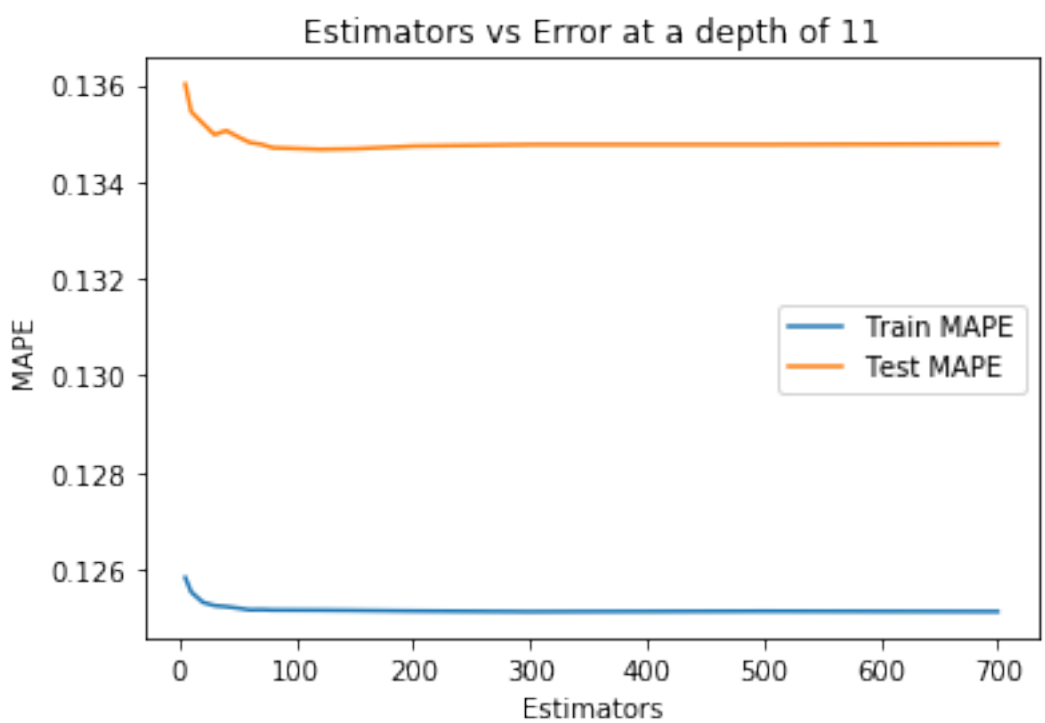
In [115]:
```
train_errors = []
test_errors = []
```

```
estimators = [5,10,20,30,40,50,60,70,80,120,150,200,300,500,700]
for i in estimators:
    rf_reg = RandomForestRegressor(n_estimators=i,max_depth=11,random_state=25,n_jobs
    rf_reg.fit(df_train, tsne_train_output)
    rndf_test_predictions = rf_reg.predict(df_test)
    rndf_train_predictions = rf_reg.predict(df_train)
    train_err = MAPE(tsne_train_output,rndf_train_predictions)
    test_err = MAPE(tsne_test_output, rndf_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('Estimators = ',i)
    print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
    print('---')
plt.figure()
plt.plot(estimators,train_errors,label='Train MAPE')
plt.plot(estimators,test_errors,label='Test MAPE')
plt.xlabel('Estimators')
plt.ylabel('MAPE')
plt.title('Estimators vs Error at a depth of 11')
plt.legend()
```

```
Estimators =  5
Train MAPE =  0.12582812866724316 Test MAPE =  0.1360186208135663
---
Estimators =  10
Train MAPE =  0.12553184315058252 Test MAPE =  0.13545007626467473
---
Estimators =  20
Train MAPE =  0.12531737112489558 Test MAPE =  0.1352079024729633
---
Estimators =  30
Train MAPE =  0.12524940026363113 Test MAPE =  0.13497404325105414
---
Estimators =  40
Train MAPE =  0.12522814827763526 Test MAPE =  0.1350576667979158
---
Estimators =  50
Train MAPE =  0.1251975347696245 Test MAPE =  0.13493027229738253
---
Estimators =  60
Train MAPE =  0.1251615084384582 Test MAPE =  0.13481348604107968
---
Estimators =  70
Train MAPE =  0.12516536840251172 Test MAPE =  0.1347733467385861
---
Estimators =  80
Train MAPE =  0.12515689422855514 Test MAPE =  0.13470386751736502
---
```

```
Estimators =  120
Train MAPE =  0.1251540103473657 Test MAPE =  0.1346693668311741
---
Estimators =  150
Train MAPE =  0.1251482869523898 Test MAPE =  0.1346810263428508
---
Estimators =  200
Train MAPE =  0.12513732820387 Test MAPE =  0.13473703022680616
---
Estimators =  300
Train MAPE =  0.12512432855481617 Test MAPE =  0.13476847268042616
---
Estimators =  500
Train MAPE =  0.12513395628124852 Test MAPE =  0.1347690460990332
---
Estimators =  700
Train MAPE =  0.12512628072054438 Test MAPE =  0.13478347713397731
---
```

```
Out[115]: <matplotlib.legend.Legend at 0x1510e31538d0>
```




```
In [135]: param_dist = {"max_depth": sp_randint(8,14),
                        "n_estimators":sp_randint(90,160),
```

```
                    "max_features": sp_randint(3,9),
                    "min_samples_split": sp_randint(2, 8),
                    "min_samples_leaf": sp_randint(1, 12),
                    "bootstrap": [True, False]}
model_rs_rf_fd = RandomizedSearchCV(RandomForestRegressor(n_jobs=-1),
                                    param_distributions=param_dist,
                                    n_iter=50,scoring=mape_score,
                                    cv=cv_custom,n_jobs=-1)
model_rs_rf_fd.fit(df_train, np.array(tsne_train_output).reshape(-1,1))
pickle.dump(model_rs_rf_fd,open('model_rs_rf_fd.p','wb'))
```

In [136]:
```
dict_scores = []
idx = 0
for i in model_rs_rf_fd.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['max_features'])
    dict_score.append(i[0]['min_samples_leaf'])
    dict_score.append(i[0]['min_samples_split'])
    dict_score.append(i[0]['bootstrap'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_rf_fd.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df = pd.DataFrame(dict_scores,columns=['n_estimators','depth','max_features',
                                              'min_samples_split','bootstrap','Test_
                                              'Test_std','Train_score'])
```

In [139]:
```
scores_df.sort_values('Test_score').head(10)
```

Out[139]:

	n_estimators	depth	max_features	min_samples_leaf	min_samples_split	\
10	142	13	3	4	2	
25	148	13	3	10	2	
35	105	13	3	2	2	
45	122	13	4	2	7	
11	125	12	3	6	4	
31	136	13	6	4	6	
2	94	13	7	10	2	
3	112	11	3	6	7	
9	140	12	7	4	4	
13	91	11	3	9	7	

	bootstrap	Test_score	Test_std	Train_score
10	True	0.128380	0.002400	0.116656
25	False	0.128453	0.002460	0.116979
35	False	0.128492	0.002525	0.110787

```
45       True    0.128520  0.002457     0.115801
11      False    0.128580  0.002423     0.118743
31       True    0.128603  0.002533     0.116353
2        True    0.128619  0.002485     0.119601
3        True    0.128831  0.002186     0.124180
9        True    0.128844  0.002504     0.120044
13       True    0.128848  0.002230     0.124886
```

```
In [97]: regr1 = RandomForestRegressor(max_features=3,min_samples_leaf=4,
                                       min_samples_split=2,n_estimators=142,
                                       bootstrap = True,max_depth = 13,n_jobs=-1)
         regr1.fit(df_train, tsne_train_output)
```

```
Out[97]: RandomForestRegressor(bootstrap=True, criterion='mse', max_depth=13,
                    max_features=3, max_leaf_nodes=None, min_impurity_decrease=0.0,
                    min_impurity_split=None, min_samples_leaf=4,
                    min_samples_split=2, min_weight_fraction_leaf=0.0,
                    n_estimators=142, n_jobs=-1, oob_score=False, random_state=None,
                    verbose=0, warm_start=False)
```

```
In [98]: rndf_test_predictions = regr1.predict(df_test)
         rndf_train_predictions = regr1.predict(df_train)
         print('Train MAPE for Random Forest',MAPE(tsne_train_output,rndf_train_predictions))
         print('Test MAPE for Random Forest',MAPE(tsne_test_output,rndf_test_predictions))
```

```
Train MAPE for Random Forest 0.12014275655561656
Test MAPE for Random Forest 0.13062953717472828
```

```
In [99]: print(df_train.columns)
         print(regr1.feature_importances_)
```

```
Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
       'exp_avg', 'fourier_decom'],
      dtype='object')
[3.15421085e-02 6.37782561e-02 1.07422849e-01 1.99216923e-01
 2.58368465e-01 2.30382953e-03 1.08840568e-03 3.11120457e-04
 3.28983210e-01 6.98483286e-03]
```

XGBoost:

```
In [123]: param_dist = {"learning_rate":uniform(0,0.4),
                        "n_estimators":sp_randint(900,1400),
                        "gamma": uniform(1,5),
                        "max_depth":sp_randint(2,5),
                        "min_child_weight":sp_randint(3,10),
                        "subsample":uniform(0.8,0.2),
                        "colsample_bytree": uniform(0.8,0.2),
```

```
                    "reg_alpha":uniform(60,300),
                    "reg_lambda":uniform(60,300)}
model_rs_xgb_fd = RandomizedSearchCV(xgb.XGBRegressor(n_jobs=-1,random_state=25),
                                     param_distributions=param_dist,
                                   n_iter=50,scoring=mape_score,cv=cv_custom,n_jobs=
model_rs_xgb_fd.fit(df_train, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_xgb_fd,open('model_rs_xgb_fd.p','wb'))
```

In [128]:
```
dict_scores = []
idx = 0
for i in model_rs_xgb_fd.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['subsample'])
    dict_score.append(i[0]['min_child_weight'])
    dict_score.append(i[0]['learning_rate'])
    dict_score.append(i[0]['reg_alpha'])
    dict_score.append(i[0]['reg_lambda'])
    dict_score.append(i[0]['gamma'])
    dict_score.append(i[0]['colsample_bytree'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_xgb_fd.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df1 = pd.DataFrame(dict_scores,columns=['n_estimators','depth','subsample','m
                                               'learning_rate','reg_alpha','reg_lamb
                                              'colsample_bytree','Test_score',
                                               'Test_std','Train_score'])
```

In [130]:
```
scores_df1.sort_values('Test_score').head(10)
```

Out[130]:

	n_estimators	depth	subsample	min_child_weight	learning_rate	\
46	1095	4	0.850206	3	0.049602	
36	1128	4	0.952067	8	0.075744	
18	949	4	0.901559	6	0.062635	
0	1289	4	0.886377	9	0.098831	
1	1235	3	0.931745	8	0.059308	
40	925	4	0.889576	7	0.105524	
9	973	3	0.971690	6	0.058182	
35	1337	4	0.843928	9	0.110045	
2	1204	4	0.948913	8	0.143786	
34	1159	4	0.953729	7	0.136508	

	reg_alpha	reg_lambda	gamma	colsample_bytree	Test_score	Test_std	\
46	180.459898	154.947350	5.790371	0.822383	0.129703	0.004196	

```
36  115.390252  246.410363  1.023009          0.946372    0.129777  0.004983
18  304.497039  161.793246  5.783577          0.987440    0.129857  0.004796
0   310.871741  102.326289  4.592620          0.928766    0.129979  0.005775
1   136.535256   99.040945  3.913000          0.920730    0.130119  0.003970
40  152.192937  337.337267  1.196343          0.908711    0.130154  0.005488
9   279.760074  165.013872  2.103654          0.856333    0.130167  0.003338
35   90.992150  113.570674  3.786735          0.820471    0.130256  0.005774
2   237.767302  134.165719  4.539142          0.851575    0.130260  0.006323
34  245.655238  247.288400  4.349678          0.988563    0.130270  0.006270

    Train_score
46     0.126990
36     0.125588
18     0.126656
0      0.123361
1      0.128281
40     0.125586
9      0.129449
35     0.122285
2      0.121866
34     0.123063
```

In [100]:
```
regr_xgb = xgb.XGBRegressor(
 learning_rate =  0.049602,
 n_estimators=1095,
 max_depth=4,
 min_child_weight=3,
 gamma=5.790371,
 subsample=0.850206,
 reg_alpha=180.459898, reg_lambda=154.947350,
 colsample_bytree=0.822383,njobs=-1,random_state=25)
regr_xgb.fit(df_train, tsne_train_output)
```

Out[100]:
```
XGBRegressor(base_score=0.5, booster='gbtree', colsample_bylevel=1,
       colsample_bytree=0.822383, gamma=5.790371, learning_rate=0.049602,
       max_delta_step=0, max_depth=4, min_child_weight=3, missing=None,
       n_estimators=1095, n_jobs=1, njobs=-1, nthread=None,
       objective='reg:linear', random_state=25, reg_alpha=180.459898,
       reg_lambda=154.94735, scale_pos_weight=1, seed=None, silent=True,
       subsample=0.850206)
```

In [101]:
```
xgb_test_predictions = regr_xgb.predict(df_test)
xgb_train_predictions = regr_xgb.predict(df_train)
print('Train MAPE for Random Forest',MAPE(tsne_train_output,xgb_train_predictions))
print('Test MAPE for Random Forest',MAPE(tsne_test_output,xgb_test_predictions))
```

```
Train MAPE for Random Forest 0.12644239536058943
Test MAPE for Random Forest 0.13697059127103967
```

```
In [102]: print(df_train.columns)
          print(regr_xgb.feature_importances_)
```

```
Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
       'exp_avg', 'fourier_decom'],
      dtype='object')
[0.11961907 0.07837334 0.08699569 0.10913069 0.12907793 0.08532269
 0.08242713 0.03693456 0.08956952 0.18254939]
```

6.7.7 Error Metric

```
In [105]: print ("Error Metric Matrix (Tree Based Regression Methods With Some fourier Feature)
          print ("-------------------------------------------------------------------------------
          print ("XGBoost Regression -                    Train: ",MAPE(tsne_train_output,xgb_tr
          print ("Random Forest Regression -              Train: ",MAPE(tsne_train_output,rndf_t
          print ("-------------------------------------------------------------------------------
```

```
Error Metric Matrix (Tree Based Regression Methods With Some fourier Feature) -  MAPE
-------------------------------------------------------------------------------------------
XGBoost Regression -                    Train:  0.12644239536058943      Test:  0.13697059127103
Random Forest Regression -              Train:  0.12014275655561656       Test:  0.130629537174
-------------------------------------------------------------------------------------------
```

6.7.8 Some Transformations and Extra Features:

```
In [106]: df_train = df_train.drop('fourier_decom',axis=1)
          df_test = df_test.drop('fourier_decom',axis=1)
```

```
In [107]: df_train.columns
```

```
Out[107]: Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
                 'exp_avg'],
                dtype='object')
```

Got Weather data from https://www.wunderground.com/

```
In [108]: #2016 weather data(hourly data)
          df_weather = pd.read_csv('Weather-2016.csv')
```

```
In [109]: #columns
          df_weather.columns
```

```
Out[109]: Index(['pickup_datetime', 'tempm', 'tempi', 'dewptm', 'dewpti', 'hum', 'wspdm',
                 'wspdi', 'wgustm', 'wgusti', 'wdird', 'wdire', 'vism', 'visi',
                 'pressurem', 'pressurei', 'windchillm', 'windchilli', 'heatindexm',
                 'heatindexi', 'precipm', 'precipi', 'conds', 'icon', 'fog', 'rain',
                 'snow', 'hail', 'thunder', 'tornado'],
                dtype='object')
```

Column Descriptions : tempm -- Temp in C
tempi -- Temp in F
dewptm -- Dewpoint in C
dewpti -- Duepoint in F
hum -- Humidity %
wspdm -- WindSpeed kph
wspdi -- Windspeed in mph
wgustm -- Wind gust in kph
wgusti -- Wind gust in mph
wdird -- Wind direction in degrees
wdire -- Wind direction description (ie, SW, NNE)
vism -- Visibility in Km
visi -- Visability in Miles
pressurem -- Pressure in mBar
pressurei -- Pressure in inHg
windchillm -- Wind chill in C
windchilli -- Wind chill in F
heatindexm -- Heat index C
heatindexi -- Heat Index F
precipm -- Precipitation in mm
precipi -- Precipitation in inches

In [110]:
```
#converting to unix time in EST format
def convert_to_unix_from_date(s):
    return time.mktime(s.timetuple())
from datetime import date
```

In [111]:
```
#converting to date time obj
df_weather['pickup_datetime'] = pd.to_datetime(df_weather['pickup_datetime'])
```

In [112]:
```
#unix time stamp
df_weather['date_unix'] = df_weather.pickup_datetime.apply(convert_to_unix_from_date
```

In [113]:
```
#getting only jan feb mar data
df_weather_req = df_weather[df_weather.date_unix < 1459486800]
```

In [114]:
```
#sorting by time to use for time binning
df_weather_req = df_weather_req.sort_values('date_unix')
```

In [115]:
```
#adding extra feature  as month
df_weather_req['Month'] = df_weather_req.pickup_datetime.dt.month
```

In [116]:
```
#adding extra feature as Hour no 1-24
df_weather_req['Hour'] = df_weather_req.pickup_datetime.dt.hour
```

In [117]:
```
##converting to bins for weather data
def convert_weather_to_bins(df,month,year):
    unix_times = np.array([[1420070400,1422748800,1425168000,1427846400,1430438400,14
                            [1451606400,1454284800,1456790400,1459468800,1462060800,14647392
```

```
        unix_times = list(unix_times + 18000)
        start_time = unix_times[year-2015][month-1]
        df_temp_month = df[df.Month == month]
        unix_time_array = df_temp_month.date_unix.values
        final_array = []
        n = (date(year,month+1, 1) - date(year,month, 1)).days
        for i in range(n*144):
            time_req = start_time + 600*i
            #print(i)
            unix_time_temp = np.min(unix_time_array[unix_time_array>time_req])
            val_temp = list(df_temp_month[df_temp_month.date_unix==unix_time_temp].values
            val_temp = np.append(val_temp,[i])
            final_array.append(val_temp)
        cols = list(df_temp_month.columns)
        cols.append('Time_bin')
        return pd.DataFrame(final_array,columns=cols)
```

```
In [118]: jan_weather_2016 = convert_weather_to_bins(df_weather_req,1,2016)
          feb_weather_2016 = convert_weather_to_bins(df_weather_req,2,2016)
          mar_weather_2016 = convert_weather_to_bins(df_weather_req,3,2016)
```

```
In [119]: #stacking all df
          df_concat_weather = pd.concat([jan_weather_2016.iloc[5:],feb_weather_2016,mar_weathe
```

```
In [120]: #train and test
          train_weather_df = df_concat_weather[:9169]
          test_weather_df = df_concat_weather[9169:]
```

```
In [121]: #for each cluster train and test
          train_weather_final = train_weather_df.copy()
          test_weather_final = test_weather_df.copy()
          for i in range(39):
              train_weather_final = pd.concat([train_weather_final,train_weather_df],axis=0)
              test_weather_final = pd.concat([test_weather_final,test_weather_df],axis=0)
```

```
In [122]: #test and train size
          print('Length of train data',len(train_weather_final))
          print('Length of test data',len(test_weather_final))
```

```
Length of train data 366760
Length of test data 157200
```

```
In [123]: df_train_new = df_train.copy()
          df_test_new = df_test.copy()
```

```
In [124]: #Mean of values in last 5 values
          df_train_new['roll_mean'] = df_train[['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1']].mean(a
          df_test_new['roll_mean'] = df_test[['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1']].mean(axi
```

```
In [125]: #Variance of values in last 5 values
          df_train_new['roll_variance'] = df_train[['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1']].va
          df_test_new['roll_variance'] = df_test[['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1']].var
```

```
In [126]: #temparature
          df_train_new['temp'] = train_weather_final['tempi'].values
          df_test_new['temp'] = test_weather_final['tempi'].values
```

```
In [127]: #Hour no 1-24
          df_train_new['hour'] = train_weather_final['Hour'].values
          df_test_new['hour'] = test_weather_final['Hour'].values
```

```
In [128]: #rain or no rain 0 or 1
          df_train_new['rain'] = train_weather_final['rain'].values
          df_test_new['rain'] = test_weather_final['rain'].values
```

```
In [129]: ### final df columns
          df_train_new.columns
```

```
Out[129]: Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
                 'exp_avg', 'roll_mean', 'roll_variance', 'temp', 'hour', 'rain'],
                dtype='object')
```

Random Forests

```
In [133]: ###stardard model
          rf_reg = RandomForestRegressor(random_state=25,n_jobs=-1)
          rf_reg.fit(df_train_new, tsne_train_output)
          y_pred = rf_reg.predict(df_test_new)
          rndf_test_predictions = [round(value) for value in y_pred]
          y_pred = rf_reg.predict(df_train_new)
          rndf_train_predictions = [round(value) for value in y_pred]
          print('Train MAPE = ',MAPE(tsne_train_output,rndf_train_predictions))
          print('Test MAPE = ',MAPE(tsne_test_output, rndf_test_predictions))
```

```
Train MAPE =  0.05137496800001066
Test MAPE =  0.12582180443367266
```

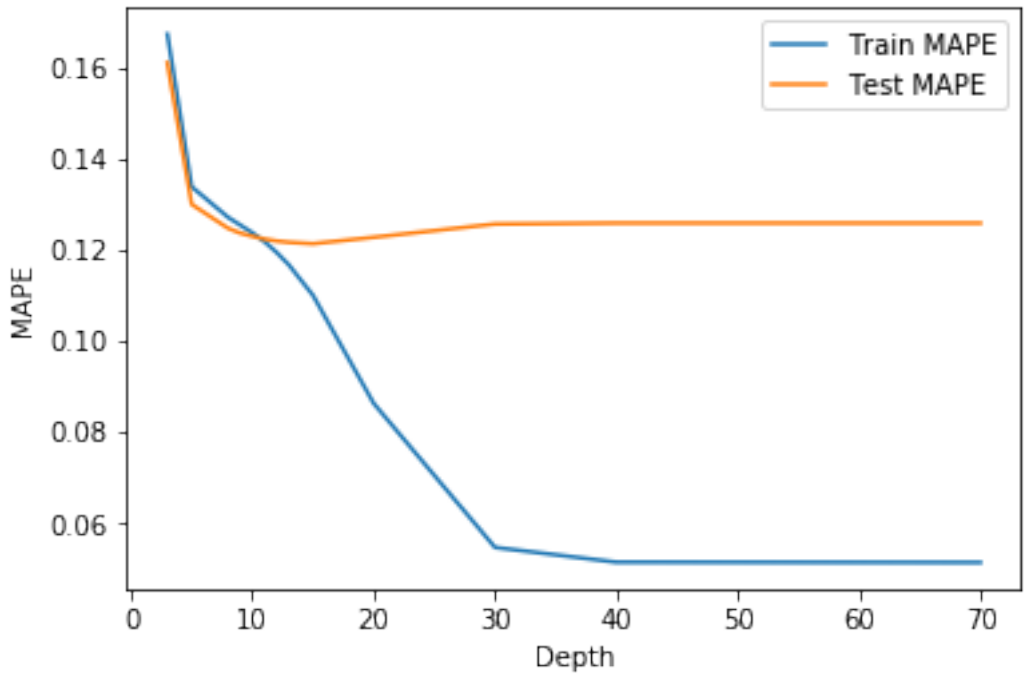
```
In [134]: %matplotlib inline
          train_errors = []
          test_errors = []
          depths = [3,5,8,9,10,11,12,13,15,20,30,40,70]
          for depth in depths:
              rf_reg = RandomForestRegressor(max_depth=depth,random_state=25,n_jobs=-1)
              rf_reg.fit(df_train_new, tsne_train_output)
              y_pred = rf_reg.predict(df_test_new)
              rndf_test_predictions = [round(value) for value in y_pred]
              y_pred = rf_reg.predict(df_train_new)
```

```
            rndf_train_predictions = [round(value) for value in y_pred]
            train_err = MAPE(tsne_train_output,rndf_train_predictions)
            test_err = MAPE(tsne_test_output, rndf_test_predictions)
            train_errors.append(train_err)
            test_errors.append(test_err)
            print('Depth = ',depth)
            print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
            print('---')
        plt.figure()
        plt.plot(depths,train_errors,label='Train MAPE')
        plt.plot(depths,test_errors,label='Test MAPE')
        plt.xlabel('Depth')
        plt.ylabel('MAPE')
        plt.legend()
```

```
Depth =  3
Train MAPE =  0.16728436443917163 Test MAPE =  0.16101068852283543
---
Depth =  5
Train MAPE =  0.13380756171394828 Test MAPE =  0.12994172155557
---
Depth =  8
Train MAPE =  0.12709996463917989 Test MAPE =  0.12469322104522743
---
Depth =  9
Train MAPE =  0.12540331078434028 Test MAPE =  0.12363430801753895
---
Depth =  10
Train MAPE =  0.12370417235493746 Test MAPE =  0.12292957654947265
---
Depth =  11
Train MAPE =  0.12175710887805043 Test MAPE =  0.12232815600044344
---
Depth =  12
Train MAPE =  0.11940239770318829 Test MAPE =  0.12188691038231729
---
Depth =  13
Train MAPE =  0.11670162078559587 Test MAPE =  0.12161405869512862
---
Depth =  15
Train MAPE =  0.10998120153138528 Test MAPE =  0.12130450821708862
---
Depth =  20
Train MAPE =  0.08629348948292898 Test MAPE =  0.12270928823459701
---
Depth =  30
Train MAPE =  0.05467812114696836 Test MAPE =  0.12562120349097247
---
```

```
Depth =  40
Train MAPE =  0.05143814718176039 Test MAPE =  0.1258316481197604
---
Depth =  70
Train MAPE =  0.05137496800001066 Test MAPE =  0.12582180443367266
---
```

Out[134]: <matplotlib.legend.Legend at 0x1547b3a0bbe0>



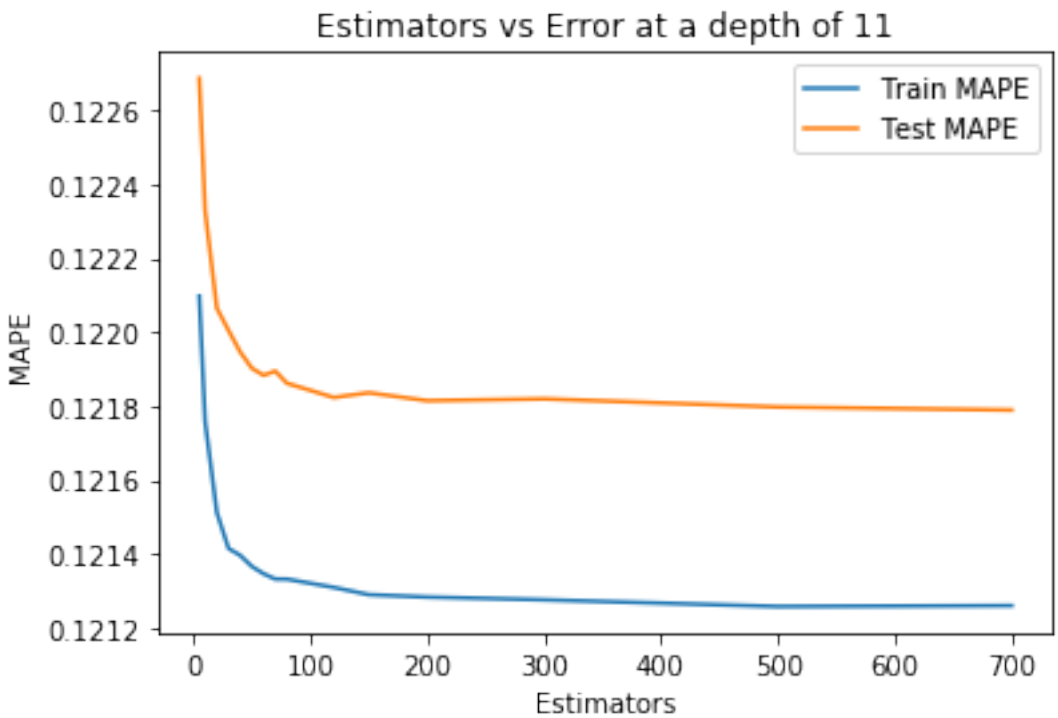
In [135]:
```
train_errors = []
test_errors = []
estimators = [5,10,20,30,40,50,60,70,80,120,150,200,300,500,700]
for i in estimators:
    rf_reg = RandomForestRegressor(n_estimators=i,max_depth=11,random_state=25,n_jobs
    rf_reg.fit(df_train_new, tsne_train_output)
    rndf_test_predictions = rf_reg.predict(df_test_new)
    rndf_train_predictions = rf_reg.predict(df_train_new)
    train_err = MAPE(tsne_train_output,rndf_train_predictions)
    test_err = MAPE(tsne_test_output, rndf_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('Estimators = ',i)
    print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
    print('---')
```

```
plt.figure()
plt.plot(estimators,train_errors,label='Train MAPE')
plt.plot(estimators,test_errors,label='Test MAPE')
plt.xlabel('Estimators')
plt.ylabel('MAPE')
plt.title('Estimators vs Error at a depth of 11')
plt.legend()
```

```
Estimators =  5
Train MAPE =  0.12209731812359606 Test MAPE =  0.12268702047868983
---
Estimators =  10
Train MAPE =  0.12175710887805043 Test MAPE =  0.12232815600044344
---
Estimators =  20
Train MAPE =  0.1215116684051294 Test MAPE =  0.12206467015050326
---
Estimators =  30
Train MAPE =  0.12141428195573288 Test MAPE =  0.12200446119676285
---
Estimators =  40
Train MAPE =  0.12139547017689728 Test MAPE =  0.12194597249884359
---
Estimators =  50
Train MAPE =  0.1213654334451244 Test MAPE =  0.12190191483586849
---
Estimators =  60
Train MAPE =  0.12134577868563344 Test MAPE =  0.12188280088230008
---
Estimators =  70
Train MAPE =  0.12133131491228344 Test MAPE =  0.12189417368467326
---
Estimators =  80
Train MAPE =  0.12133113744267178 Test MAPE =  0.12186129768453566
---
Estimators =  120
Train MAPE =  0.12130877627160305 Test MAPE =  0.12182278306809535
---
Estimators =  150
Train MAPE =  0.12128903277730627 Test MAPE =  0.12183558941698618
---
Estimators =  200
Train MAPE =  0.12128264387128662 Test MAPE =  0.12181408621922175
---
Estimators =  300
Train MAPE =  0.12127523451499997 Test MAPE =  0.12181896027738168
---
Estimators =  500
```

```
Train MAPE =  0.121257132614611 Test MAPE =  0.12179707480054588
---
Estimators =  700
Train MAPE =  0.12125943971956253 Test MAPE =  0.12178885580051148
---
```

Out[135]: <matplotlib.legend.Legend at 0x1547af8d8eb8>



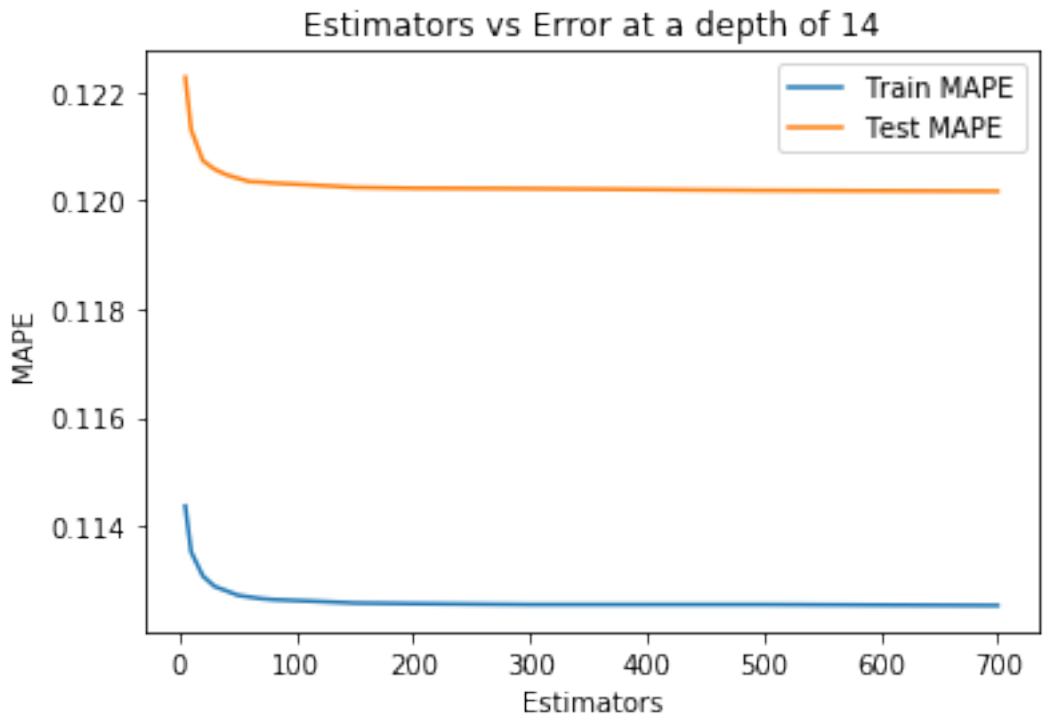
In [136]:
```
train_errors = []
test_errors = []
estimators = [5,10,20,30,40,50,60,70,80,120,150,200,300,500,700]
for i in estimators:
    rf_reg = RandomForestRegressor(n_estimators=i,max_depth=14,random_state=25,n_jobs
    rf_reg.fit(df_train_new, tsne_train_output)
    rndf_test_predictions = rf_reg.predict(df_test_new)
    rndf_train_predictions = rf_reg.predict(df_train_new)
    train_err = MAPE(tsne_train_output,rndf_train_predictions)
    test_err = MAPE(tsne_test_output, rndf_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('Estimators = ',i)
    print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
    print('---')
```

```
plt.figure()
plt.plot(estimators,train_errors,label='Train MAPE')
plt.plot(estimators,test_errors,label='Test MAPE')
plt.xlabel('Estimators')
plt.ylabel('MAPE')
plt.title('Estimators vs Error at a depth of 14')
plt.legend()
```

```
Estimators =  5
Train MAPE =  0.11436297061058864 Test MAPE =  0.12228304707002205
---
Estimators =  10
Train MAPE =  0.11351493207128778 Test MAPE =  0.12131396962410497
---
Estimators =  20
Train MAPE =  0.1130703706940881 Test MAPE =  0.12074083772635698
---
Estimators =  30
Train MAPE =  0.11287746122621742 Test MAPE =  0.1205823830512751
---
Estimators =  40
Train MAPE =  0.1127964463484962 Test MAPE =  0.12048174808573756
---
Estimators =  50
Train MAPE =  0.11271365677465842 Test MAPE =  0.12041637836453366
---
Estimators =  60
Train MAPE =  0.1126812685705311 Test MAPE =  0.1203537801665972
---
Estimators =  70
Train MAPE =  0.11265176424759318 Test MAPE =  0.12034565673633066
---
Estimators =  80
Train MAPE =  0.11263383981681588 Test MAPE =  0.12032635164322657
---
Estimators =  120
Train MAPE =  0.11259847899669331 Test MAPE =  0.12028363195700124
---
Estimators =  150
Train MAPE =  0.11257030569584282 Test MAPE =  0.12024731544522131
---
Estimators =  200
Train MAPE =  0.11255877017108515 Test MAPE =  0.12023173757306309
---
Estimators =  300
Train MAPE =  0.11254244296681275 Test MAPE =  0.12022409199163574
---
Estimators =  500
```

```
Train MAPE =  0.11254284227343897 Test MAPE =  0.1201914071310338
---
Estimators =  700
Train MAPE =  0.1125250509448704 Test MAPE =  0.12017678495655397
---
```

Out[136]: <matplotlib.legend.Legend at 0x1547af9bc860>



In [153]:
```
param_dist = {"max_depth": sp_randint(8,14),
              "n_estimators":sp_randint(90,160),
              "max_features": sp_randint(3,14),
              "min_samples_split": sp_randint(2, 8),
              "min_samples_leaf": sp_randint(1, 12),
              "bootstrap": [True, False]}
model_rs_rf_new = RandomizedSearchCV(RandomForestRegressor(n_jobs=-1),
                                     param_distributions=param_dist,
                                     n_iter=60,scoring=mape_score,
                                     cv=cv_custom,n_jobs=-1)
model_rs_rf_new.fit(df_train_new, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_rf_new,open('model_rs_rf_new.p','wb'))
```

In [156]:
```
dict_scores = []
idx = 0
```

```
for i in model_rs_rf_new.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['max_features'])
    dict_score.append(i[0]['min_samples_leaf'])
    dict_score.append(i[0]['min_samples_split'])
    dict_score.append(i[0]['bootstrap'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_rf_new.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df = pd.DataFrame(dict_scores,columns=['n_estimators','depth','max_features',
                                              'min_samples_split','bootstrap','Test_
                                              'Test_std','Train_score'])
```

In [158]:
```
scores_df.sort_values('Test_score').head(15)
```

Out[158]:

	n_estimators	depth	max_features	min_samples_leaf	min_samples_split	\
6	107	13	8	5	6	
15	156	13	8	4	6	
39	131	13	8	11	7	
46	96	13	7	9	2	
11	101	12	11	6	7	
50	112	12	7	8	2	
57	148	13	4	5	2	
35	137	11	12	11	3	
34	139	13	4	8	7	
44	103	12	7	6	3	
38	129	12	6	9	2	
1	133	11	9	11	5	
45	138	11	12	8	7	
0	97	11	11	7	4	
3	156	11	9	6	5	

	bootstrap	Test_score	Test_std	Train_score
6	True	0.125383	0.002714	0.113323
15	True	0.125401	0.002706	0.112598
39	False	0.125413	0.002947	0.112946
46	True	0.125433	0.002691	0.115951
11	True	0.125562	0.002636	0.117205
50	False	0.125670	0.002723	0.115739
57	False	0.125718	0.002631	0.110677
35	True	0.125720	0.002577	0.121396
34	False	0.125735	0.002674	0.113055
44	False	0.125754	0.002745	0.114898
38	True	0.125762	0.002594	0.119046

```
1        True    0.125765  0.002567     0.121588
45       True    0.125797  0.002614     0.120847
0        True    0.125836  0.002576     0.120742
3        True    0.125845  0.002535     0.120583
```

Got good cv scores at n_estimators = 107, depth = 13, max_features = 8, min_samples_leaf = 5, min_samples_split = 6, bootstrap = True and the Test Score =0.125383, test std = 0.002714, Train score = 0.113323.

```
In [130]: regr1 = RandomForestRegressor(max_features=8,min_samples_leaf=5,
                                        min_samples_split=6,n_estimators=107,
                                        bootstrap = True,max_depth =13,n_jobs=-1)
          regr1.fit(df_train_new, tsne_train_output)
```

```
Out[130]: RandomForestRegressor(bootstrap=True, criterion='mse', max_depth=13,
                     max_features=8, max_leaf_nodes=None, min_impurity_decrease=0.0,
                     min_impurity_split=None, min_samples_leaf=5,
                     min_samples_split=6, min_weight_fraction_leaf=0.0,
                     n_estimators=107, n_jobs=-1, oob_score=False, random_state=None,
                     verbose=0, warm_start=False)
```

```
In [131]: rndf_test_predictions = regr1.predict(df_test_new)
          rndf_train_predictions = regr1.predict(df_train_new)
          print('Train MAPE for Random Forest',MAPE(tsne_train_output,rndf_train_predictions))
          print('Test MAPE for Random Forest',MAPE(tsne_test_output,rndf_test_predictions))
```

```
Train MAPE for Random Forest 0.11621908091150168
Test MAPE for Random Forest 0.12085752841289196
```

```
In [132]: regr1.feature_importances_
```

```
Out[132]: array([8.28583401e-04, 7.54277782e-04, 1.59839695e-02, 3.13384496e-02,
                 2.76114140e-01, 4.65444360e-04, 5.36476313e-04, 2.34988906e-04,
                 5.63100612e-01, 1.06142785e-01, 1.03588264e-03, 3.77136722e-04,
                 3.07513427e-03, 1.21201399e-05])
```

6.7.9 XGBoost

```
In [163]: #basic xgboost without any change
          xgb_reg = xgb.XGBRegressor(random_state=25,n_jobs=-1)
          xgb_reg.fit(df_train_new, tsne_train_output)
          xgb_test_predictions = xgb_reg.predict(df_test_new)
          xgb_train_predictions = xgb_reg.predict(df_train_new)
          print('Train MAPE = ',MAPE(tsne_train_output,xgb_train_predictions))
          print('Test MAPE = ',MAPE(tsne_test_output, xgb_test_predictions))
```

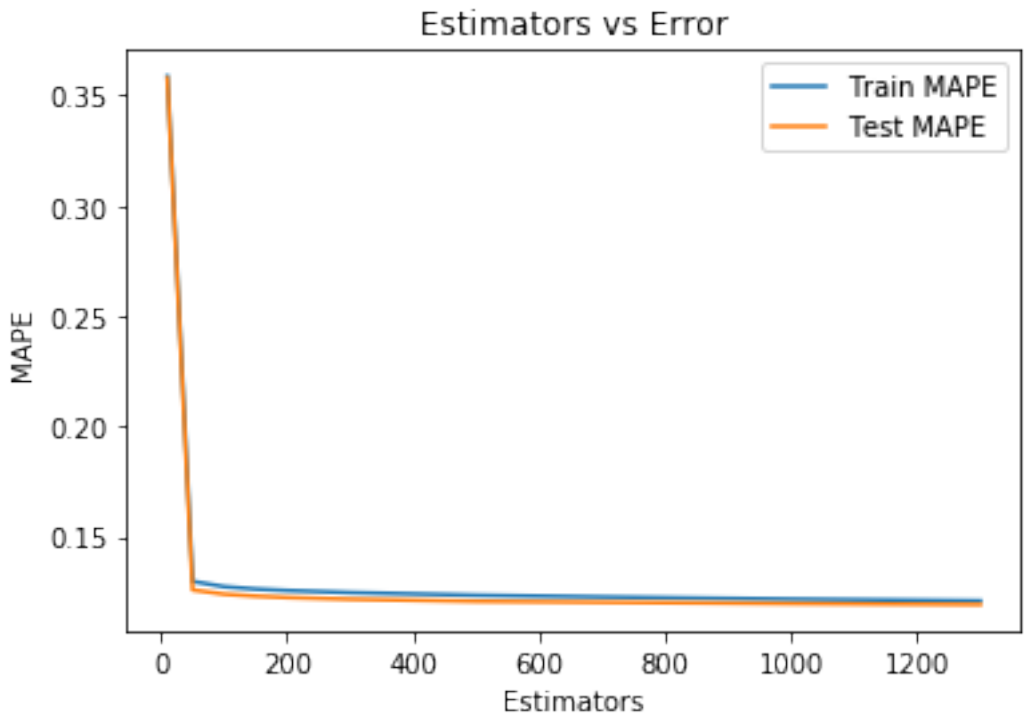
```
Train MAPE =  0.12733697530554722
Test MAPE =  0.12393554392577667
```

```
In [166]: %matplotlib inline
          train_errors = []
          test_errors = []
          estimators = [10,50,100,150,200,300,500,700,1000,1300]
          for i in estimators:
              xgb_reg = xgb.XGBRegressor(n_estimators=i,random_state=25,n_jobs=-1)
              xgb_reg.fit(df_train_new, tsne_train_output)
              xgb_test_predictions = xgb_reg.predict(df_test_new)
              xgb_train_predictions = xgb_reg.predict(df_train_new)
              train_err = MAPE(tsne_train_output,xgb_train_predictions)
              test_err = MAPE(tsne_test_output, xgb_test_predictions)
              train_errors.append(train_err)
              test_errors.append(test_err)
              print('Estimators = ',i)
              print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
              print('---')
          plt.figure()
          plt.plot(estimators,train_errors,label='Train MAPE')
          plt.plot(estimators,test_errors,label='Test MAPE')
          plt.xlabel('Estimators')
          plt.ylabel('MAPE')
          plt.title('Estimators vs Error')
          plt.legend()
```

```
Estimators =  10
Train MAPE =  0.35878827298553134 Test MAPE =  0.3578414994514295
---
Estimators =  50
Train MAPE =  0.1297407568280324 Test MAPE =  0.12585162220123933
---
Estimators =  100
Train MAPE =  0.12733697530554722 Test MAPE =  0.12393554392577667
---
Estimators =  150
Train MAPE =  0.12625587479873837 Test MAPE =  0.12303957735225869
---
Estimators =  200
Train MAPE =  0.12559626461961387 Test MAPE =  0.12242038082641088
---
Estimators =  300
Train MAPE =  0.12465727290433902 Test MAPE =  0.12164120050919572
---
Estimators =  500
Train MAPE =  0.12345260918041429 Test MAPE =  0.12069009017963292
---
Estimators =  700
Train MAPE =  0.12267019003002343 Test MAPE =  0.12027359713137785
---
```

```
Estimators =  1000
Train MAPE =  0.12167853420748949 Test MAPE =  0.11967322784979491
---
Estimators =  1300
Train MAPE =  0.12093356114515817 Test MAPE =  0.11926428481319933
---
```

Out[166]: <matplotlib.legend.Legend at 0x1547b3798438>

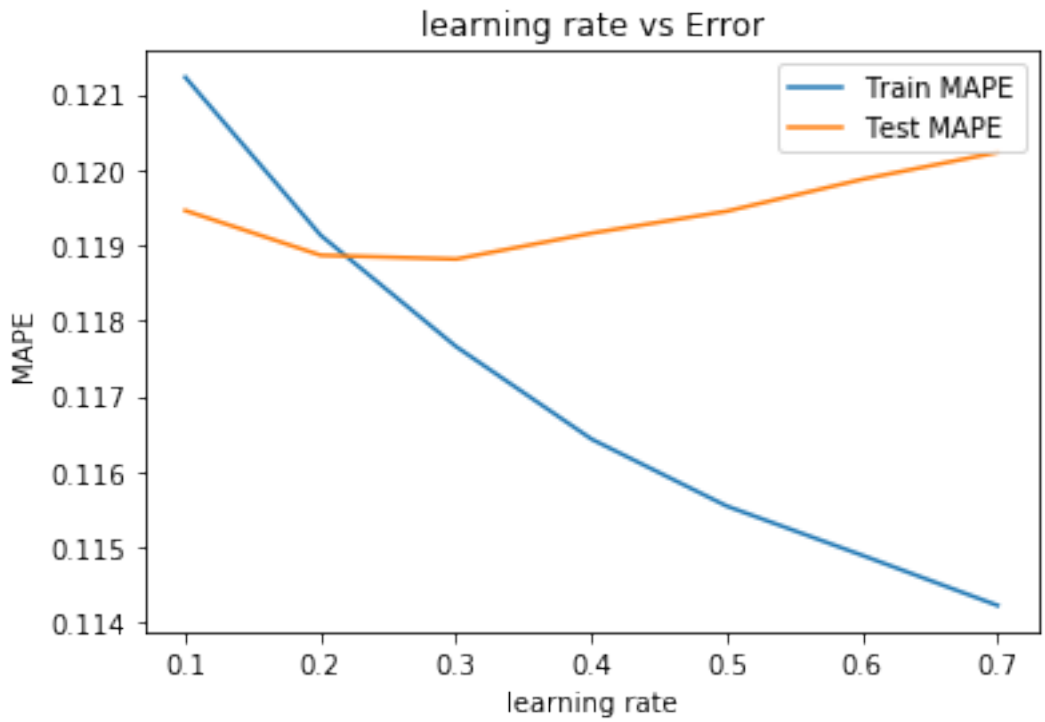


In [167]:
```
%matplotlib inline
train_errors = []
test_errors = []
learning_rates = [0.1,0.2,0.3,0.4,0.5,0.6,0.7]
for i in learning_rates:
    xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=3,learning_rate=i,random_s
    xgb_reg.fit(df_train_new, tsne_train_output)
    xgb_test_predictions = xgb_reg.predict(df_test_new)
    xgb_train_predictions = xgb_reg.predict(df_train_new)
    train_err = MAPE(tsne_train_output,xgb_train_predictions)
    test_err = MAPE(tsne_test_output, xgb_test_predictions)
    train_errors.append(train_err)
    test_errors.append(test_err)
    print('learning rate = ',i)
```

```
            print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
            print('---')
        plt.figure()
        plt.plot(learning_rates,train_errors,label='Train MAPE')
        plt.plot(learning_rates,test_errors,label='Test MAPE')
        plt.xlabel('learning rate')
        plt.ylabel('MAPE')
        plt.title('learning rate vs Error')
        plt.legend()
```

```
learning rate =  0.1
Train MAPE =  0.12122172742708551 Test MAPE =  0.11945580662795453
---
learning rate =  0.2
Train MAPE =  0.11912869519461096 Test MAPE =  0.1188641341952452
---
learning rate =  0.3
Train MAPE =  0.11765538684604113 Test MAPE =  0.11881367335782467
---
learning rate =  0.4
Train MAPE =  0.11643071342340212 Test MAPE =  0.11915629097553795
---
learning rate =  0.5
Train MAPE =  0.11554269985407563 Test MAPE =  0.11944720534884876
---
learning rate =  0.6
Train MAPE =  0.11489169695111645 Test MAPE =  0.11986971929247789
---
learning rate =  0.7
Train MAPE =  0.11422747256208886 Test MAPE =  0.12022657680559962
---
```

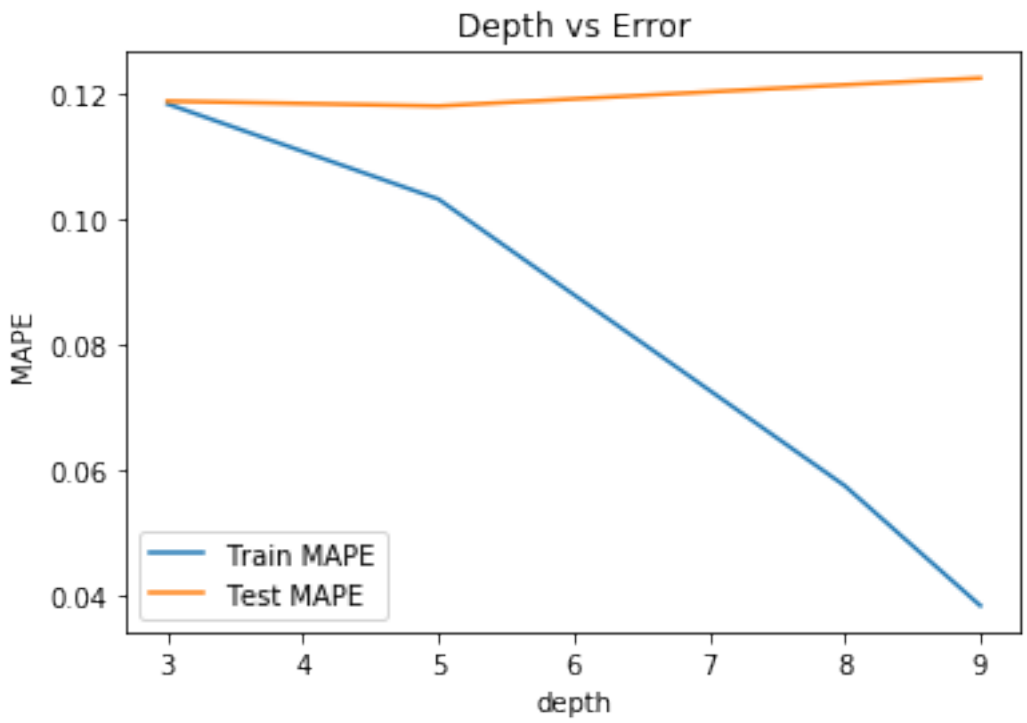
```
Out[167]: <matplotlib.legend.Legend at 0x1547b32366d8>
```




```
In [169]: %matplotlib inline
          train_errors = []
          test_errors = []
          depths = [3,5,8,9]
          for i in depths:
              xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=i,learning_rate=0.25,rando
              xgb_reg.fit(df_train_new, tsne_train_output)
              xgb_test_predictions = xgb_reg.predict(df_test_new)
              xgb_train_predictions = xgb_reg.predict(df_train_new)
              train_err = MAPE(tsne_train_output,xgb_train_predictions)
              test_err = MAPE(tsne_test_output, xgb_test_predictions)
              train_errors.append(train_err)
              test_errors.append(test_err)
              print('depth = ',i)
              print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
              print('---')
          plt.figure()
          plt.plot(depths,train_errors,label='Train MAPE')
          plt.plot(depths,test_errors,label='Test MAPE')
          plt.xlabel('depth')
          plt.ylabel('MAPE')
          plt.title('Depth vs Error')
          plt.legend()
```

```
depth =  3
Train MAPE =  0.11833766876805478 Test MAPE =  0.11877028468322443
---
depth =  5
Train MAPE =  0.10318841904302174 Test MAPE =  0.11799234677299121
---
depth =  8
Train MAPE =  0.057591772863742834 Test MAPE =  0.12136318805454355
---
depth =  9
Train MAPE =  0.03852625685088161 Test MAPE =  0.12243815680322949
---
```

Out[169]: <matplotlib.legend.Legend at 0x1547bfa4c630>

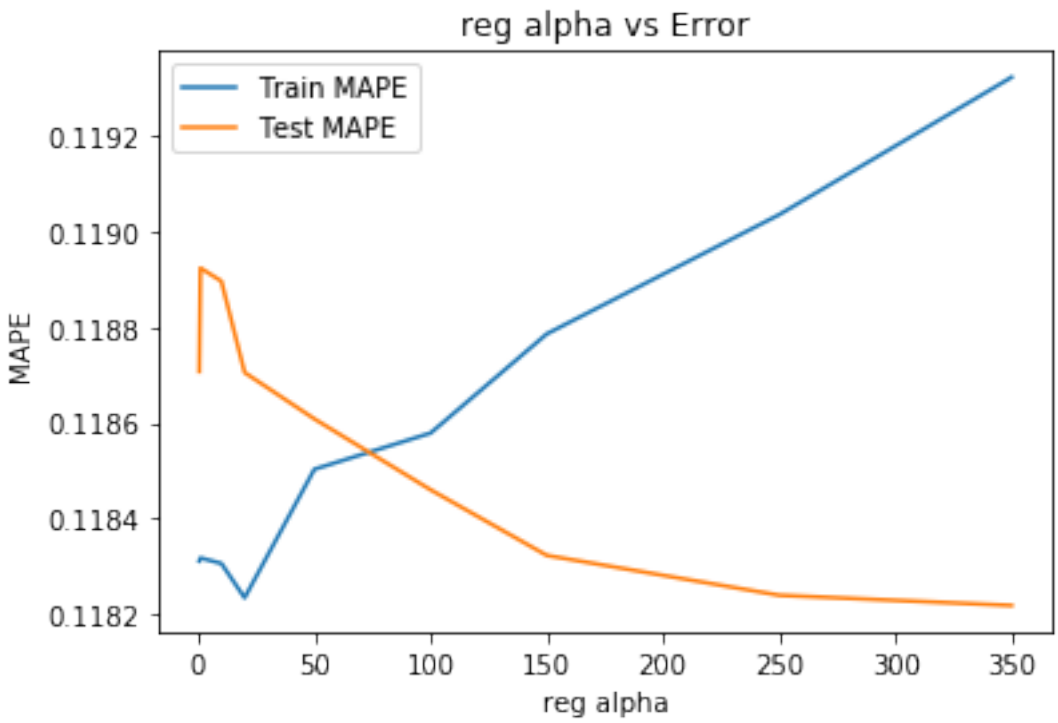


In [171]:
```
%matplotlib inline
train_errors = []
test_errors = []
alphas = [0.5,1,10,20,50,100,150,250,350]
for i in alphas:
    xgb_reg = xgb.XGBRegressor(n_estimators=1200,max_depth=3,learning_rate=0.25,reg_a
    xgb_reg.fit(df_train_new, tsne_train_output)
    xgb_test_predictions = xgb_reg.predict(df_test_new)
```

```
            xgb_train_predictions = xgb_reg.predict(df_train_new)
            train_err = MAPE(tsne_train_output,xgb_train_predictions)
            test_err = MAPE(tsne_test_output, xgb_test_predictions)
            train_errors.append(train_err)
            test_errors.append(test_err)
            print('alpha = ',i)
            print('Train MAPE = ',train_err,'Test MAPE = ',test_err)
            print('---')
        plt.figure()
        plt.plot(alphas,train_errors,label='Train MAPE')
        plt.plot(alphas,test_errors,label='Test MAPE')
        plt.xlabel('reg alpha')
        plt.ylabel('MAPE')
        plt.title('reg alpha vs Error')
        plt.legend()
```

```
alpha =  0.5
Train MAPE =  0.11831087085669463 Test MAPE =  0.11870768648528798
---
alpha =  1
Train MAPE =  0.11831748159972884 Test MAPE =  0.11892443871875345
---
alpha =  10
Train MAPE =  0.11830585734016533 Test MAPE =  0.11889691462561498
---
alpha =  20
Train MAPE =  0.11823393778004151 Test MAPE =  0.1187056795201633
---
alpha =  50
Train MAPE =  0.11850307044611869 Test MAPE =  0.11860858063603591
---
alpha =  100
Train MAPE =  0.11857884997029602 Test MAPE =  0.11846016078657741
---
alpha =  150
Train MAPE =  0.11878644504853128 Test MAPE =  0.11832273146042073
---
alpha =  250
Train MAPE =  0.119036189159535 Test MAPE =  0.11823929905309473
---
alpha =  350
Train MAPE =  0.11932400050223901 Test MAPE =  0.11821789142509814
---
```

```
Out[171]: <matplotlib.legend.Legend at 0x1547b3950a58>
```




```
In [101]: param_dist = {"max_depth": sp_randint(2,5),
                        "learning_rate":uniform(0,0.4),
                        "n_estimators":sp_randint(500,1500),
                        "min_child_weight": sp_randint(2, 8),
                        "gamma": uniform(0,4),
                        "subsample":uniform(0.8,0.2),
                        "colsample_bytree": uniform(0.7,0.3),
                        "reg_alpha":uniform(50,300),
                        "reg_lambda":uniform(50,300)}
          model_rs_xgb_new = RandomizedSearchCV(xgb.XGBRegressor(n_jobs=-1,random_state=25), pa
                                                n_iter=60,scoring=mape_score,cv=cv_cus,n_jobs=-1,
          model_rs_xgb_new.fit(df_train_new, np.array(tsne_train_output).reshape(-1,1))

          pickle.dump(model_rs_xgb_new,open('model_rs_xgb_new.p','wb'))

In [104]: dict_scores = []
          idx = 0
          for i in model_rs_xgb_new.grid_scores_:
              dict_score = []
              dict_score.append(i[0]['n_estimators'])
              dict_score.append(i[0]['max_depth'])
              dict_score.append(i[0]['subsample'])
              dict_score.append(i[0]['min_child_weight'])
              dict_score.append(i[0]['learning_rate'])
```

```
        dict_score.append(i[0]['reg_alpha'])
        dict_score.append(i[0]['reg_lambda'])
        dict_score.append(i[0]['gamma'])
        dict_score.append(i[0]['colsample_bytree'])
        dict_score.append(-i[1])
        dict_score.append((np.abs(i[2]).std()))
        dict_score.append(-model_rs_xgb_new.cv_results_['mean_train_score'][idx])
        dict_scores.append(dict_score)
        idx = idx + 1
    scores_df1 = pd.DataFrame(dict_scores,columns=['n_estimators','depth','subsample','mi
                                                   'learning_rate','reg_alpha','reg_lambd
                                                  'colsample_bytree','Test_score',
                                                   'Test_std','Train_score'])
```

In [106]:
```
scores_df1.sort_values('Test_score').head(15)
```

Out[106]:

	n_estimators	depth	subsample	min_child_weight	learning_rate	\
6	1316	4	0.842178	4	0.057302	
31	774	4	0.895843	3	0.072612	
41	963	4	0.958552	7	0.053748	
44	1259	4	0.914763	4	0.090186	
25	1057	4	0.893215	2	0.089675	
12	1338	4	0.850788	4	0.106222	
35	1205	4	0.893485	3	0.103720	
40	976	4	0.991355	2	0.099804	
2	1139	3	0.931902	3	0.079522	
45	1172	3	0.962525	4	0.065034	
18	1486	3	0.938193	4	0.071167	
28	1272	4	0.815958	2	0.109780	
23	883	3	0.871986	6	0.081953	
43	1315	4	0.836418	7	0.134539	
32	1064	3	0.820826	6	0.156207	

	reg_alpha	reg_lambda	gamma	colsample_bytree	Test_score	Test_std	\
6	273.050386	84.261124	3.683958	0.861056	0.124538	0.004169	
31	266.242736	163.834405	0.772127	0.705739	0.124626	0.003585	
41	138.866715	344.123186	1.755301	0.945922	0.124819	0.003734	
44	190.422001	128.879774	2.219924	0.861579	0.124878	0.004657	
25	336.481407	315.245742	3.830071	0.748091	0.124890	0.004723	
12	341.693610	190.218137	2.600864	0.782034	0.124910	0.004503	
35	341.470386	238.515595	1.460080	0.820331	0.124912	0.004818	
40	73.057919	344.787223	0.444643	0.783801	0.125070	0.004933	
2	337.335268	195.960869	1.365512	0.881720	0.125224	0.003258	
45	308.163289	207.963717	0.688847	0.933498	0.125276	0.003097	
18	140.522831	313.159977	0.796910	0.958962	0.125427	0.003519	
28	170.124582	287.404926	1.091602	0.719427	0.125470	0.005322	
23	234.482717	111.506303	2.754981	0.776214	0.125471	0.002968	
43	161.557205	211.436723	3.210371	0.868878	0.125535	0.004980	

```
32  295.609720  275.285258  0.676589          0.894995    0.125585  0.003661

    Train_score
6      0.120288
31     0.122253
41     0.123021
44     0.118398
25     0.120825
12     0.118775
35     0.119306
40     0.120578
2      0.123742
45     0.124263
18     0.123556
28     0.119124
23     0.124578
43     0.117183
32     0.122656
```

In [142]:
```
param_dist = {"max_depth": sp_randint(2,5),
              "learning_rate":uniform(0.2,0.4),
              "n_estimators":sp_randint(900,1600),
              "min_child_weight": sp_randint(2, 8),
              "gamma": uniform(0,4),
              "subsample":uniform(0.8,0.2),
              "colsample_bytree": uniform(0.8,0.2),
              "reg_alpha":uniform(50,300),
              "reg_lambda":uniform(50,300)}
model_rs_xgb_new1 = RandomizedSearchCV(xgb.XGBRegressor(n_jobs=-1,random_state=25), p
                                       n_iter=60,scoring=mape_score,cv=cv_custom,n_jobs=-
model_rs_xgb_new1.fit(df_train_new, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_xgb_new1,open('model_rs_xgb_new1.p','wb'))
```

In [144]:
```
dict_scores = []
idx = 0
for i in model_rs_xgb_new1.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['max_depth'])
    dict_score.append(i[0]['subsample'])
    dict_score.append(i[0]['min_child_weight'])
    dict_score.append(i[0]['learning_rate'])
    dict_score.append(i[0]['reg_alpha'])
    dict_score.append(i[0]['reg_lambda'])
    dict_score.append(i[0]['gamma'])
    dict_score.append(i[0]['colsample_bytree'])
    dict_score.append(-i[1])
```

```
        dict_score.append((np.abs(i[2]).std()))
        dict_score.append(-model_rs_xgb_new1.cv_results_['mean_train_score'][idx])
        dict_scores.append(dict_score)
        idx = idx + 1
    scores_df1 = pd.DataFrame(dict_scores,columns=['n_estimators','depth','subsample','m
                                                    'learning_rate','reg_alpha','reg_lamb
                                                   'colsample_bytree','Test_score',
                                                    'Test_std','Train_score'])
```

In [146]:
```
scores_df1.sort_values('Test_score').head()
```

Out[146]:
```
    n_estimators  depth  subsample  min_child weight  learning rate  \
13          1293      3   0.935721                 6       0.238606
10           964      3   0.823203                 4       0.288716
1            936      4   0.857527                 2       0.267529
28           949      3   0.913689                 6       0.351802
22          1142      3   0.943972                 5       0.276006

     reg_alpha  reg_lambda     gamma  colsample_bytree  Test_score  Test_std  \
13  131.101072  125.089120  3.486901          0.897799    0.126437  0.004755
10  197.908653  212.185805  0.614218          0.879060    0.126532  0.004807
1   324.277975  257.261088  0.187194          0.911496    0.126618  0.006069
28  204.852790  185.866971  1.214635          0.846680    0.126661  0.005094
22  171.311130  103.686593  3.467762          0.948933    0.126750  0.005086

    Train_score
13     0.118760
10     0.120626
1      0.116400
28     0.119369
22     0.118513
```

i tried with some models which i got best above with anoter grid search

In [169]:
```
param_dist = {"learning_rate":uniform(0,0.2),
              "n_estimators":sp_randint(1250,1400),
              "gamma": uniform(1,5),
              "subsample":uniform(0.8,0.2),
              "colsample_bytree": uniform(0.8,0.2),
              "reg_alpha":uniform(60,300),
              "reg_lambda":uniform(60,300)}
model_rs_xgb_new = RandomizedSearchCV(xgb.XGBRegressor(max_depth=4,min_child_weight=
                                      param_distributions=param_dist,
                                  n_iter=50,scoring=mape_score,cv=cv_custom,n_jobs=
model_rs_xgb_new.fit(df_train_new, np.array(tsne_train_output).reshape(-1,1))

pickle.dump(model_rs_xgb_new,open('model_rs_xgb_new2.p','wb'))
```

In [172]:
```
dict_scores = []
idx = 0
```

```
for i in model_rs_xgb_new.grid_scores_:
    dict_score = []
    dict_score.append(i[0]['n_estimators'])
    dict_score.append(i[0]['subsample'])
    dict_score.append(i[0]['learning_rate'])
    dict_score.append(i[0]['reg_alpha'])
    dict_score.append(i[0]['reg_lambda'])
    dict_score.append(i[0]['gamma'])
    dict_score.append(i[0]['colsample_bytree'])
    dict_score.append(-i[1])
    dict_score.append((np.abs(i[2]).std()))
    dict_score.append(-model_rs_xgb_new.cv_results_['mean_train_score'][idx])
    dict_scores.append(dict_score)
    idx = idx + 1
scores_df = pd.DataFrame(dict_scores,columns=['n_estimators','subsample',
                                              'learning_rate','reg_alpha','reg_lambd
                                              'colsample_bytree','Test_score',
                                              'Test_std','Train_score'])
```

In [179]:
```
scores_df.sort_values('Test_score').head(15)
```

Out[179]:

	n_estimators	subsample	learning_rate	reg_alpha	reg_lambda	gamma	\
44	1386	0.949161	0.040684	195.885297	116.788343	2.753796	
48	1322	0.917723	0.039795	130.175836	65.360413	1.659372	
2	1338	0.850493	0.051658	265.886586	119.270517	5.567378	
13	1353	0.820879	0.054719	150.610498	223.669957	2.910631	
29	1380	0.977296	0.085404	231.868072	202.971487	5.506059	
46	1373	0.898217	0.052024	326.747005	240.010386	5.060630	
6	1323	0.844394	0.040253	192.627763	221.719652	3.028794	
17	1286	0.991152	0.102631	289.538655	180.378760	3.748313	
18	1250	0.855049	0.089251	288.767520	269.148419	5.049806	
3	1279	0.957445	0.083952	244.565312	65.617164	4.692559	
34	1344	0.814841	0.025335	162.525218	185.955949	5.438559	
23	1379	0.895193	0.050555	162.494825	324.540639	4.785130	
1	1293	0.868781	0.086114	78.769599	91.868768	4.717780	
40	1289	0.812960	0.127935	342.008892	187.546503	5.845458	
35	1318	0.820046	0.085085	154.072737	359.612608	5.649383	

	colsample_bytree	Test_score	Test_std	Train_score
44	0.821209	0.124367	0.003538	0.121500
48	0.842337	0.124441	0.003454	0.121353
2	0.891632	0.124447	0.003811	0.120904
13	0.835675	0.124562	0.003627	0.121400
29	0.838655	0.124587	0.004547	0.118759
46	0.930818	0.124598	0.004083	0.121298
6	0.967625	0.124603	0.003495	0.122479
17	0.837826	0.124671	0.004747	0.118171
18	0.896539	0.124746	0.004338	0.119837

```
3           0.969939    0.124864  0.004773     0.117890
34          0.999185    0.124913  0.003275     0.124100
23          0.955013    0.124928  0.004529     0.121733
1           0.974852    0.124987  0.004404     0.117950
40          0.856565    0.125013  0.004469     0.118192
35          0.925100    0.125017  0.004514     0.120233
```

```
In [133]: regr_xgb = xgb.XGBRegressor(
           learning_rate = 0.040684,
           n_estimators=1386,
           max_depth=4,
           min_child_weight=4,
           gamma=2.753796,
           subsample=0.949161,
           reg_alpha=195.885297, reg_lambda=116.788343,
           colsample_bytree=0.821209,njobs=-1,random_state=25)
          regr_xgb.fit(df_train_new, tsne_train_output)

          xgb_test_predictions = regr_xgb.predict(df_test_new)
          xgb_train_predictions = regr_xgb.predict(df_train_new)
          print('Train MAPE for Random Forest',MAPE(tsne_train_output,xgb_train_predictions))
          print('Test MAPE for Random Forest',MAPE(tsne_test_output,xgb_test_predictions))
```

```
Train MAPE for Random Forest 0.11862352794503056
Test MAPE for Random Forest 0.11776966921391954
```

```
In [134]: print(df_train_new.columns)
          regr_xgb.feature_importances_
```

```
Index(['ft_5', 'ft_4', 'ft_3', 'ft_2', 'ft_1', 'lat', 'lon', 'weekday',
       'exp_avg', 'roll_mean', 'roll_variance', 'temp', 'hour', 'rain'],
      dtype='object')
```

```
Out[134]: array([0.08062341, 0.07103252, 0.07238124, 0.08247165, 0.09880614,
                 0.06773565, 0.07083271, 0.0333683 , 0.05954343, 0.06583746,
                 0.10899645, 0.04605625, 0.1392677 , 0.00304711], dtype=float32)
```

6.7.10 Error Metric:

```
In [136]: print ("Error Metric Matrix (Tree Based Regression Methods With Extra Weather Featur
          print ("-----------------------------------------------------------------------------
          print ("XGBoost Regression -                    Train: ",MAPE(tsne_train_output,xgb_tr
          print ("Random Forest Regression -              Train: ",MAPE(tsne_train_output,rndf_
          print ("-----------------------------------------------------------------------------
```

```
Error Metric Matrix (Tree Based Regression Methods With Extra Weather Features) -  MAPE
-------------------------------------------------------------------------------------------
```

```
XGBoost Regression -                    Train:  0.11862352794503056      Test:  0.1177696692139
Random Forest Regression -              Train:  0.11621908091150168       Test:  0.120857528412
-----------------------------------------------------------------------------------------------
```